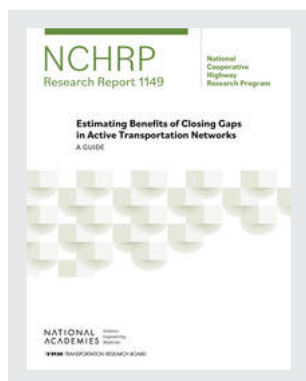


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CONTRIBUTORS

Mike Sellinger, David Wasserman, Kim Voros, Elizabeth Yarnall, Rohan Oprisko, Joseph Broach, Jennifer Dill, Jessica Schoner, Rebecca Sanders, David Von Stroh, Chris Porter; National Cooperative Highway Research Program; Transportation Research Board; National Academies of Sciences, Engineering, and Medicine

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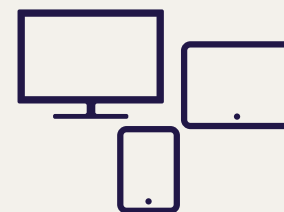
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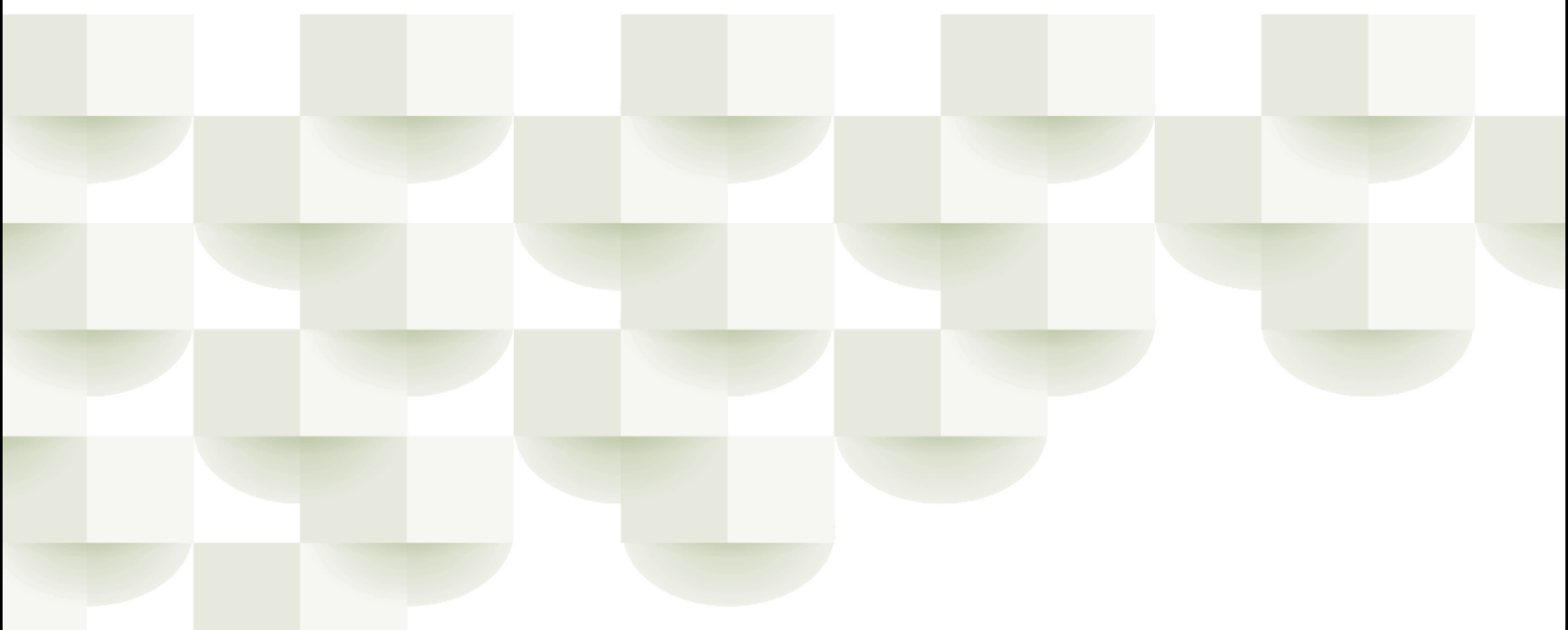
NCHRP

Research Report 1149

National
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Research Program

Estimating Benefits of Closing Gaps in Active Transportation Networks

A GUIDE



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NATIONAL COOPERATIVE HIGHWAY RESEARCH PROGRAM

NCHRP RESEARCH REPORT 1149

**Estimating Benefits of Closing Gaps
in Active Transportation Networks**
A GUIDE

Mike Sellinger
ALTA PLANNING + DESIGN
Portland, OR

David Wasserman
Kim Voros
ALTA PLANNING + DESIGN
Seattle, WA

Elizabeth Yarnall
ALTA PLANNING + DESIGN
Atlanta, GA

Rohan Oprisko
ALTA PLANNING + DESIGN
San Diego, CA

Joseph Broach
Jennifer Dill
PORTLAND STATE UNIVERSITY
Portland, OR

Jessica Schoner
SAFE STREETS RESEARCH + CONSULTING
Minneapolis, MN

Rebecca Sanders
SAFE STREETS RESEARCH + CONSULTING
Portland, OR

David Von Stroh
CAMBRIDGE SYSTEMATICS
Seattle, WA

Chris Porter
CAMBRIDGE SYSTEMATICS
Medford, MA

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Systematic, well-designed, and implementable research is the most effective way to solve many problems facing state department of transportation (DOT) administrators and engineers. Often, highway problems are of local or regional interest and can best be studied by state DOTs individually or in cooperation with their state universities and others. However, the accelerating growth of highway transportation results in increasingly complex problems of wide interest to highway authorities. These problems are best studied through a coordinated program of cooperative research.

Recognizing this need, the leadership of the American Association of State Highway and Transportation Officials (AASHTO) in 1962 initiated an objective national highway research program using modern scientific techniques—the National Cooperative Highway Research Program (NCHRP). NCHRP is supported on a continuing basis by funds from participating member states of AASHTO and receives the full cooperation and support of the Federal Highway Administration (FHWA), United States Department of Transportation.

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The needs for highway research are many, and NCHRP can make significant contributions to solving highway transportation problems of mutual concern to many responsible groups. The program, however, is intended to complement, rather than to substitute for or duplicate, other highway research programs.

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CRP STAFF FOR NCHRP RESEARCH REPORT 1149

Monique R. Evans, *Director, Cooperative Research Programs*

Waseem Dekelbab, *Deputy Director, Cooperative Research Programs, and Manager, National Cooperative Highway Research Program*

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FOREWORD

By **Michael Brooks**

Staff Officer

Transportation Research Board

NCHRP Research Report 1149: Estimating Benefits of Closing Gaps in Active Transportation Networks: A Guide presents methods and strategies to quantify the benefits of closing gaps in active transportation networks. The guide provides case studies, gap closure benefit quantification, and practitioner vetting and usability testing to support its findings. It was developed through a scan of applied methods, interviews, practical experience, and original research. The findings will prove useful to practitioners at state departments of transportation (DOTs) as well as regional and local governments.

Active transportation users include pedestrians, bicyclists, e-bike users, and those who use personal conveyances. For active transportation users, these modes not only address mobility and accessibility needs but also increase levels of physical exercise, improve access to transit, and reduce out-of-pocket travel costs. For communities, the benefits from active transportation include more mobility opportunities, healthy and active lifestyles, and local and regional development opportunities. The presence of any gaps in the network reduces the accessibility of valued destinations for system users.

Under NCHRP Project 08-149, “Estimating Benefits of Closing Gaps in Active Transportation Networks,” Alta Planning + Design was tasked to develop a guide for state DOTs and other transportation agencies to help them estimate the benefits of closing gaps in active transportation networks. The research summarized methods of identifying gaps, produced a gap typology reflecting varying user groups and contexts, developed a methodology that estimates the economic, health, and social benefits of closing gaps, provided effective approaches to prioritizing gaps for long-range planning, programming and/or project development activities, and crafted recommendations on how to effectively communicate the value of closing gaps to transportation decision-makers, local governments, communities, and other stakeholders.

In addition to the guide, the following materials can be found on the National Academies Press website (nationalacademies.org/publications) by searching for *NCHRP Research Report 1149: Estimating Benefits of Closing Gaps in Active Transportation Networks: A Guide*:

- *Web-Only Document 426: Developing a Guide for Estimating Benefits of Closing Gaps in Active Transportation Networks*, which is the research agency’s report documenting the development of the guide and the entire research effort. The following materials can also be found on the webpage for this publication:
 - *Web-Only Document 426 Appendix A (Phase I Report)*.
 - *Web-Only Document 426 Appendix B (Phase III Report)*.
 - A stand-alone Implementation Plan.



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CHAPTER 1

Introduction

About This Guide

Communities are increasingly asked to justify and evaluate investments of all types, including those to improve networks for active travel. This guide is a resource on best practices for estimating the benefits of closing gaps in active transportation networks. It has been designed for use by practitioners, including at state departments of transportation (DOTs), as well as regional and local governments. It provides guidance on identifying gaps, estimating the benefits of closing gaps, and communicating the results.

What Is a Gap in the Active Transportation Network?

The system of trails, sidewalks, bike lanes, and roads intended for walking, bicycling, or rolling is known as the active transportation network. In the United States, active transportation networks often have gaps, as when networks are interrupted by missing facilities, restricted by vehicular traffic, blocked by human-built infrastructure, or fail to cross natural features.

These gaps in the network can make active travel less desirable or make it feel less safe. People are more likely to walk, bicycle, or roll to nearby destinations if the network is complete and comfortable. Therefore, many communities are focused on increasing the connectivity of active transportation networks, including the 40,000 miles of multiuse trails across the nation. Recent estimates indicate that communities nationwide have over \$7 billion of ideas to fill trail gaps and only need the funding to implement them (Mills 2021).

Active Transportation refers to modes of travel that are human-powered and human-scale, for example, walking, bicycling, rolling a wheelchair, or operating other micromobility vehicles, such as e-scooters (FHWA 2022a).

This guide provides specific steps for estimating the benefits to society from closing a physical gap in the local active transportation network. Active transportation gaps can be caused by missing infrastructure or substandard infrastructure. The methods described in this guide are intended to be applied for one gap closure project at a time. However, once benefits are estimated for each individual project, they can then be combined to understand the total benefits of a program of multiple projects.

Why Focus on Active Travel?

Walking, bicycling, and rolling are important and desirable modes of travel. Especially for short trips, shifting to active transportation is one step toward creating more active and healthier communities, reducing emissions, and enhancing the vibrancy and resilience of local economies. Increasing physical activity is one of the most important health promotion and disease prevention

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actions for our society today, as it reduces risk factors for disease such as high blood pressure and weight gain and protects against chronic diseases including heart disease, stroke, many cancers, Type 2 diabetes, and depression (Deenihan and Caulfield 2014; U.S. Surgeon General 2015).

Shifting some of our travel to human-powered modes not only allows for additional exercise through our daily activities, it also reduces emissions and air quality pollutants. The transportation sector, including freight, accounted for 28% of U.S. greenhouse gas emissions in 2021—more than any other sector. Cars and light-duty trucks, including sport utility vehicles, pickup trucks, and minivans, make up about 16% of those emissions, emitting one billion metric tons of carbon dioxide (CO₂) yearly. The main fuel used to power transportation was petroleum based, including gasoline and diesel (U.S. EPA 2023a).

More than half of all trips in the United States are less than 3 miles in length (U.S. DOT 2024)—a distance that could potentially be traveled by bicycling, walking, or rolling under supportive conditions. Mode shift to walking and bicycling decreases both tailpipe emissions and life-cycle emissions. Substituting car trips with active transportation trips can decrease vehicle miles traveled (VMT) and associated greenhouse gas emissions, as the average passenger car releases 400 grams (0.88 pounds) of CO₂ per mile driven (U.S. EPA 2023a).

Increasingly, economists are encouraging investments in active transportation networks as part of a larger economic strategy to embrace placemaking (McKinsey & Company 2022). Bikeable and walkable communities are in demand and support vibrant economies. Properties located adjacent to trails sell quicker and for more money (Rail Trail Conversion Advisory Committee 2015). Large businesses also often look at the health of a community when deciding where to locate because of insurance costs (Lawrie et al. 2004).

Nonmotorized user safety is a growing concern as pedestrian fatalities reached a 40-year high in 2022 (Smart Growth America 2024). The Infrastructure Investment and Jobs Act (IIJA) called for all states to conduct a Vulnerable Road User Safety Assessment and integrate the analysis and results into their regularly updated Strategic Highway Safety Plans. The IIJA also created the Safe Streets and Roads for All (SS4A) funding program, allowing agencies across the country to plan for and implement projects that prioritize pedestrian and bicyclist safety. Closing active transportation gaps can be part of these proactive safety strategies and generate significant, measurable safety benefits for all road users. Streets that are designed for the comfort and safety of people walking, bicycling, and rolling also show lower rates and severity of crashes for people driving (Marshall and Ferencak 2019).

Human-powered travel also still works even when systems for power and fuel are disrupted by storms or supply failures. Facilities for active travel can be designed to help mitigate the effects of climate shocks such as flooding events and can be repurposed as important gathering spaces during public health emergencies, such as during the COVID-19 pandemic.

Benefits Covered in This Guide

This guide provides methods for estimating the benefits of projects that close gaps in active transportation networks. To estimate most types of benefits, it is necessary to first estimate the expected behavior change caused by closing the gap. The methods for this are explained in *Chapter 3: Step 2. Estimating Behavior Change*.

The benefits covered by this guide in *Chapter 4: Step 3. Estimating Benefits* include:

- **Health.** The expected reductions in mortality that can be attributed to improved health from increased physical activity, and an associated monetary value of this mortality reduction.
- **Safety.** The expected reduction in number or severity of crashes, and an associated monetary value of those avoided crashes due to safer infrastructure.

- **Emissions.** The expected reduction in pollutants that impact air quality, such as CO₂, NO_x, SO₂, and PM_{2.5}, and the monetary value of the resulting improvements in air quality.
- **Enhanced travel options.** The expected reductions in travel costs for users of active transportation networks, which typically include out-of-pocket expenditures and time costs. Other benefits may include quality improvements experienced by the user, such as reduced crowding on transit or the installation of lower-stress active transportation facilities.
- **Local economic activity.** Estimated spending from nonlocal visitors to the newly connected facility on goods, services, and lodging or increased property values due to proximity to a new facility.

Research Behind This Guide

To inform the guide, the research team relied on existing research and input from technical experts and practitioners. The effort began with a review of recent research and practice focused on estimating the impacts of closing gaps in active transportation networks. The review focused on research that could best be directly applied, including research applicable to the U.S. context, studies using revealed preference data (in contrast to stated preference or hypothetical behavior), and research with specific measures that could be applied to different contexts. A starting point was a comprehensive review of active transportation research that members of the team conducted for the American Association of State Highway and Transportation Officials' Council on Active Transportation Research Roadmap (Dill et al. 2021). Four additional sources were particularly valuable in this work. First is the *Benefit Cost Analysis Guidance for Discretionary Grant Programs*, hereafter referred to as BCA Guidance (U.S. DOT 2023a), which relies on a range of research sources to provide estimates of benefits of transportation improvements. Three other key sources are from researchers at the University of California, Davis; one documenting their efforts to create the Active Transportation Benefit–Cost Tool for Caltrans (Fitch et al. 2022) and two providing information on how to improve estimates of VMT reductions from new infrastructure for biking and walking (Volker et al. 2019a, 2019b). We supplemented these key sources with additional research, largely from peer-reviewed journals and technical reports, along with unpublished research conducted by team members. *Appendix A* includes details of how we used specific research findings to inform quantitative estimates in the guide, while *NCHRP Web-Only Document 426: Developing a Guide for Estimating Benefits of Closing Gaps in Active Transportation Networks* (which can be found on nationalacademies.org/publications) provides additional documentation of the research process.

The existing research identified several useful frameworks for estimating a range of benefits to active travel network gap closure, although most existing applications have considered entire systems or networks instead of individual projects or gaps. Most existing benefit estimation efforts have taken active travel behavior change as either given (e.g., assume that bicycling accounts for 5% of all trips) or calculated based on crude multipliers (e.g., triple the number of pedestrian trips); exceptions have relied on regional travel model outputs or primary survey data collection. The review also suggested that health and travel cost/travel option benefits to users make up a large share of overall benefits.

We engaged several experts and likely users of the guide to provide input at two key stages of the project. During the first phase, we held interviews with subject matter experts and focus groups with regional agency and state DOT staff to supplement the review of the research. That input helped in defining gaps and gap types, considering how to define benefits, and understanding current practices and how agencies would use the guide. During the second phase, to verify the research methods were technically sound, the team convened a panel of seven experts. The panel reviewed a detailed report outlining the team's research methods and

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provided written feedback, and discussed key feedback topics in a virtual meeting. The team also invited eight transportation planning practitioners to review various sections of the draft guide and provide feedback on its usability. This input was used to shape the final guide's content, methods, look, and feel.

The research phases revealed several challenging topics that are difficult to directly address through the methods presented in the guide:

- **Differences in evidence for modal behavior change.** The existing research on behavior change is much more developed for conventional bicycling than for other modes. Research on walking, e-biking, and micromobility behavior-change estimates is sparse, and research on responses to Americans with Disabilities Act accessibility improvements is even more limited.
- **Uncertainty.** The estimates in this guide are just that—estimates. There is an unknown level of uncertainty or error associated with every estimate that users should consider when applying the methods and conveying results. Users should check for updates in documents referenced, such as the U.S. DOT BCA Guidance. Users can also conduct both conservative and optimistic evaluations using lower and higher estimates of behavior change or other benefits estimates. *Chapter 5: Step 4. Communicating the Results* provides some ideas for communicating uncertainty in analysis results.
- **Destination shifts.** The guide's process for estimating behavior change includes the generation of new trips, trips shifted from other modes, and trips shifted from other routes. In general, the guide does not account for the potential that people will choose different destination locations as a result of the gap closure. For example, a person who would normally drive to a grocery store several miles away might bike to a closer store after a key gap is closed. Alternatively, someone who normally walks to a coffee shop might walk to a different one that is actually closer but is now more accessible because of the improvement in a key crossing. We did not find any research clearly related to estimating such changes and, with input from the technical review panel, concluded that such shifts were likely to be marginal. Therefore, such behavior change is not included in the guide's estimation methods.
- **Primary vs. secondary effects.** The guide focuses on the primary benefits of closing gaps within each category. For example, closing a gap with safer infrastructure has a primary benefit of reducing crashes for existing users. However, that improvement in safety may lead to other secondary benefits, such as an improved perception of safety, leading to behavior change (i.e., increased active travel), leading to health benefits. Such increases in active travel may feed back negatively into safety via increased exposure to motor vehicle traffic, and so on. Reviewers felt that limiting benefit calculations to the most primary effects was generally the correct approach. However, they felt it would be helpful to users to make this more explicit, especially in cases where a user might only apply a selection of the methods. For example, our measure of safety benefits covers only crash avoidance, but a full accounting of safety impacts spills over into health and, via behavior change due to safety perceptions, into other benefit categories. It should be clear to users that the methods presented are simplified and narrowed by necessity, and users should approach them with that understanding. Users applying only a subset of benefit categories might find it useful to note the potential unquantified, often secondary, impacts in their results. For example, after noting the estimated reduction in crashes, the user can explain that this reduction could improve people's perception of safety and, therefore, increase use of the new facility. Another example is that active transportation network improvements might also improve access to transit and thus increase transit-related active travel. We do not explicitly consider this secondary effect in our guidance, although transit impacts on walking are included in some behavior-change estimates drawn from other sources. Research on active transportation network facility improvements on transit use is even more limited than on the primary behavior shift impacts.

- **Benefit accrual.** Estimation and allocation of benefits by user type (existing vs. new) have limited research support. Some methods capture only existing or only new user benefits.
- **Overlap management.** Depending on how different benefit categories are captured, care has to be taken to avoid inadvertently double counting. For instance, when travel benefits are derived from choice models, some portion of health benefits and route safety are likely considered in individual decisions to walk, bicycle or roll, and this portion would need to be subtracted from total social benefits of public health and safety calculated separately. Pending further research, such adjustments remain largely a matter of qualitative judgment.
- **Benefit timing.** Behavior-change and associated follow-on benefits from increased active travel may have lags between project implementation and realization of benefits. Some tools referenced by this guidebook—such as the Health Economic Assessment Tool (HEAT)—explicitly consider these lags, but in general it will be up to the user to specify timing of benefits.
- **Monetization.** Which benefits should be monetized? The U.S. DOT BCA Guidance provides some estimates of monetized benefits that could be used (U.S. DOT 2023a). However, using some of them may introduce issues of overlap, and not all benefits can be monetized. Presenting only some outcomes this way might risk devaluing important non-monetized benefits. In some cases, intermediate, non-monetized benefit estimates (e.g., crashes or premature deaths avoided instead of dollar values of lives saved) might be preferable.
- **Crash modification factors (CMFs).** The safety benefit calculations rely on CMFs, which have known limitations. However, our team and technical reviewers concluded that there are no viable substitutes for this effort. Use of this guide for safety benefit estimation will probably require some degree of prior experience to properly interpret crash data and apply CMFs.
- **Level of traffic stress (LTS).** The term *traffic stress* is defined as the user's perception of risk when traveling in proximity to motor vehicle traffic (Mekuria et al. 2012). This guide relies on LTS as the primary metric to quantify the amount of change in the transportation network to estimate behavior change. However, the empirical evidence linking behavior change to LTS is limited, and the team made several assumptions in linking evidence based on other measures of network characteristics to LTS. Future research and evolutions of the LTS framework could be used to improve the methods. In particular, there is growing evidence that LTS 1 (least stress) may be too broad of a category that does not work for all user types. See *Chapter 3: Step 2. Estimating Behavior Change* for a more detailed explanation of the LTS framework.
- **Context transfer.** The guide recommends developing an estimate of travel behavior change informed by activity levels at existing active transportation facilities. When there is a lack of baseline information, such as in the case of new facilities, one solution is to use other existing facilities with similar features and in contexts resembling the gap closure project. This process of identifying similar facilities and extrapolating activity on those facilities to the project, based on other datasets such as demographics or travel information, is referred to as *context transfer*.

Analysis Approach

The general approach to estimating the benefits of closing a gap in an active transportation network involves estimating the change in existing bicyclist and pedestrian behavior and safety impacts due to the gap closure and then quantifying the benefits to society.

The general process is shown in Figure 1 and comprises the following steps:

Step 1. Understanding the Gap Closure

This step involves identifying the gap for analysis, determining the gap type, and describing the gap closure. During this initial step, it is also important to think ahead to the final step of communicating the results of the analysis.

6 Estimating Benefits of Closing Gaps in Active Transportation Networks: A Guide

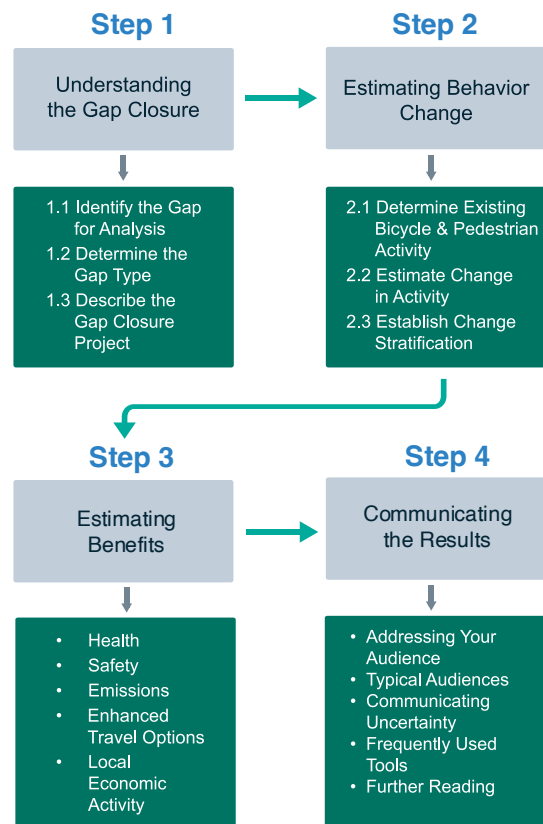


Figure 1. General analysis approach.

Step 2. Estimating Behavior Change

This step involves choosing an appropriate methodology and assembling data to quantify the change in active travel due to the gap closure. This includes estimating the impacts on route shift, mode shift, and trip generation.

Step 3. Estimating Benefits

This step involves choosing specific benefits that apply to the project and quantifying the benefits that can be expected from the gap closure. We describe each type of benefit, and then provide an overview of the methods to choose, data needed, and detailed steps to follow for analysis.

Step 4. Communicating the Results

This step involves properly communicating the estimated benefits of the gap closure, taking into consideration the needs and interests of different audiences.

How to Use This Guide

For each part of the estimation process, the guide offers two approaches: Basic or Augmented Methods. The Basic Methods are presented in the body of this guide, and the Augmented Methods are presented in the appendices. The choice between the two methods will depend on available resources (e.g., time, data, software, and knowledge).

The **Basic Methods** are analysis methods that can be used across a wide variety of geographies and project types. They are based on current best practices and research available at the time of this guide's development. Basic Methods use existing publicly available/universal datasets and

multipliers drawn from research. These methods are intended to be widely accessible and depend on national level datasets.

The **Augmented Methods** are analysis methods that provide a more informed analysis using the specific local conditions of the gap. For example, the Augmented Methods for estimating behavior change, described in Appendix C, use GIS-based network analysis to gain a better understanding of the network accessibility impacts of closing a gap. This technique better accounts for local contextual factors, like the density of destinations and existing network connectivity, when estimating new bicycling or walking activity. The Augmented Methods for estimating the benefits from enhanced travel options, described in Appendix J, also modify estimates of behavior change to better account for network effects. However, Augmented Methods require more technical expertise and effort to develop. In most cases, these methods use localized information rather than national data sources.

The guide also presents potential methods or examples of benefits estimation that require substantial investment of time and resources and result in sophisticated original analysis. Examples of these **Advanced Applications** include customized regional travel demand model runs or direct survey efforts to understand the impact of a single project. Case study examples of Advanced Applications are provided throughout *Chapter 4: Step 3. Estimating Benefits*, but specific methods are not outlined in this guide due to the highly customized nature of this type of analysis.

The following considerations can be used to select methods for a given benefit type. Basic and Augmented Methods can be mixed and matched within one project for different benefit types based on available data, resources, and need.

- Use Basic Methods if:
 - Local data for one or more required inputs are not available.
 - Time, staff expertise, or resources is a limiting factor.
- Use Augmented Methods if:
 - Staff expertise and availability of tools make network analysis using GIS software possible.
 - Local data inputs are readily available or can be generated in time to meet project deadlines.
- Consider methods similar to the approaches in Advanced Applications if:
 - A regional travel demand model exists that is capable of estimating trips for pedestrian, bicycle, or other forms of micromobility to estimate behavior change.
 - The jurisdiction has access to Location-Based Services (LBS) data.
 - Local safety performance functions (SPFs) exist that can be used in safety analysis.
 - Localized emissions factors are available.
 - Funding is available for detailed project-specific analysis.
 - Staff expertise exists to conduct original analysis.
 - The gap closure is a high-profile project.

This guide has been built to allow any user to quickly understand the needed data, assumptions, and limitations as well as final outcomes of each type of benefit estimation. It is intended to be used sequentially, starting with *Chapter 2: Step 1. Understanding the Gap Closure* and continuing through *Chapter 5: Step 4. Communicating the Results*. The *Chapter 6: Steps in Action (Analysis Walk-Throughs)* section provides walk-throughs of the methods with example calculations.

We present step-by-step guidance for **Basic Methods** in the body of the guide to make the document easy to use for most practitioners. The appendices present additional technical details behind the guidance for all methods, as well as specific guidance for using **Augmented** and **Advanced Methods** for estimating the benefits of closing gaps in active transportation networks. A references section and list of abbreviations and acronyms follow the appendices.



CHAPTER 2

Step 1. Understanding the Gap Closure

Different types of network gap closures and surrounding contexts can affect the chosen methods for estimating behavior change and the resulting benefits. This chapter provides guidance on how to identify gaps and the aspects of a gap that are important to understand prior to analysis.

There are three parts to *Step 1. Understanding the Gap Closure*:

- Step 1.1. Identify the Gap for Analysis
- Step 1.2. Determine the Gap Type
- Step 1.3. Describe the Gap Closure Project

Methods

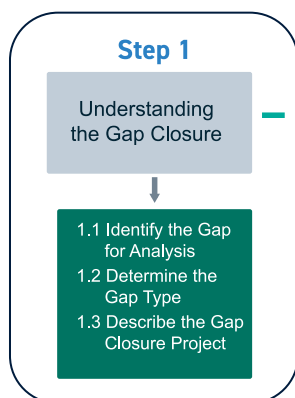
Following these steps will provide an understanding of the gap closure project prior to analyzing the benefits.

Step 1.1. Identify the Gap for Analysis

First, identify the network gap. You can identify infrastructure gaps through a variety of methods, including GIS-based analysis; public engagement; review of long-range and capital improvement plans; review of strategic safety plans or safety action plans; and field visits. Gaps in active transportation networks typically fall into two categories:

- **Missing infrastructure.** A lack of transportation infrastructure that prevents people walking, bicycling, or rolling to reach destinations safely and conveniently. Examples include missing bike lanes or bike routes, missing sidewalks, missing connections (e.g., trail gaps), and inaccessible crossings due to natural features (e.g., water bodies or steep hillsides) or human-constructed infrastructure (e.g., highways or cul-de-sac neighborhood design).
- **Substandard infrastructure.** A location where infrastructure exists but does not meet the needs of transportation users. This might be a sidewalk with missing curb ramps, or a standard bike lane on a busy roadway with inadequate separation from motor vehicle traffic. It could also be unmaintained infrastructure such as sidewalks broken up by tree roots or bike lanes with potholes or frequent drainage grates. It could even be a non-intuitive route that weaves through city streets to avoid tough hills and simply needs better wayfinding signage to direct those walking, bicycling, or rolling.

It is also common for gaps to be identified and defined by a locally adopted policy. For example, a jurisdiction might identify a list of high-priority trail gaps as part of a master planning process or a Vision Zero Action Plan. While these gaps represent physical locations where infrastructure



improvements are needed, they also have a basis in policy—such as a policy of prioritizing connections to schools—that makes their closure a priority for the jurisdiction.

To the extent possible, this guide relies on available methods and literature to quantify the benefits of physical gap closures of active transportation networks. There are other intangible, non-quantifiable benefits of gap closure that are not covered in this guide. The quantification of benefits in this guide is limited to evaluating gap closure of active transportation networks (i.e., physical infrastructure). Programmatic improvements, such as temporary street closures or changes in traffic enforcement, are outside of the scope of this guide.

Step 1.2. Determine the Gap Type

Next, determine what type of gap is being assessed. What is the barrier form and whom does the gap affect? These factors can be used to describe the circumstances and concerns associated with each gap. The categories of barrier forms and user types are not mutually exclusive for a given location or gap; a particular location can have multiple types of users and barrier forms.

Barrier Form(s): Why Is It a Gap?

Physical obstacles to active travel are often called barriers and can differ in form. Specific methods for estimating additional active travel due to removing a physical barrier are covered in *Chapter 3: Step 2. Estimating Behavior Change*, and these methods depend on the type of barrier form involved. If the chosen gap crosses a particular barrier, describe the barrier form which is associated with the gap.

Barrier forms include:

- Missing facilities (e.g., no pedestrian or bicycle facility).
- Natural features (e.g., water bodies or steep hillsides).
- Human-constructed infrastructure (e.g., freeways).
- Corridor design and operation: conditions that create high-stress or high-risk intersections and segments (e.g., extra-wide roads without pedestrian refuges, crosswalks at skewed-angle intersections, or high traffic arterials with blind corners and sparse crossings).
- Improperly constructed facilities (e.g., narrow bike lanes).
- Lack of maintenance on an existing facility (e.g., sections of broken pavement, debris build-up blocking facilities, chronically flooded facilities, or eroded trails).
- Issues that result from network design (e.g., cul-de-sacs or dead ends).

To identify a barrier form, review local topography, network data, maintenance data, and data related to natural features that could act as physical obstacles.

User Type(s): Who Is Affected by the Gap?

Determine which users will be affected by the gap closure to properly estimate behavior change in later steps. User type may be described by transportation mode, age, confidence level, or reliance on wheelchairs or other mobility devices.

Transportation modes to consider include:

- Walking and rolling (including the use of wheelchairs, strollers, or other mobility devices).
- Bicycling.
- E-micromobility devices (including e-scooters).
- Rollerblading.
- Skateboarding.

State of Research for Micromobility

Micromobility refers to travel achieved by means of small, low-speed transportation devices, including bicycles, e-bicycles, and scooters. The state of research for micromobility is evolving, and our guidance to practitioners is to use the research and methods developed for bicycling when evaluating other micromobility modes. Future research should refine methods to better account for differences between e-bikes, scooters, traditional bicycles, and other personal mobility devices.

Similarities between bicycling and other micromobility modes support the application of findings regarding bicycling to the other modes. For example, studies have shown that e-scooter users prefer riding in bike lanes, on lower-traffic streets, and on sidewalks (Guo and Zhang 2021; Yang et al. 2022; Zhang et al. 2021). Additionally, recent work, funded by the Oregon Department of Transportation (ODOT) and conducted by Metro [a metropolitan planning organization (MPO) in Portland, OR] and the City of Portland, developed a model of e-scooter use that included a measure of network quality (i.e., how well the network linked origin and destination zones) (Broach et al. 2022). Lacking scooter user preferences, the analysis relied on measures of bicycle network quality taken from the region's bicycle model and revealed preference route data. Both network quality and distance proved to be significant predictors of e-scooter levels, even after controlling for additional land-use context characteristics. This line of research indicates that micromobility would be influenced similarly by facilities' level of traffic stress, with some potential changes in how sensitive they are to hilly terrain and other variables.

Resolving gaps may look markedly different depending on the user mode: a stairway may connect downhill and uphill neighborhoods for able-bodied/ambulatory pedestrians, but not for bicyclists or pedestrians using wheelchairs or other assistive devices, thus constituting a gap. An on-street protected bikeway will not serve pedestrians in the same way it serves bicyclists, nor does it serve all ages and types of bicyclists equally well.

Related to this are distinct concerns for specific age groups: children and older adults. Some corridors or crossings may be fine for people with higher risk tolerances (e.g., strong and fearless cyclists), but are experienced as barriers by less confident cyclists and people with different abilities.

Some gap closures located through or leading to park systems may better serve recreational users, who may also prefer longer routes for exercise. However, this guide focuses on the shared interests of utilitarian and recreational users and does not differentiate between trip purposes.

Follow the steps outlined in this section to determine the gap type and develop a project narrative. Prior to undertaking analysis, we recommend you take the following steps:

1. Identify the project name and physical extents.
List the roadway name and cross streets bounding the project.
2. Determine the gap type.
Answer the questions listed earlier in Step 1.2 to determine the barrier form and user types affected by the gap. The factors related to barrier form and user type described in this section are not mutually exclusive or an exhaustive list, but rather are a guide for characteristics to consider when determining the gap type.

Step 1.3. Describe the Gap Closure Project

The specific gap types and users affected should help guide the gap closure project. Understanding the characteristics of the gap closure project is important to properly estimate behavior change and benefits.

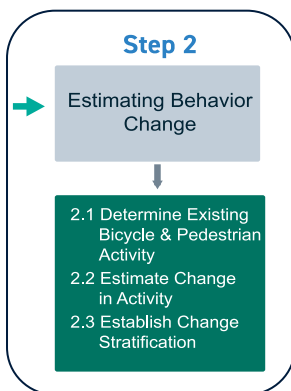
For a bicycle project, the gap closure project might be a trail, protected bike lane, bicycle boulevard, or standard bike lane. For a pedestrian project, it might be a trail or sidewalk infill. Crossing infrastructure (e.g., bridges, new traffic signals, or crosswalks) may also be considered for either mode.



CHAPTER 3

Step 2. Estimating Behavior Change

This chapter presents methods for estimating the changes in travel behavior due to the gap closure.



For any given gap closure, we present a collection of methodologies that identify how many active travelers will use the new or upgraded facility, and whether those travelers are making new trips, shifting from vehicle trips, shifting from public transit or another active mode, or shifting routes. To the extent possible, this guide uses the LTS framework (i.e., perceived safety due to physical proximity to motor vehicle traffic) to quantify changes in user behavior due to network gap closure.

Background on Behavior Change

The behavior-change estimates are a primary input used in calculating health, emissions, and enhanced travel options benefits. Note that while expected travel behavior changes, such as modal shift, often underlie the safety benefits methods, projects may have safety benefits to existing users that can be calculated independent of any new travel by active transportation.

Why Model Behavior Change

Behavioral pathways are important because they speak to how different kinds of benefits accrue. A behavioral pathway is typically defined as a set or series of actions, behaviors, or decisions that follow a specific set of circumstances that tend to be repeated or followed consistently. For example, a new shared-use path is built directly between a neighborhood and a nearby park, so a bicyclist is likely to use that path to get to their destination instead of more circuitous local roads.

A network gap closure can trigger a change in a user's behavior (see Figure 2), and the behavior change observed is related to how benefits accrue. When gaps in the active transportation network are filled, existing active transportation road users also benefit through improved quality. For example, a person walking along a roadway without a sidewalk will benefit after a sidewalk is installed.

For people walking, bicycling, and rolling, benefits can be realized through reduced risk of crashes and severe crashes, as well as a trip that feels safer and more comfortable. For example, a bicyclist may have been riding in mixed traffic, but once the gap is filled with a shared-use path, they use that path. Many of the countermeasures that increase safety for active travelers, such as reduced speeds and roadway reallocation, also make streets measurably safer for motorists, so the benefits may be broadly shared among all road users. A broader mix of benefits is triggered

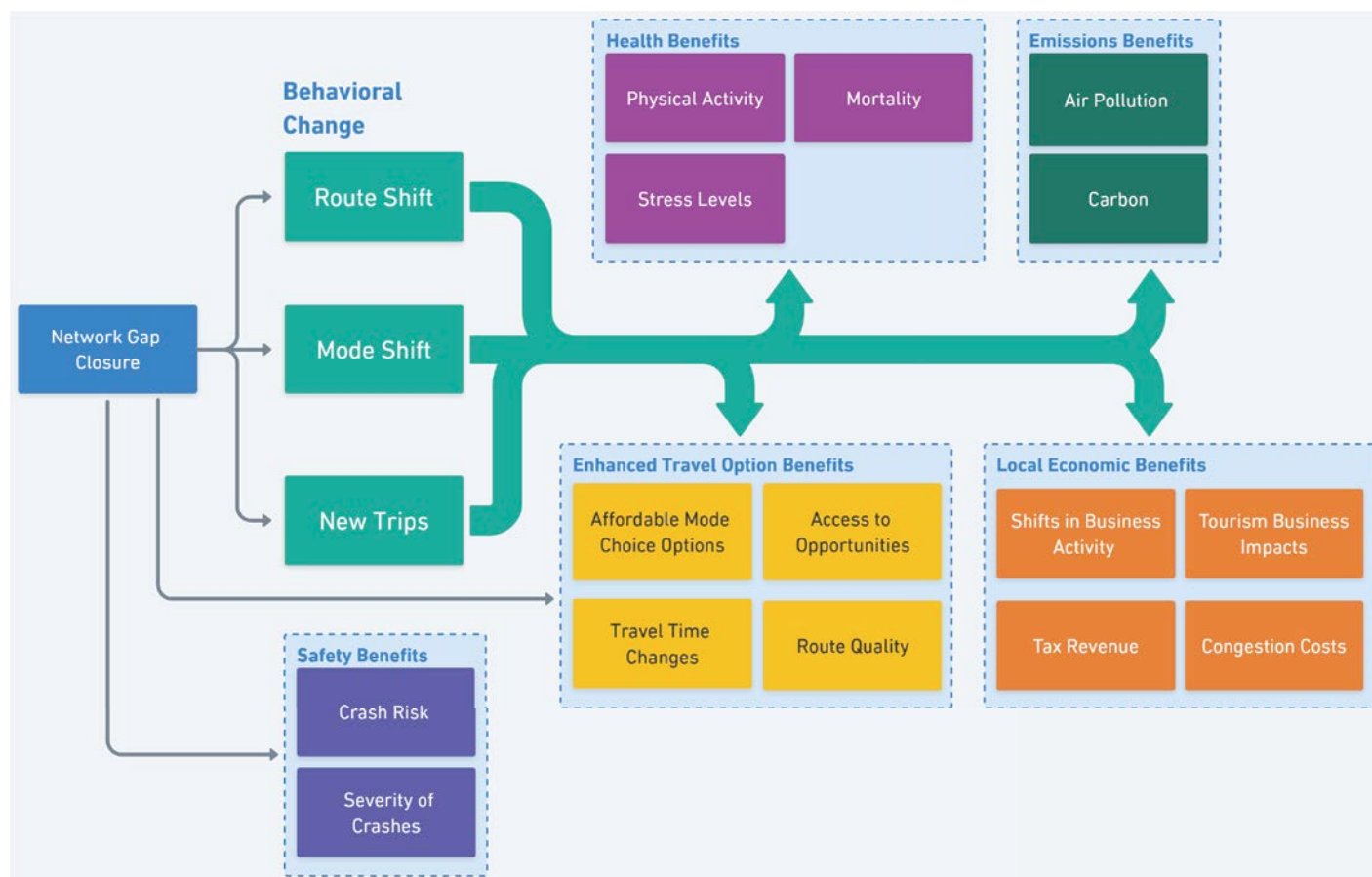


Figure 2. Network gap closure as a behavior-change trigger.

when the active transportation gap closure induces changes in travel behavior. For example, health benefits can accrue when a gap closure leads more people to choose active travel options such as walking or bicycling.

Potential behavior changes that can occur due to a network gap closure are shown in Figure 2. While this guide offers methods for monetizing certain benefits illustrated in this figure, it is important to note that some of these benefits are best explored qualitatively through narrative. In some cases, this is due to the lack of sufficient evidence in current literature to support development of standard benefit calculation methods for some types of benefits. In others, the benefits simply do not fit within traditional framework of benefit-cost analysis (BCA). On the other hand, there are well-established monetization methods for other benefits such as reduced emissions and improved health.

It is possible for a gap closure to result in one or more types of behavior change based in part on the gap type. For example, construction of several miles of separated bikeway may result in route shift, mode shift, or activity shift/trip generation:

- **Route shift.** Active transportation users may switch from a parallel route to the route with the new gap closure improvement. The new route may be safer and more comfortable. It may provide a faster, more direct option, if they no longer must detour around the gap.
- **Mode shift.** People may switch from driving to walking, bicycling, or micromobility. Any reductions in VMT can result in less air pollution and noise pollution, and these reductions are enjoyed by active transportation users along the network and by people who live, work,

and play nearby. Increased active travel also means increased physical activity and reduced burden of chronic disease among people whose modes change.

- **Activity shift/trip generation (new trips).** Because there is a safer, more convenient, or more comfortable option now that a network gap has been closed, people may take new active trips. For example, they may visit friends more often; make more frequent quick trips to the store; start walking, running, or biking for exercise; or venture out for coffee in the afternoon instead of staying at their desk. Similar to mode shift changes, these new trips increase physical activity, bringing health benefits. They may additionally generate economic benefits, such as increased tourism.

As highlighted in the Federal Highway Administration (FHWA) *Guidebook for Measuring Multimodal Connectivity* (FHWA 2018a), the preferences and sensitivities of people who walk, bicycle, or use micromobility devices are influenced by various factors related to the conditions they encounter while traveling on roadways. Among these factors, the speed and volume of motor vehicle traffic on the same routes are particularly significant. Understanding transportation patterns requires recognizing that individuals possess diverse physical capabilities and personal thresholds and constraints when it comes to the conditions they find acceptable for their travel experiences.

Level of Traffic Stress

This guide uses LTS to quantify behavior change to estimate the benefits of gap closure. LTS provides a way to measure the effect of user sensitivity to network quality. It is used in both the Basic and Augmented Methods. LTS is a four-point scale that estimates the level of comfort for people biking (Figure 3) or walking (Figure 4) on a given roadway segment. An LTS 1 facility is considered comfortable for people of all ages and abilities, whereas LTS 4 represents a high-stress facility, the least comfortable for all users. Most people avoid LTS 4 facilities.

The original bicycle LTS was developed by researchers at the Mineta Transportation Institute and Northeastern University (Mekuria et al. 2012); since its inception in 2012, the method has become an industry standard metric and has been adapted in a variety of ways to consider both pedestrians and bicyclists. While it is commonly accepted that bicycle LTS measures the user's perceived danger from riding in proximity to motor vehicle traffic, the 1- to 4-point scale used to measure pedestrian LTS can additionally be used to measure the amount of attention required to safely interact with motor vehicles in a given environment (ODOT 2023).

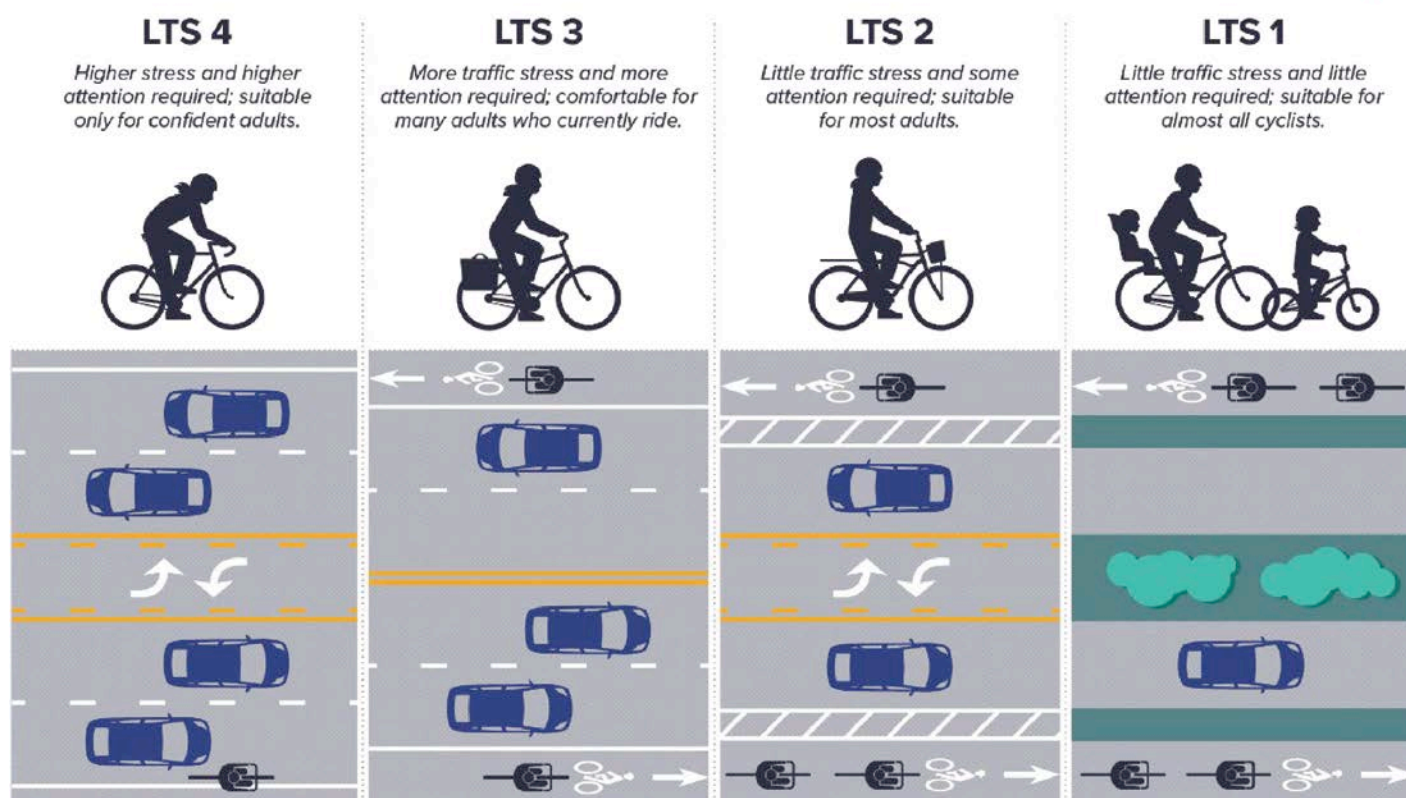
Data Needs

Data sources for behavior-change analysis may come from a variety of sources. The following types of data may be required:

- Data used to estimate general travel demand:
 - Recent traffic counts
 - Ongoing bicycle and pedestrian count programs
 - Mobile/cell phone data sources
 - Travel demand model outputs
- Data used to understand how many trips can access the new facility:
 - Transportation network (roadway functional classifications, bicycle and pedestrian facility data, etc.)
 - Destinations
 - Demographic data

BICYCLE LEVEL OF TRAFFIC STRESS

INCREASING LEVEL OF COMFORT, SAFETY, AND INTEREST IN BICYCLING FOR TRANSPORTATION



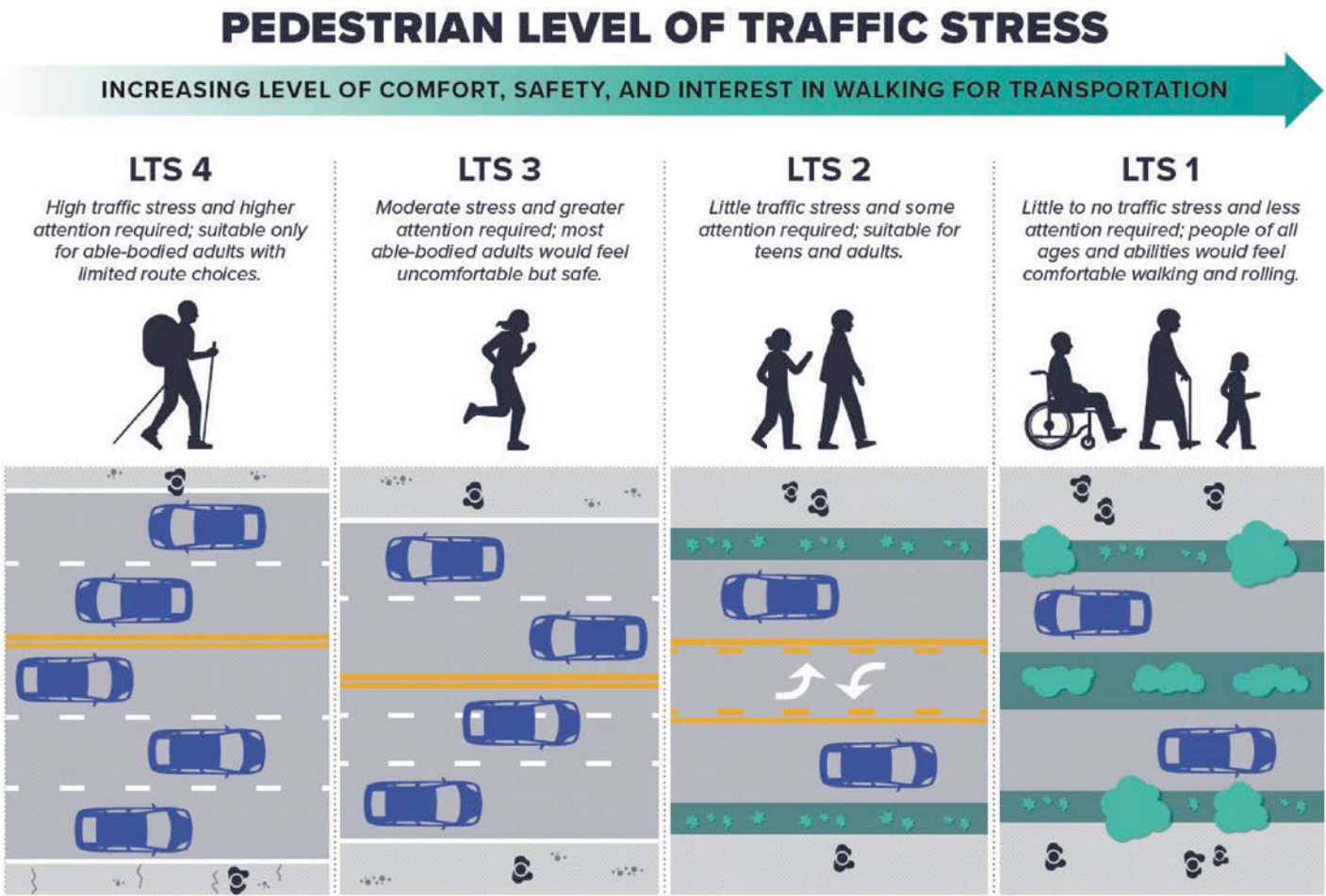
Source: Adapted from Mekuria et al. (2012).

Figure 3. Bicycle level of traffic stress.

Method Selection

The methods primarily focus on benefits estimation for pedestrians and bicyclists, drawing on assumptions and findings from pedestrian- and bicyclist-focused research. Additionally, the research support for behavior-change and benefit estimation is better established for bicycling than walking. This is due to a variety of factors, including limitations on available data at a scale appropriate for pedestrian travel and additional complexities of modeling pedestrian behavior. Where appropriate, we adapted guidance to estimate impacts on walking. We anticipate these methods and numbers will be updated as pedestrian research advances, particularly research focused on quantifying behavioral responses to specific network enhancements.

This guide presents two methods for estimating behavior change: Basic and Augmented. An example of an Advanced Application is also provided in Appendix D. As a reminder, the Basic Method includes analysis methods that can be used across a wide variety of geographies and project types. The **Basic Method for estimating behavior change** uses either percent change or context transfer processes depending on whether the project has a baseline of walking and bicycling activity, as illustrated in Figure 5 and Figure 6. Key terms and concepts from Figures 5 and 6 are explained in the sections that follow.



Source: Adapted from Oregon DOT (2023).

Figure 4. Pedestrian level of traffic stress.

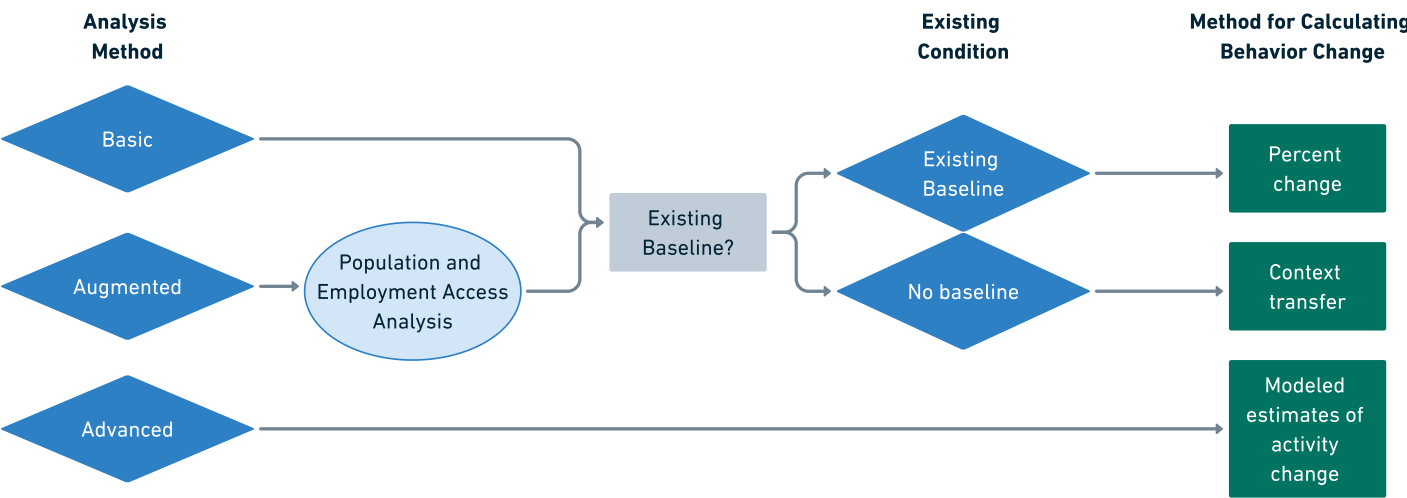
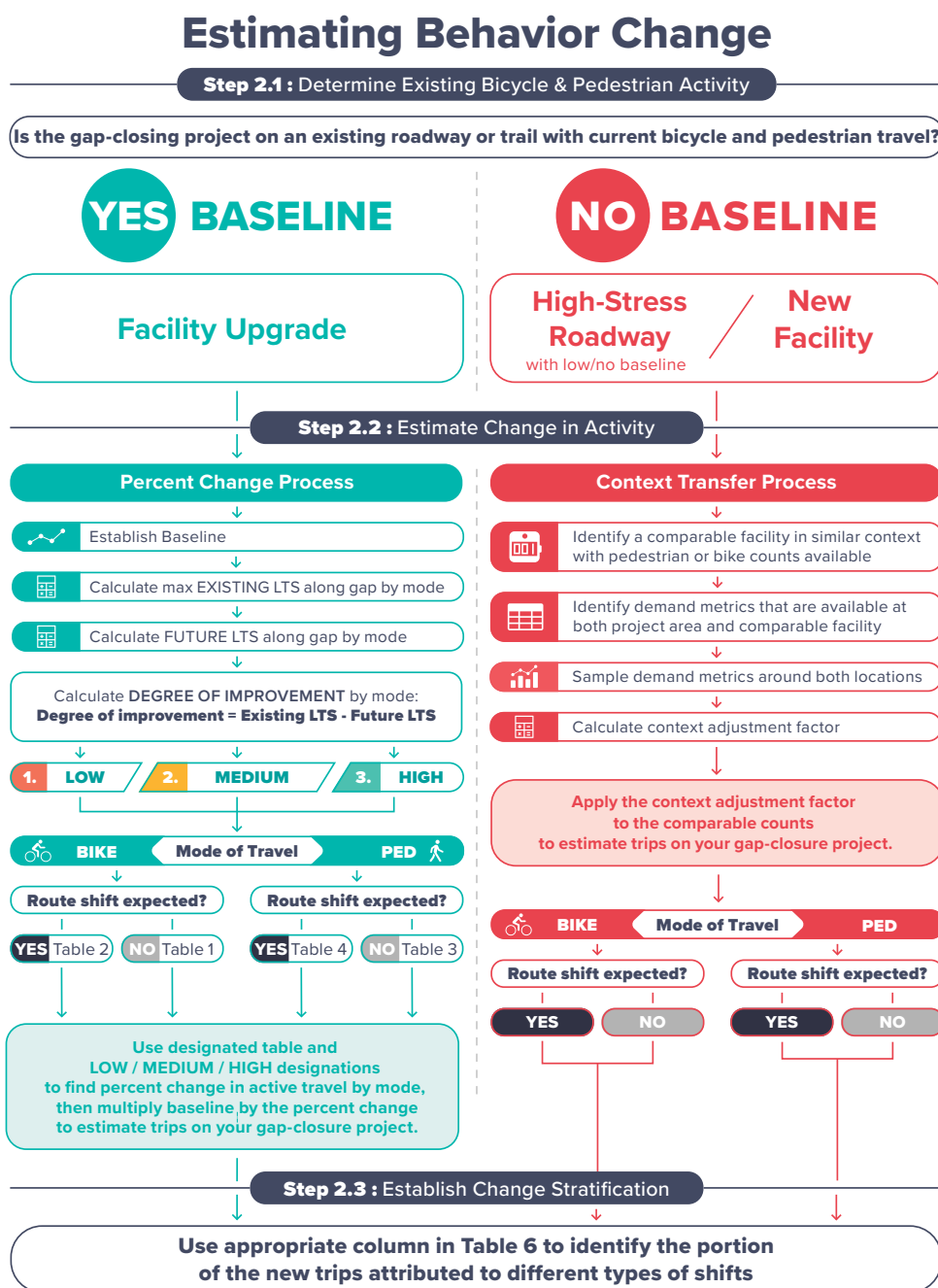


Figure 5. Basic, Augmented, and Advanced Application Methods for calculating behavior change.



Basic Method Steps

The general process to estimate behavior change using the Basic Methods consists of a few steps:

- Step 2.1. Determine Existing Bicycle and Pedestrian Activity. Decide whether the project type should have a baseline of walking and bicycling activity estimate (i.e., volume of trips per day), and, if applicable, determine what that baseline estimate would be for the project.
- Step 2.2. Estimate Change in Activity (i.e., volume of trips per day).
- Step 2.3. Establish Change Stratification (i.e., mode shift/route shift/new travel proportions).

These steps are illustrated in Figure 6.

Steps in Action

Refer to *Chapter 6: Steps in Action (Analysis Walk-Throughs)* for detailed walkthroughs of the methods with example calculations. The examples include walkthroughs of both the percent change process and context transfer process.

Step 2.1. Determine Existing Bicycle and Pedestrian Activity

The purpose of this step is to identify whether the analysis should start by using an estimate of the current number of active transportation trips at the project location, also known as a baseline. In some cases, there are no existing trips, and an estimate of new trips will be made based on similar projects in other locations. Whether there should be a baseline estimate depends on the type of project and whether there would be any existing trips:

- **New facilities.** Completely new infrastructure, such as bridges and trails where no access currently exists, will not have baseline activity.
 - For these projects, proceed to *Step 2.2* and use the **Context Transfer Process** guidance.
- **Roadway or trail upgrades.** A project that upgrades a facility that has existing active transportation trips should always have a baseline activity estimate. Upgrade of a standard bike lane to a protected bike lane or sidewalk infill are examples of projects that should have an activity baseline.
 - For these projects, proceed to *Step 2.2* and use the **Percent Change Process** guidance.
- **Upgrades on high-stress roadways.** High-stress roadways often have very low existing volumes of active transportation trips as they are not comfortable for people walking, bicycling, and rolling. For projects that will upgrade high-stress roads with no or very low volumes of existing active transportation trips (approximately less than 20 combined walking and bicycling trips per day), the analysis could assume no baseline activity. You should justify why this approach was taken, and it should generally be used only on facilities with an existing LTS of 3 or 4. For example, while an extra-wide shoulder on a high-speed and high-volume highway is an existing facility, it does not function as an existing facility for most active transportation trips.
 - For these projects, proceed to *Step 2.2* and use the **Context Transfer Process** guidance.

Step 2.2. Estimate Change in Activity

Use one of the following two processes to estimate changes in walking and bicycling activity; skip the steps for the process that you do not choose.

- **Percent Change Process.** Use percent change when a baseline estimate of walking and bicycling activity is available and is greater than zero. In the Basic Method, a range of percent change

estimates for different projects and conditions are provided in Tables 1 to 4. Note for the Augmented Method, apply network analysis to further refine the percent change estimate. See *Appendix C* for an explanation of network analysis and related steps for this Augmented Method.

- **Context Transfer Process.** Use context transfer when the baseline activity estimate is assumed to be zero or very low (typically for new facilities or high-stress existing facilities). For the Basic Methods, develop a post-gap closure activity estimate of trips based on sample counts along similar corridors in a similar context that already have similar improvements. For the Augmented Method, apply network analysis to further refine the activity estimate.

See Figure 6 for an overview of both the **Percent Change** and **Context Transfer** processes.

Percent Change Process

To use the **Percent Change Process**, first develop a baseline estimate. Then apply an estimated percentage change in behavior to your baseline estimate of the number of pedestrian and bicycle trips on the current facility. You will select an appropriate percentage change depending on a few factors: the type of project, the degree of improvement expected, and whether or not the project is expected to induce existing trips to shift routes to use the new project.

The dominant facility improvement type (e.g., protected bike lane or signalized pedestrian crossing) will influence the degree of improvement and expected behavior response. The estimates in Tables 1 to 4 cover a range of improvement types for both bicycle- and pedestrian-focused projects, including different types of bike lanes, road diets, bicycle boulevards, sidewalks, crossing improvements, and more. Where a gap closure involves multiple types of facility improvement, segment the project by improvement type and analyze separately. If segmenting is not possible (e.g., a road diet including new bike lanes), you could choose the primary change with the highest expected impact (bike lane), and qualitatively note the additional improvement along with the results. Underlying research was not available on combined effects of different improvements.

The **degree of improvement** will determine the magnitude of behavior changes, either low, medium, or high. The degree of improvement is based on the change in LTS expected from the gap closure project. Whether route shift is expected will determine the proportion of additional active travel on the improved facility that will be attributable to new trips and trips shifted from other modes.

The following list provides the steps for estimating change in activity using the **Percent Change Process**:

1. Establish baseline.

The goal of this step is to estimate the existing active travel trips at the project location (annual average daily trips). The recommended approaches for establishing a baseline activity estimate of current use at the gap site are provided in the following list, in order of preference:

a. Existing, On-Site Trips

First, use any existing trip counts of bicyclists and pedestrians at the site that can be treated as a direct observation of existing behavior as the baseline estimate. For guidance on current best practices in nonmotorized counting, refer to *NCHRP Report 797: Guidebook on Pedestrian and Bicycle Volume Data Collection* (Ryus et al. 2014) and National Association of City Transportation Officials *Making Bikes Count: Effective Data Collection, Metrics & Storytelling* (NACTO 2022). Generally, higher quality counts are taken from

automatic counters with a recording period of at least 72 hours. More recent research recommends at least one week of data for accurate annual activity estimates, where possible (Nordback et al. 2019).

b. Modeled Expansion

Establish a baseline estimate using established direct demand models, a travel demand model, or other locally calibrated models that apply understood statistical and behavioral methods. This approach either employs direct demand models or fit-to-purpose travel demand models to estimate existing demand at the project location.

Direct demand modeling would analyze multiple count locations over a large area (e.g., a whole city) to develop a predictive count model that correlates relationships in the independent variables (e.g., facility type or surrounding land-use characteristics) with the observed bicycle and pedestrian volumes from the counts. The direct demand model can be used to model activity at the site or at other locations with different geographic characteristics. Guidance on the application and development of direct demand models is identified in *NCHRP Report 770: Estimating Bicycling and Walking for Planning and Project Development: A Guidebook* (Kuzmyak et al. 2014).

Travel demand models attempt to simulate a typical day of activity and can be either trip-based or activity-based. These models can provide a baseline estimate of travel behavior in a region at a relatively disaggregated level, and some of them can represent bicycle and pedestrian facilities sufficiently to estimate these travel modes' travel costs accurately. Activity-based models tend to represent active travel needs more accurately (through accounting for behavioral theories on how people make travel decisions), but trip-based models with appropriate accounting of routing options can also create reasonable estimates of active travel (Castiglione et al. 2015). Most cities and regions do not currently have travel demand models that estimate walking and bicycling travel for the network, though more agencies are developing such models.

c. Comparable Counts

Develop a baseline estimate based on trips counted at locations similar to the baseline condition of the gap closure project in terms of existing facilities and nearby roadway functional classification. Comparable facilities can be used to estimate existing baseline behavior for the project being evaluated. Facilities that are considered comparable would be those with similar functional classifications and land-use, demographic, and built contexts; ideally, they would be in the same region as the project under evaluation. For a more nuanced approach, consider following the **Context Transfer Process** to establish a baseline, then return to the **Percent Change Process**.

d. Mobile/Cell Phone Data Sources

If nothing else is available, use large mobile/cell phone datasets to establish a baseline estimate of walking and bicycling activity. These can be thought of as uncalibrated direct application of mobile/cell phone trace observations. If using mobile/cell phone data sources, refer to documentation of the dataset in question for detailed information about data development and expected levels of accuracy. Mobile/cell phone data can be obtained in a variety of formats, and developing a baseline activity estimate might range in complexity. While it may be as easy as selecting the area from a web portal and downloading the appropriate data, at a minimum, you will likely need to select an area of influence and a time range for the activity estimate. When using mobile/cell phone data-based estimates, be aware of potential coverage bias in any chosen product.

2. Calculate degree of improvement by mode.

For improvements along a corridor, calculate existing LTS for facilities that currently exist in the gap. If there are multiple segments that make up the gap, use the highest stress-level LTS score for the gap. Then determine the future LTS along the project span given

the new facility. The LTS analysis method used should match the recommendations of the relevant state DOT, if available. If there are no state guidelines for LTS calculations, this guide recommends either the Mineta 1.0 or 2.0 for bikes (Mekuria et al. 2012) and ODOT evaluation procedures for pedestrians (ODOT 2023).

To calculate the degree of improvement, subtract the future LTS from the existing LTS, resulting in an improvement of 1, 2, or 3 points on the 4-point LTS scale. For example, moving from LTS 4 to LTS 2 is an improvement of 2. If there is no significant change in the LTS, use a 1 for degree of improvement; you should justify why the improvement may enable any behavior change at all.

For crossing/intersection gap improvements for pedestrians, determine the degree of improvement by the extent to which they create new appropriate crossing opportunities. An appropriate crossing opportunity would be one contextually appropriate to the type of facility it is intended to cross. For example, a high visibility crosswalk on a 6-lane arterial with a speed limit of 35 mph is likely not sufficient to qualify as an appropriate crossing opportunity, but the same arterial with advanced yield signs and a pedestrian hybrid beacon (PHB) would be appropriate (FHWA 2018b). Behavior-change literature does suggest that the provision of signalized crossings show a positive impact (for roads with a functional classification of minor arterial or higher) even when LTS does not change, but other crossing types are not well studied, much less inventoried in many regions (Broach and Dill 2015). Given these understudied crossing types, this guide recommends deferring to FHWA's *Guide for Improving Pedestrian Safety at Uncontrolled Crossing Locations* (FHWA 2018b) and NCHRP *Research Report 1036: Roadway Cross-Section Reallocation: A Guide* to identify contextually appropriate crossing opportunities (Semler et al. 2023). When a new contextually appropriate crossing is added, use the following crossing spacing distance after the improvement for the degree of improvement number:

- **Level 1—Low improvement.** New crossings are more than 800 feet from one another.
- **Level 2—Medium improvement.** New crossings are 400 to 800 feet from one another.
- **Level 3—High improvement.** New crossings are less than 400 feet from one another.

For example, if a signalized crossing is installed in the midpoint between two existing signalized crossings 1,000 feet apart, the new crossing spacing would be 500 feet, and a level of improvement of 2 (medium) should be used.

Use the degree of improvement number (from the LTS shift or the new crossing distance) to determine whether the expected behavior change is low (1), medium (2), or high (3).

3. Determine whether the gap closure project is expected to induce significant route shift behavior.

Once the degree of improvement is established, you then need to determine if the gap closure project is expected to induce significant **route shift** behavior. Research has shown that for some new active transportation facilities, a portion of the trips are shifting from other, less desirable nearby routes.

Determining if significant route shift is expected is primarily a qualitative assessment. If the gap closure is in an area with reasonable but inferior route options paralleling the gap closure, a larger share of the estimated trips on the new gap closure project are likely to be the result of route shift, rather than mode shift or new trips. This would particularly be the case if the new project provides a more direct path to many destinations.

On the other hand, a gap-filling project on an arterial in an area with low adjacent street connectivity would not likely induce measurable route shift behavior. You should keep in mind that active travelers' observed detours typically add 10% or less to the most direct route, and rarely more than 20% (Sevtsuk and Kalvo 2022; Broach et al. 2010). Limited research on the effect of e-micromobility on this observed detouring behavior to date has not yet reached a consensus on whether electrification has significantly changed detour rates. Overall,

you must determine whether significant route shift is expected after the gap closure project is implemented and defend your reasoning if you feel it is significant.

4. Calculate changes in number of walking and bicycling trips due to the gap closure project.

After completing the previous three steps in the Percent Change Process, use the appropriate table for the given mode and expectation of route shift to calculate travel behavior change due to the gap closure project. In general, we recommend the tables that do not assume significant route shift: **Table 1** and **Table 3**, thus including only new trips. We make this recommendation because most benefit streams only calculate benefits that accrue to new trips: a common trait of best practice cost–benefit analysis for active travel. **Table 2** and **Table 4** account for both route shift and mode shift behavior change.

In summary, if you are estimating:

- Bicycle travel and **no** significant route shift expected, then use: **Table 1**.
- Bicycle travel and significant route shift expected, then use: **Table 2**.
- Pedestrian travel and **no** significant route shift expected, then use: **Table 3**.
- Pedestrian travel and significant route shift expected, then use: **Table 4**.

The percent change is then applied to the baseline to determine the change in active travel on the facility post-implementation.

Table 5 provides a hypothetical example of how this would be used to determine a before-and-after estimate of new trips.

Research quantifying behavior change from specific facility improvements remains limited, and quality, methods, and context vary widely from study to study. The numbers in Table 1 through Table 4 were developed by revising and augmenting estimates developed for the Caltrans Active Transportation Benefit-Cost Tool (Fitch et al. 2022) using detailed results from the following research studies: Broach et al. (2012), Broach and Dill (2015, 2016, 2017), McNeil et al. (2015), and Sevtsuk et al. (2021). Caltrans estimates included both before-and-after and cross-sectional studies, including both revealed preference and stated preference results, and studies for a given infrastructure intervention were summarized in the original work based on a qualitative weighting exercise. The six studies used in this research project to revise and augment the Caltrans estimates used cross-sectional, revealed preference and stated preference methodologies that quantified responses to a range of key facility improvement types. The focused selection of research studies included key components important for this research, including separating estimates for mode versus route shift and quantifying behavior change in a standardized way (elasticities or marginal rates of substitution) in the North American context.

Where Caltrans and selected detailed research estimates were both available, we weighted each estimate equally and averaged the expected impact. Before averaging, we inflated research mode shift results to account for new demand (using Caltrans shift allocation estimates), since the underlying studies had not included new trip generation.

Additional details of how we arrived at specific behavior-change estimates are provided in *Appendix A*.

5. Adjust for trip distance for facilities greater than 2 miles in length.

Percentage changes based on intercept surveys or similar studies fundamentally assume some total net change in travel behavior to active modes. At longer facility lengths, the count can be inflated by some percentage and is therefore not likely to be representative of the facility's impact. In cases where the facility is greater than 2 miles, multiple counts should be taken to account for inflation, taking approximately one count for each mile. Each of these 1-mile segments should be treated as separate projects with their own percent increase in counts. The projected increases in counts should then be adjusted for trip distance per *Appendix B*.

Table 1. Percent change in bicycle travel—no route shift.

Project Element	Class	Increase in Bicycle Travel <i>Where NO Significant Route Shift Expected, %</i>		
		Degree of Improvement (Number)		
		Low (1)	Medium (2)	High (3)
Trails/Off-street paths	Roadway	40%	73%	114%
Conventional bike lane	Roadway	9%	19%	38%
Buffered/Protected bike lane*	Roadway	31%	91%	113%
Bicycle boulevard	Roadway	29%	61%	104%
Road diet**	Roadway	5%	8%	18%

Notes:

* Varying and sometimes overlapping definitions of buffered and protected bike lane made it difficult to distinguish the two categories. Best available research finds buffered bike lane impacts exceed conventional bike lanes but fall short of protected lanes. We suggest users apply these values but consider the degree of improvement a step lower for a buffered lane than for a protected bike lane, given a similar existing alternative.

** Limited research support; applied average factors to impute mode shift portion; includes only the primary road diet elements related to auto travel lanes and not any specific change to cycling facilities.

Table 2. Percent change in bicycle travel—when route shift is expected.

Project Element	Class	Increase in Bicycle Travel <i>Where Significant Route Shift Expected, %</i>		
		Degree of Improvement (Number)		
		Low (1)	Medium (2)	High (3)
Trails/off-street paths	Roadway	92%	254%	411%
Conventional bike lane	Roadway	22%	131%	301%
Buffered/Protected bike lane*	Roadway	65%	174%	328%
Bicycle boulevard	Roadway	68%	229%	387%
Road diet**	Roadway	7%	12%	25%

Notes:

* Varying and sometimes overlapping definitions of buffered and protected bike lane made it difficult to distinguish the two categories. Best available research finds buffered bike lane impacts exceed conventional bike lanes but fall short of protected lanes. We suggest users apply these values but consider the degree of improvement a step lower for a buffered lane than for a protected bike lane, given a similar existing alternative.

** Limited research support; applied average factors to impute mode shift portion; includes only the primary road diet elements related to auto travel lanes and not any specific change to cycling facilities.

Table 3. Percent change in pedestrian travel—no route shift expected.

Project Element	Class	Increase in Pedestrian Travel <i>Where NO Significant Route Shift Expected, %</i>		
		Degree of Improvement (Number)		
		Low (1)	Medium (2)	High (3)
Crossing island*	Intersection	3%	5%	8%
Midblock/collector marked crossing*	Roadway	6%	13%	31%
Signalized arterial crossing	Intersection	7%	14%	21%
Road diet**	Roadway	4%	8%	16%
Sidewalk	Roadway	13%	19%	25%
Trails/Off-street paths	Roadway	69%	87%	138%

Notes:

* Limited research support; applied average factors to impute mode shift portion. This is an evolving area of research, see forthcoming research under NCHRP 17-115, "Guide for Marked Crosswalk Design, Spacing, Placement, and Safety."

** Limited research support; applied average factors to impute mode shift portion; includes only the primary road diet elements related to auto travel lanes and not any specific change to walking facilities.

Table 4. Percent change in pedestrian travel—when route shift is expected.

Project Element	Class	Increase in Pedestrian Travel Where Significant Route Shift Expected, %		
		Degree of Improvement (Number)		
		Low (1)	Medium (2)	High (3)
Crossing island*	Intersection	5%	10%	15%
Midblock/collector marked crossing*	Roadway/Intersection	11%	24%	57%
Signalized arterial crossing	Intersection	21%	41%	62%
Road diet**	Roadway	8%	15%	30%
Sidewalk	Roadway	18%	32%	46%
Trails/off-street paths***	Roadway	129%	162%	255%

* Limited research support; applied average factors to impute mode shift portion. This is an evolving area of research, see forthcoming research from the authors of this guide.

** Limited research support; applied average factors to impute mode shift portion; includes only the primary road diet elements related to auto travel lanes and not any specific change to walking facilities.

*** Limited research support; applied average factors to impute route shift portion.

Table 5. Example of calculating estimated increase in existing trips.

	Walking Trips	Bicycling Trips
<i>Inputs</i>		
Existing baseline count (before project)	100	100
Previous LTS (before project)	3	4
Post-project LTS (after implementation of project)	1	1
Improvement in LTS	2	3
<i>Calculation</i>		
Road diet % change (route shift included)	15%	25%
Estimated count (post-project)	115	125

Refer to the Northeast Case Study in *Chapter 6: Steps in Action (Analysis Walk-Throughs)* for an example of percent change calculations.

Context Transfer Process

Context transfer uses trip counts at comparable facilities to estimate behavior for the project being evaluated. Facilities that are considered comparable would be those with similar functional classifications and land-use, demographic, and built contexts; ideally, they would be in the same region as the project under evaluation.

This method extrapolates a context-sensitive estimate of volumes in a location without existing trip counts based on the context around a location or set of locations with existing trip counts. This approach is primarily recommended to estimate future volumes when a baseline cannot be established (i.e., new facilities or roadways without existing bicycle and pedestrian activity), but it can also be used to establish a baseline if no other data are available.

The steps for the **Context Transfer Process** for estimating either baseline or future trips are presented in the following list.

1. Find a comparable facility (or set of facilities) in similar context with pedestrian or bike trip counts available.

Identify a location or set of locations comparable to your facility and in a similar context to your project. Using at least three to five comparable count locations can help provide a more

normalized comparison than using only one or two. If a set of facilities with similar contexts with existing trip counts are available, they should be averaged to create an estimate of trips for your project. For example, a new trail in a rural community could estimate the future volumes of pedestrians and bicyclists based on existing trails in other rural communities in the region with similar land-use, demographic, and built contexts.

2. Identify demand metrics that are available at both project area and comparable facility.

Demand data that can be used for context could consist of demographics or travel information, including active commute mode share from the census (or American Community Survey), or model-derived estimates of either short trip rates or walking and biking trips. Model-derived estimates are typically summarized to geographic areas known as transportation analysis zones (TAZs). The demand data should be in units (i.e., trips or people) and not a percentile (i.e., that a given zone has higher active commute mode share than 60% of comparable zones).

3. Sample demand metrics around both locations.

Sample (i.e., summarize) the chosen demand data around both the comparable facility trip count location and your project. If you do not have access to GIS software, use entire non-overlapping demand data geographic units in proximity to the comparable facility trip count locations and your project [e.g., census block groups (CBGs)]. If you do have access to GIS software, we recommend using proportional allocations of demand data within a set distance of the comparable facility trip count locations (see the *Proportional Allocation* section for details).

The evaluation extent buffer size should be selected based on the mode whose volumes are being estimated. We recommend basing the extent on an assumed 10-minute access range of the gap closure project. Thus, while a typical area of influence for a project is subject to local conditions such as land use, topography, and network connectivity, the recommended settings for identifying modal influence areas for estimating the benefits of a gap closure project are as follows:

- **Bicycle influence areas.** We recommended using an **extent of a 1.25 miles** as-the-crow-flies buffer.
- **Pedestrian influence areas.** We recommended using an **extent of a 0.5 miles** as-the-crow-flies buffer.

For example, to develop a demand metric for walking, you could use the proportional allocation methods and American Community Survey (ACS) data to calculate the number of walking commute trips within both the influence area (0.5 mile in this example) of your project location and the comparable facility with existing trip counts.

If your project is over 2 miles in length, we recommend splitting it into 1-mile segments and collecting demand data for each segment.

4. Calculate a context adjustment factor using the values of the demand metric at the count location relative to the demand metric at the new location.

To calculate the context adjustment factor, divide the demand metric for your project location by the demand metric for the comparable facility or facilities. Using the previously mentioned example, this would mean dividing the number of estimated walking commute trips within 0.5 mile of the project location by the estimated number of walking commute trips within 0.5 mile of the comparable facility. The resulting factor will be used to adjust the comparable facility counts to the context of gap-closing project location.

5. Estimate the number of trips on your gap-closing project.

Apply the context adjustment factor to the counts at the comparable facility or facilities. Multiply the demand rate by the counts (bicycle or pedestrian counts collected from comparable facilities) to generate a synthetic count (the expected number of trips on the gap-closing project). For facilities shorter than 2 miles, this synthetic trip count should be used as the trip estimate for the project. For projects that are longer than 2 miles, we recommend consulting

Appendix B to adjust for the typical length of bicycle and pedestrian trips. If the project consists of multiple small segments in different locations, examine each as individual segments.

If you used the **Context Transfer Process** for a project without a baseline (new facility or low existing bicycle and pedestrian activity) continue to *Step 2.3. Establish Change Stratification*.

If you used the **Context Transfer Process** to establish a baseline for a project, return to the **Percent Change Process** section.

Refer to the Southwest Case Study in *Chapter 6: Steps in Action (Analysis Walk-Throughs)* for an example of context transfer calculations.

Proportional Allocation

Generally, proportional allocations refer to a sampling method where a sampling geography (influence area) is used to sample demographic base data (CBGs) based on their degree of overlap. A conceptual diagram of the process is seen in Figure 7. Throughout the guide, there are references to proportional allocation procedures for the purposes of sampling data such as CBGs or travel model-derived TAZs.

There are two assumed types of statistics that can be computed through proportional allocations, and their application depends on what type of variable is being sampled:

- **Proportional sums.** This refers to the sampling of population totals based on the overlap with a sampling geography for the purposes of quantifying how many people are within a study

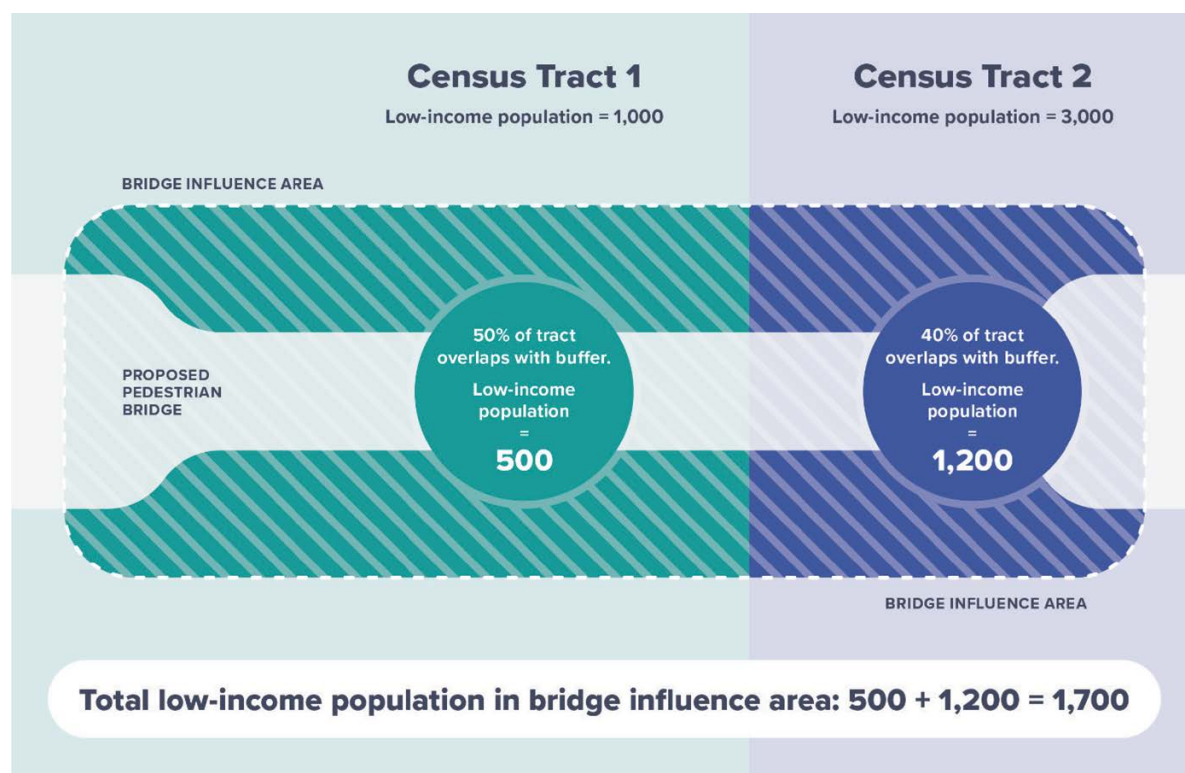


Figure 7. Proportional allocation process illustration.

geography. This is done by computing the weighted sum of the zone totals with the fraction of overlap each base data geography has with the sampling geography.

- For example, if a sampling geography covers 50% of zone 1 with 1,000 people, and 40% of zone 2 with 3,000 people, the proportional sum would be 1,700 people.
- **Proportional averages.** This refers generally to the sampling of rates or percentages based on the area-weighted-average overlap of a sampling geography with a base dataset. This is done by computing area-weighted-averages of the base data rates or percentages within the sampling geography.
 - For example, if a sampling geography covers 15 acres of zone 1 with a walking mode split of 20%, and 10 acres of zone 2 with a walking mode split of 10%, the proportional-average percentage would be 16%.

Proportional allocations provide a consistent method to estimate underlying populations within a sampling geography, and they can reduce bias associated with large zones common in demographic base data when using conventional spatial relationships such as intersect. However, they are limited by the assumption of equal density throughout a base geography. This is an assumption we know is routinely violated based on known distributions of populations within zones such as CBGs. This limitation can be mitigated, but not entirely eliminated, by using more disaggregated units of analysis where possible (fusing parcel or address points data with demographic data for example).

Step 2.3. Establish Change Stratification

For certain benefit analyses it is important to know more about the behavior shifts. For example, emissions reductions will be higher if people who are now biking on the new facility previously drove a passenger car than if they previously traveled by transit. Some travelers may be induced to make a completely new trip, such as walking to a nearby coffee shop, because of your project. This shift will not reduce emissions, but it will have health benefits from the increased physical activity.

You should therefore use **Table 6** to identify the portion of the behavior-change estimate generated from the **Percent Change Process** or the **Context Transfer Process** that can be attributed to vehicle shift (from motor vehicle to active transportation), new trip generation,

Table 6. Presumed travel shifts induced as percentage of total increase (all capacity levels).

Shift Type	Bicycle Shifts and New Travel (WITH Route Shift)	Bicycle Shifts and New Travel (NO Route Shift)	Pedestrian Shifts and New Travel (WITH Route Shift)*	Pedestrian Shifts and New Travel (NO Route Shift)
Vehicle shift**	17%	43%	18%	36%
New travel / trip generation	11%	28%	6%	12%
Route shift***	60%	0%	50%	0%
Other mode shift	12%	29%	26%	52%

Notes:

* Very limited evidence supporting pedestrian route shift potential.

** As facility improvement “gradient” increases, see larger bicycle route shift component consistent with greater elasticity of route choice.

*** Caltrans-based estimates included shifts from vehicles to walking and transit trips.

route shift, and shift from other modes (e.g., transit to bike). Mode shift without significant route shift, for example, might occur if an existing facility without useful parallel alternatives were improved.

Additional details on the methodology for determining these values are provided in *Appendix A*.

Table 7 can be used as a template for reporting the results of the behavior-change calculations.

Table 7. TEMPLATE for reporting behavior-change calculations.

Number of New Trips	Mode Shift from Motor Vehicles	Activity Shift/Trip Generation/New Travel (Number of Trips)	Route Shift	Other Mode Shift
Pedestrians				
Bicyclists				

CHAPTER 4

Step 3. Estimating Benefits

This chapter provides detailed methods used to estimate the benefits associated with gap closure. Except for safety benefits, these methods rely on the estimate of behavior change calculated in *Step 2*. Each benefit category begins with an introduction to the unit of measure and approach that frames the analysis, followed by data needs and analysis considerations. Then, a step-by-step walk-through of the Basic Methods is provided, along with sample calculations. For additional examples of the methods, refer to *Chapter 6: Steps in Action (Analysis Walk-Throughs)*.

The Basic Methods are intended for practitioners with standard levels of staff training, software resources and time. The more complex Augmented Methods are included in the appendices.

Detailed methods are provided to estimate the following types of benefits:

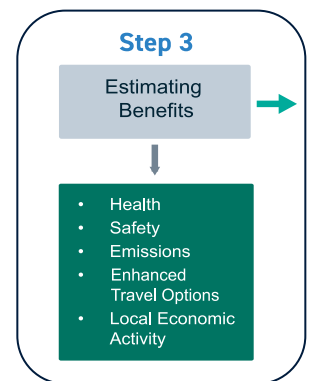
- Health
- Safety
- Emissions
- Enhanced travel options
- Local economic activity

The methods primarily focus on estimating benefits related to safety and increases in walking and bicycling, drawing on findings from pedestrian- and bicycle-focused research. Other concurrent benefits related to ecosystem services and enhanced community resilience are also described at the end of the chapter.

This guide primarily focuses on benefits estimation for pedestrians and bicycles, drawing assumptions from pedestrian- and bicycle-focused research. However, future research should refine methods to better account for differences between e-bikes, e-scooters, traditional bicycles, and other personal mobility devices.

Health Benefits

This section provides guidance on quantifying the health benefits of increased physical activity related to behavior shifts from closing gaps in the active transportation network. Both the HEAT developed by the World Health Organization and the Integrated Transportation and Health Impact Model (ITHIM) are tools that can estimate the value of improved community health through increases in walking and bicycling. The underlying calculations for these tools rely on established research linking regular physical activity at various intensity levels (metabolic equivalent of task, or METs)—including walking, bicycling, and, more recently, micromobility—to changes in health outcomes measured as risk of premature death (mortality) or living with chronic disease (morbidity). While capturing an important component of total health impacts from increased active travel, many additional pathways exist linking transportation shifts to health outcomes (e.g., safety improvements leading to reduced injuries and fatalities).



Unit of Measure

The Basic and Augmented Methods presented in this guide are based on HEAT, which measures the value of premature deaths avoided (mortality). The Advanced Applications rely on the ITHIM, which considers a wider range of outcomes affected by physical activity: cardiovascular diseases, depression, dementia, diabetes, and certain cancers (morbidity). Additionally, the tools underlying methods presented here differ in their monetization of health benefits. HEAT estimates reducing mortality risk benefits for individuals, relying on the economic concept of value of a statistical life. Intermediate outputs of HEAT are communicated in simple terms of premature deaths avoided. ITHIM includes a monetization module based on avoided health care costs for treating specific chronic diseases. The tool produces an intermediate, non-monetized health outcome measure in terms of disability-adjusted life years avoided, where one disability-adjusted life year represents a cost equivalent to 1 year of full health.

Advanced Application Case Study – Health Benefits – State of Oregon

In the report *Oregon Outdoor Recreation Metrics: Health, Physical Activity, and Value*, the Oregon State University College of Forestry estimated the public health benefits of outdoor recreation in Oregon using the Integrated Transport and Health Modeling Tool (ITHIM) framework (Rosenberger 2018).

ITHIM has the capacity to estimate the impact of active transportation on public health arising from three sources: increased physical activity, changes to injury rates, and air pollution reduction. Using ITHIM only for physical activity, the team extrapolate survey data to calculate aggregate physical activity among all Oregonians participating in the selected activities. They then used a built-in function to model the relationship between physical activity and reduced risk of certain chronic diseases such as heart disease, diabetes, and cancer. Finally, they converted reduced disease risk to Cost of Illness (COI) savings. This includes savings from costs of both direct healthcare and lost productivity.

Overall, the study estimated that Oregonians participated in the tracked activities on 794 million separate occasions and expended 503 billion calories in 2017. This resulted in an estimated COI savings of 2–3.6% of all healthcare costs in Oregon, and 9–17% of costs involved in treating heart disease, cancer, diabetes, and depression specifically.

The top two activities, by total COI savings, were walking on local streets or sidewalks and walking on trails. See *Appendix M* for more details, including the total COI savings for each activity.

Method Selection

Both the Basic and Augmented Methods for walking and for bicycling apply the methods from HEAT for walking and cycling developed by the WHO (2023). The Basic Method relies on readily available inputs from national datasets, while the Augmented Method requires additional local data (see *Appendix L*). For Advanced Application estimates of health benefits, the research team recommends using ITHIM-based tools, although some experts note that ITHIM has proven less suited to project-scale analysis as currently constructed, and has mainly been applied at the city or regional levels, pointing to this as an ongoing research need and opportunity (see *Appendix M*).

Figure 8 provides a flowchart of the guidance on quantifying the benefits of increased physical activity.

These methods can be used across project scales, land-use contexts, and gap types. For the Basic Method, the calculated benefits are restricted to reductions in mortality and do not consider morbidity impacts. The Basic Method relies on national data for mortality profiles and age distribution and applies a constant value per trip.

One limitation of the Basic, HEAT-based approach is that guidance currently excludes benefits of increased physical activity for the youngest and oldest adults. Benefits are estimated for people ages 20 to 74 years for walking, and aged 20 to 64 years for bicycling as these were the populations studied in the underlying research. Inclusion of younger groups would not have much impact, since mortality rates are so low. Older adults are excluded due to uncertainty around their participation rates in cycling and walking combined with the extreme sensitivity of the calculation to their inclusion, due to very high baseline mortality rates.

Another limitation of the Basic Method is that, for simplicity, active travel is only represented as trips, based on average trip distances, rather than the specific distance traveled using active travel modes at a site. The Augmented HEAT method relaxes that assumption and supports localized trip and mode characteristics.

Data Needs

The primary data need for the Basic, HEAT-based approach is shown in Table 8. Augmented and Advanced Application Methods require additional local values such as localized, age-specific death rates across a specific population, existing bicycle and walking mode share, and more. Sources for finding each of the data inputs for the Augmented and Advanced Application Methods are described in *Appendix L* and *Appendix M*, respectively.

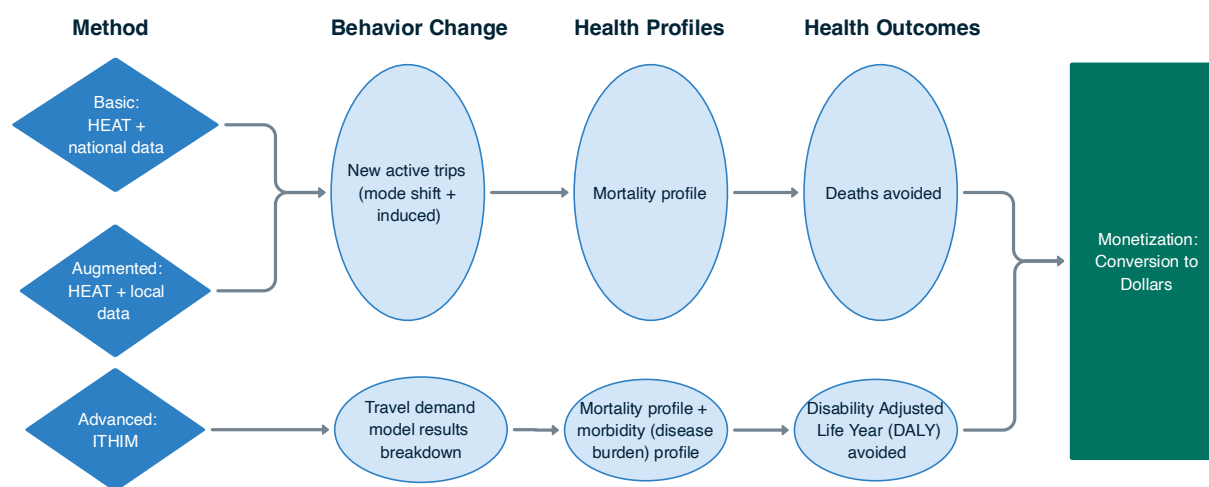


Figure 8. Basic, Augmented, and Advanced Application Methods for calculating health benefits.

Table 8. Data needs for Basic Method health benefit estimation.

Key Data Item	How to Determine
Number of new active trips.	Follow procedure outlined in <i>Chapter 3: Step 2. Estimating Behavior Change</i> .

Basic Methods

The Basic Method follows the procedure described in the U.S. DOT BCA Guidance (U.S. DOT 2023a, Table A-13). Users are encouraged to visit the U.S. DOT website for the most up-to-date guidance and values: <https://www.transportation.gov/mission/office-secretary/office-policy/transportation-policy/benefit-cost-analysis-guidance>.

1. **Estimate number of new active trips and trips shifting from vehicular modes.**
Estimates of the number of new active trips may be obtained using a variety of methods as described in *Chapter 3: Step 2. Estimating Behavior Change*. Do not include trips that were due to route shifts or shifts from non-vehicular modes.
2. **Apply multipliers.**
Multiply the number of new active trips by the appropriate age group and benefit rate multipliers (walk or bike) shown in Table 9. The benefit rate multiplier represents the estimated monetary value of premature deaths avoided per trip. The age group multiplier adjusts for the share of active trips by the relevant age group, and U.S. DOT permits substitution of localized data.

The **Estimated Annual Health Benefits from Walking** are calculated using the following equation:

Estimated health benefits from walking = New walking trips per day for all ages × Percent of population within 20–74 age range × Health benefits per trip in that age group × 365 days

The **Estimated Annual Health Benefits from Bicycling** are calculated using the following equation:

Estimated health benefits from bicycling = New bicycling trips per day for all ages × Percent of population within 20–64 age range × Health benefits per trip in that age group × 365 days

Example: In the example provided here, we use an illustrative value of 200 new walking trips per day and 100 new bicycling trips per day. Note that the results are rounded off to better reflect the uncertainty in the estimates.

Table 9. National multipliers from the U.S. DOT BCA Guidance.

Mode	Per trip benefit	% of population in age range
Walking	\$7.63	68% 20–74 years
Bicycling	\$6.80	59% 20–64 years

Source: U.S. DOT (2023a).
Note:
These multipliers are typically updated annually. This example is from the current year in 2022 dollars. Users are encouraged to visit the U.S. DOT website for the most up-to-date guidance and values: <https://www.transportation.gov/mission/office-secretary/office-policy/transportation-policy/benefit-cost-analysis-guidance>

200 walking trips × 0.68 × \$7.63 × 365 days = Estimated health benefits

per day for all ages (2022 \$) from walking of \$379,000

(2022 \$) per year

100 bicycling trips × 0.59 × \$6.80 × 365 days = Estimated health benefits

per day for all ages (2022 \$) from bicycling of \$146,000

(2022 \$) per year

Safety Benefits

Inadequate infrastructure increases crash and injury risk and discourages walking, bicycling, and rolling. When gaps in active transportation infrastructure are closed, safety benefits accrue for all roadway users through safer facilities and reduced driving. The same behavior-change principles that underlie the previous methods are also present for safety, though behavior change does not need to be separately calculated to estimate safety benefits because safety benefits are estimated using CMFs based on infrastructure. Instead, the safety method is applied to both the gap closure facility and relevant parallel corridors or facilities to capture crashes that may be avoided due to route shift onto the new facility that closes a gap.

Note that increasing pedestrian and bicycling activity also increases exposure to crash risk. This means that while gap closure should reduce crashes, there is a small probability that if volumes increase dramatically, the number of crashes may also increase. The rate of crashes per walking or biking trip should still decrease, if appropriate facilities are installed. Further, some CMFs already account for increases in volume and exposure (e.g., CMFs derived from Avelar et al. 2021). **This guide does not present a separate method for estimating potential crash increases due to increased walking, bicycling, and rolling.**

Unit of Measure

Safety benefits can be reported in two ways: the expected number of crashes avoided or the total monetary value of those avoided crashes. Consistent with the Safe System Approach, avoided crashes can be reported by mode and severity. Monetizing the crash reduction is common for BCA. Further, calculating the cost of existing crashes can be useful to allow for the comparison of the “cost of inaction” to the cost of gap closure.

Method Selection

Instructions are provided to calculate safety benefits regardless of whether the location has a crash history. The Basic Methods use crash history or crash estimates obtained via multipliers. The Augmented Methods use crash history or crash estimates derived from local crash profiles (see Appendix E); Advanced Applications use SPFs (see Appendix F).

Advanced Application Case Study: Safety Benefits—Madison, WI

Rather than using multipliers as presented in this guide, some agencies may have the exceptional capacity to use safety performance functions (SPFs) to estimate crash frequencies based on existing roadway information. Researchers in Madison, WI have developed such functions; the predicted crash numbers then are available for estimating benefits.

The safety benefits of closing an active transportation gap in downtown Madison along Highway 151 can be estimated using SPFs, crashes, and conditions on parallel roads. Appropriate crash modification factors were chosen for the gap closure, resulting in a reduction of fatal and severe crashes from 26.5 crashes to 20 crashes, and a 24% reduction in societal costs from all crash severities combined. See Appendix F for additional details.

The flowchart in Figure 9 illustrates the safety benefit calculation process for Basic, Augmented, and Advanced Application Methods, representing a variety of data availability circumstances. All methods generate a measured or estimated baseline number of crashes for the gap closure facility and one or more relevant parallel facilities, apply one or more CMFs to the baseline, and optionally apply cost or weighting factors to estimate the value of crashes reduced due to gap closure. Note that abbreviations from the KABCO crash severity scale are used throughout this section as follows: fatal injury (K), suspected serious injury (A), suspected minor injury (B), possible injury (C), and no apparent injury (O). Consistent with a Safe System Approach, we advise focusing primarily on fatal (K) and serious injury (A) crashes, though this method works for all crash severities.

There are enhanced methods of estimating safety benefits available to projects with more technical expertise. Projects with staff expertise and software available for network analysis, typically with GIS software can use Augmented Methods. Projects with staff expertise to conduct original analysis, a regional travel demand model for estimating trips, and a variety of localized factors, including local SPFs, emissions factors, and detailed crash reports can use Advanced Applications.

The Augmented Method for estimating safety benefits includes calculating baseline multipliers using a crash profile approach for use in estimating crash risk. This method is based on systemic analysis, which proactively screens the network for known safety risk factors (see *Appendix E*).

Data Needs

The data needs for safety benefit analysis are listed in Table 10.

Historic Crash Data

This guidance is primarily written for crash data sourced from police crash reports. Some agencies may also have access to other sources of safety crash injury data, such as hospital records or emergency response records. Each dataset has its strengths and limitations.

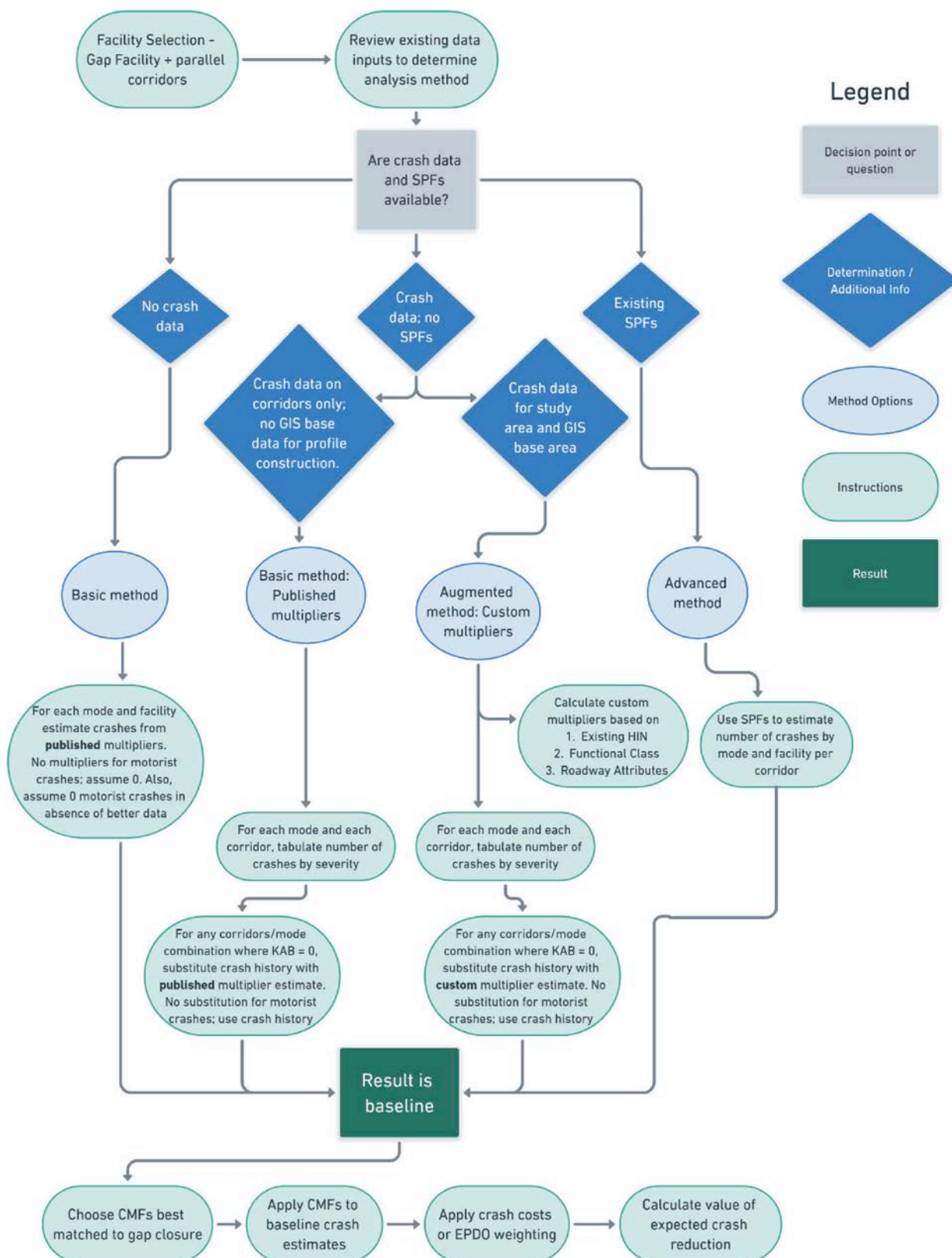
Crash data are most often sourced from a state department of public safety or DOT. Some states make their data accessible for public download (e.g., California and Texas). Police data are the main source of crashes involving vehicles, but they are prone to under-reporting, especially among lower severity crashes and crashes in BIPOC (Black, Indigenous, People of Color) communities (Sciortino et al. 2005). Some states include solo bicycle crashes in their police-reported data, while others do not. Hospital and emergency response records may include crashes not involving motorists (e.g., falls) and may be less vulnerable to under-reporting and contain more accurate assessments of injury severity. However, they are difficult to access due to privacy constraints, and they do not necessarily contain roadway or mode information.

Crash Estimation Data

If crash data are available, the Basic Method may be applied using a mix of the multipliers presented in Appendix G (see Table 13) or crash history. If local or state crash data are not available, the Basic Method may be applied using multipliers only based on functional class. If historic crash data are needed and state or local data are unavailable (e.g., due to grant or funding requirements), fatal crash data are available from NHTSA's Fatality Analysis Reporting System (FARS): <https://www.nhtsa.gov/research-data/fatality-analysis-reporting-system-fars>.

Project Area Data

Data about the project area include centerline miles or intersection count by functional class. The same information by functional class is also needed for any relevant parallel facilities. For example, a gap might be a 3-mile corridor of major arterial spanning 12 intersections, and its parallel facility might be a 3-mile corridor of minor arterial spanning 14 intersections. These mileage and intersection count totals are used with multipliers to estimate a crash baseline.



KAB: crashes with fatal, serious, or non-serious injuries. HIN: high-injury network. EPDO: equivalent property damage only.

Figure 9. Overview of safety benefits estimation approach for Basic, Augmented, and Advanced Application Methods.

Table 10. Data needs for Basic Method safety benefit estimation.

Data Type	How to Determine: Crash Rate Method
Historic crash data	For gap closure project site and any relevant parallel facilities: number of crashes by mode and severity. <i>This method can still be used even in the absence of data.</i>
Crash estimation data	Multipliers presented in Appendix G (see Table 13).
Gap closure project area data	Centerline miles and/or intersection count for gap closure project and any relevant parallel facilities, by functional class.
Gap closure improvement type(s)—to the extent known	Type of facility or countermeasures being used to close gap (e.g., adding a separated bike facility or reducing motor vehicle travel lanes).
CMFs from CMF Clearinghouse (preferred) and other sources (if needed)	CMFs matched to gap closure improvement type(s) as closely as practical.

Gap Closure Improvement Type

For the gap being closed, note the type of facility or countermeasures being used to close gap to the extent known. These countermeasures should line up with the gap closure project described in *Step 1.3*, for example, adding a separated bike facility and reducing motor vehicle travel lanes.

If the specific type of treatments that will be used to close the gap are not yet finalized, list one or more type of treatments that are being considered or would have the intended effect. For example, if the goal is to provide an LTS 1 facility along a 3-mile section of LTS 4 arterial, a separated bikeway would likely be necessary to achieve the intended target. Therefore, separated bikeway could be used as a placeholder.

CMFs from CMF Clearinghouse

CMFs are designed to be applied to police report crash data to estimate how many crashes could be avoided by implementing safety treatments and countermeasures. If hospital or emergency response data are used instead of police reports, with severity reported on the Abbreviated Injury Severity scale, refer to U.S. DOT's guidance on the valuation of a statistical life for estimating crash costs corresponding to the Abbreviated Injury Severity instead of the KABCO injury scale (U.S. DOT 2023b). Note that the transferability of CMFs to hospital or emergency response data or other sources is unknown.

CMFs can come from a variety of sources. Use CMFs from the CMF Clearinghouse (<https://www.cmfclearinghouse.org/>) for improvement types if they are well matched to the context and treatment and are rated with at least three stars by CMF Clearinghouse. Evaluate both the quality and limitations of any potential CMF to ensure that it is appropriate for the context of interest. For example, some CMFs for separated or buffered bike lanes are specific to the number of motorist lanes on the street and the amount of expected increase in bicyclist volumes from baseline.

Some popular CMFs in pedestrian and bicyclist gap closure treatments are reproduced in Table G-16 in *Appendix G*. The following list summarizes a hierarchy of recommendations for choosing a CMF for the treatment(s) of interest.

Option 1. Choose a well-matched CMF from the CMF Clearinghouse that is of sufficient quality (i.e., at least three stars). Try to match the application context of the CMF as closely as possible (e.g., traffic operations, road design, land use). This is the best option, which should be used if a high-quality, well-matched CMF can be found in CMF Clearinghouse.

Option 2. Choose an *alternate* CMF from CMF Clearinghouse that has a similar impact or mechanism of action to the treatment in question and is of sufficient quality—again, at least three stars. For example, in absence of a curb extension CMF, use a CMF for reducing pedestrian crossing distance. As with Option 1, try to align the application context of the CMF to the gap closure project as closely as possible. *This is the next best option, which should be used if a high-quality, well-matched CMF is not available, but related high-quality CMFs with similar effects or mechanisms of action are available.*

Option 3. Choose a well-matched CMF from an alternate source. Consider state DOT research, NCHRP research, and other peer-reviewed published research. *This is an acceptable backup option if an appropriate CMF or alternate CMF from CMF Clearinghouse cannot be found, but a well-matched CMF from another reputable source is available.*

Option 4. Consider a well-matched or alternative CMF from CMF Clearinghouse with a lower-quality rating (i.e., one to two stars). This option is only recommended if there is sufficient confidence in the applicability and understanding of the lower-quality rating to ensure that it is a defensible choice for the project. *This is the least preferred option, after high-quality CMF Clearinghouse options and other published research options have been exhausted.*

Using CMF Clearinghouse to Choose a CMF

For any given search query or category of pedestrian and bicyclist treatments, CMF Clearinghouse presents dozens or more unique countermeasures defined by the before and after states for each countermeasure. Each countermeasure may additionally have several CMF options, with subtle differences based on the original research study that produced the CMF.

To understand the safety potential for closing an active transportation gap, you must choose a CMF that matches the gap context and gap closure facility type as closely as possible. The following checklist outlines key steps for choosing a CMF:

- Search CMF Clearinghouse for the countermeasure of interest (e.g., “bike lane”).
- Filter on CMFs with quality star ratings of 3, 4, or 5.
- Examine the countermeasures included in the search results, and choose one that matches the treatment being considered as closely as possible. (If none of the countermeasures presented fit the project in question, refer to options 2, 3, and 4 discussed earlier for further recommendations, such as choosing a similar countermeasure or finding a CMF from another source.)
- Within the selected countermeasure, review the summary information for all CMFs returned in the search, starting with the highest ratings.
- For any CMFs that seem like a good candidate (high quality rating, recent publication year, appropriate urban/rural context), click on the CMF or use the CMF Clearinghouse’s “Compare” functionality to review detailed information about the CMF.
 - Review all the details in the “applicability” section for each CMF to see how well these conditions match the gap closure project (e.g., number of lanes, roadway type, speed limits, traffic volumes).
 - Review other details or the original study that produced the CMF for additional applicability details (e.g., some bike lane CMFs reference very specific assumptions about changes in bicyclist volume in the “other details” section and in the original publication).
 - Based on this contextual information, choose the CMF that best matches the gap closure project.
- For the chosen CMF, record the CMF value, applicable crash type, and applicable crash severity for use in this methodology.

Basic Methods

Basic Methods for estimating safety benefits are described here:

1. Identify the gap closure project facility or corridor.

As previously discussed in *Data Needs*, identify the type of facility or countermeasures being used to close the gap to the extent known.

2. Identify parallel corridors.

Identify relevant nearby parallel corridors or routes from which pedestrians and/or bicyclists may be expected to detour if an improved facility is installed to close the gap. Parallel facilities are streets, routes, or intersections that people might detour from to use the new gap closure facility. For a gap caused by a high-stress arterial, parallel facilities might include residential streets within two to three blocks of the arterial, parallel collector streets, or parallel bicycle facilities. For crossing opportunity gaps, if it seems plausible that pedestrians may detour to use the higher quality crossing once the gap is closed, then parallel facilities might be nearby intersections with lower-quality crossing facilities.

3. Measure total facility mileage.

Measure total facility mileage by functional class and/or number of intersections by maximum functional class in project area. See Table 11 for a template to fill in with the relevant information.

4. Measure crash history in the project area (if using) for all modes.

If historic crash data are not being used, skip this step and proceed to step 5 to estimate baseline crashes using multipliers.

Collect the 5 most recent years of crash data available for pedestrian, bicycle, and motorist crashes in the project area (including any applicable parallel facilities). Note: Data spanning 5 years are ideal, and data over a minimum of 3 years are required to avoid regression to the mean.

Tabulate the total number of crashes by mode (pedestrian, bicycle, motorist) and by severity (using the KABCO scale defined in Method Selection) in the project area into a table like shown in Table 12. For crashes involving multiple modes, categorize by the most vulnerable mode involved: pedestrian or other personal conveyance, bicyclist, or motorist (including motorized scooter, motorcyclist, and all other modes).

Calculate the subtotal shown in Row 6 of Table 12 specifically totaling severe and minor injury crashes ($K + A + B$).

Depending on the multipliers being used (per intersection, per mile, or both), repeat this calculation to separately tabulate the number of intersection and/or midblock/segment crashes.

Depending on the mode examined there are specific instructions for completing Table 12:

– Pedestrian and Bicycle Modes

- If the Row 6 of Table 12 subtotal of severe and minor injury crashes ($K + A + B$) of all (intersection + midblock) crashes is greater than 0, then proceed to step 6 and populate the appropriate columns in Table 14 with the actual crash history.

Table 11. TEMPLATE: Summary of project area mileage by functional class and intersections by maximum functional class.

Functional Class (Midblock/Segment) or Maximum Functional Class (Intersection)	Miles in Project Area	Intersections in Project Area
Major arterial		
Minor arterial		
Major collector		
Minor collector		
Local road		

Table 12. TEMPLATE: Summary of crash history in project area over 5 years.

Row ID	Crash Severity	Crashes with 1+ Pedestrians	Crashes with 1+ Bicyclists (and 0 Pedestrians)	Crashes with Motorists Only (0 Pedestrians and 0 Bicyclists)
1	Fatal Injury (K)			
2	Suspected Serious Injury (A)			
3	Suspected Minor Injury (B)			
4	Possible Injury (C)			
5	No Apparent Injury (O)			
6	SUBTOTAL Fatal, Serious, and Minor Injury Crashes K + A + B	If >0: Continue using crash history and skip to step 6. If =0: Proceed to step 5 to calculate estimate.	If >0: Continue using crash history and skip to step 6. If =0: Proceed to step 5 to calculate estimate.	

- However, if the Row 6 of Table 12 subtotal of severe and minor injury crashes (K + A + B) of all (intersection + midblock) crashes is equal to 0, then proceed to step 5 to calculate a crash estimate.

– Motorist Crashes

- Populate Table 12 with actual crash history.

5. Use crash multiplier to calculate crash estimate for project area for pedestrians or bicyclists where history is not being used.

Choose a set of multipliers from *Appendix G* to use (see Table 13 for guidance on selecting the appropriate multiplier set). Multipliers are provided using national data (fatal crashes only). State-level data from Minnesota (all severities) and data from counties within California (fatal and injury crashes only) were also provided, as these datasets included readily available data on both crashes and roadway miles. Minnesota's multipliers represent a lower-than-average crash state, split into urban and rural specific multipliers. California's multipliers are

Table 13. Multiplier sets.

Multiplier Set	Relevant Tables in Appendix G
Crash Multiplier Set 1: All (Intersection + Midblock/Segment) Crash Rates per Mile for All Areas (Urban + Rural) from Six Counties within CA State Database	Table G-1 Table G-2
Crash Multiplier Set 2: All (Intersection + Midblock/Segment) Crash Rates per Mile for Urban Areas from MN State Database	Table G-3 Table G-4
Crash Multiplier Set 3: All (Intersection + Midblock/Segment) Crash Rates per Mile for Rural Areas from MN State Database	Table G-5 Table G-6
Crash Multiplier Set 4: Fatal Crash Rates from FARS (Fatal Crashes Only; Not Preferred)	Table G-7
Crash Multiplier Set 5: Intersection Subset Crash Rates per Intersection for Urban Areas from MN State Database	Table G-8 Table G-9
Crash Multiplier Set 6: Intersection Subset Crash Rates per Intersection for Rural Areas from MN State Database	Table G-10 Table G-11
Crash Multiplier Set 7: Midblock/Segment Subset Crash Rates per Mile for Urban Areas from MN State Database	Table G-12 Table G-13
Crash Multiplier Set 8: Midblock/Segment Subset Crash Rates per Mile for Rural Areas from MN State Database	Table G-14 Table G-15

higher than Minnesota's and may be more representative of Sunbelt cities. All crash multipliers represent 5 years of data.

If another time period is desired (e.g., 3 years), then divide the multipliers by 5 to calculate an annual rate, and then multiply the resulting value by the number of years desired.

Estimate the total number of crashes occurring on each roadway type by multiplying the project mileage or intersection count for each roadway functional class type by the corresponding crash rates from the chosen multiplier set. Each multiplier set will have a different rate for each crash severity type and mode, resulting in a total estimate of crashes for each roadway functional class type by severity and mode.

Estimate how many crashes are associated with each travel mode (i.e., pedestrians and other personal conveyance, bicyclists, or motorists/all other modes) by summing the functional class estimates of crashes for each mode and severity.

Note: If Crash Multiplier Set 4: Fatal Crash Rates from FARS is used, then there will only be an estimate for fatal crashes. Assume 0 or observed crash history for other severity crashes, but know that this will likely underestimate crash risk. Minnesota-based options contain estimates for all severities. California-based options contain estimates for all fatal and injury crash severities but not property damage only crashes.

Populate the appropriate cells in **Table 14** with these values.

6. Summarize crash baseline (history or estimate) for all modes.

Rows 1 through 5 for all modes should have been completed in steps 4 and 5. Next, calculate totals (Row 6) in **Table 14**. If analyzing intersection and midblock crashes separately, repeat this step for each crash type.

7. Choose CMFs that correspond to the gap closure improvement type(s), to the extent known at this stage.

Use high-quality (three stars or higher) CMF from CMF Clearinghouse for improvement types if they exist. Try to match CMF to project context and modes as closely as possible. Some popular CMFs in pedestrian and bicycle gap closure treatments are reproduced in **Table G-16** in *Appendix G*.

If there is no CMF in CMF Clearinghouse for the treatment(s) of interest, use the options for selecting an alternative CMF from the *CMFs from CMF Clearinghouse* section.

8. Apply chosen CMF(s) to the applicable crash baseline data in Table 14.

Match CMF to corresponding modes and severities (e.g., CMF for all pedestrian crashes, or CMF for severe motorist crashes).

Table 14. TEMPLATE: Summary of crash baselines (history OR estimate) in project area.

Row ID	Crash Severity	Pedestrians Use ESTIMATE unless HISTORY is available and Subtotal in Table 12 is greater than 0	Bicyclists Use ESTIMATE unless HISTORY is available and Subtotal in Table 12 is greater than 0	Motorists Use HISTORY if available or 0 if no data
1	Fatal Injury (K)			
2	Suspected Serious Injury (A)			
3	Suspected Minor Injury (B)			
4	Possible Injury (C)			
5	No Injury (O)			
6	TOTAL K + A + B + C + O			

Try to apply CMFs to all modes and severities if possible. For example, if a single treatment type has CMFs for severe crashes of all modes, and all crashes for pedestrians and bicyclists, use the “severe/all modes” version for KA crashes of all modes, and use the “all ped/bicycle” version for BCO ped/bicycle crashes.

If a certain combination of mode and severity is not covered by the CMF(s), use 0 for the number of crashes reduced for that mode and severity. In the previous example, BCO motorist crashes would show 0 crashes reduced because the other CMFs cover KA for all modes and BCO for ped/bicycle.

Where multiple treatments are chosen and multiple CMFs are selected which apply to the same modes and severities, there are several possible methods for combining CMFs. The *Highway Safety Manual* guidance is to multiply them. Choose whichever method is most appropriate for your context (e.g., requirements for grant proposal, methods provided by state DOT) (Vanasse Hangen Brustlin, Inc. 2011).

Summarize the number of crashes reduced by mode and severity (from applying CMFs to the values in Table 14) in Table 15.

9. Calculate total safety benefit by mode, and then combine for overall safety benefits of the gap closure project.

Multiply crashes reduced by monetized crash costs presented in Table G-17 in *Appendix G* and populate outputs into Table 16. Sum rows 1 through 5 to calculate subtotals in Row 6. Sum modal totals from Row 6 to calculate total benefit in Row 7.

Table 15. TEMPLATE: Summary of crash reductions from gap closure.

Row ID	Crash Severity	Number of Pedestrian Crashes Avoided by Gap Closure	Number of Bicycle Crashes Avoided by Gap Closure	Number of Motor-Vehicle-Only Crashes Avoided by Gap Closure
1	Fatal Injury (K)			
2	Suspected Serious Injury (A)			
3	Suspected Minor Injury (B)			
4	Possible Injury (C)			
5	No Apparent Injury (O)			

Table 16. TEMPLATE: Summary of crash costs reduced.

Row ID	Severity	Monetary Value of Avoided Pedestrian Crashes	Monetary Value of Avoided Bicycle Crashes	Monetary Value of Avoided Motor-Vehicle-Only Crashes
1	Fatal Injury (K)			
2	Suspected Serious Injury (A)			
3	Suspected Minor Injury (B)			
4	Possible Injury (C)			
5	No Apparent Injury (O)			
6	SUBTOTAL Monetary Benefit by Mode			
7	TOTAL Monetary Benefits of Gap Closure			

Advanced Application Case Study—Emissions Benefits—Portland, OR

The Metro Active Transportation Return on Investment (ATROI) study (Metro 2022) used a travel demand model (TDM) to estimate changes in bicycle trips, bicycle miles traveled, auto trips, and auto vehicle miles traveled (VMT). The most impactful project—the Eastbank Esplanade, a 1.5-mile off-street path and bridge in Portland, OR—was estimated to reduce 2,700 daily VMT.

A VMT change can be used directly to estimate emission reductions. The EPA MOVES model may be run with TDM-generated inputs such as VMT by vehicle type, speed distributions, and road type distributions. When using TDM inputs MOVES should be run at the county scale, rather than the project scale. Practitioners with exceptional capacity that wish to calculate benefits with MOVES, are referred to *Appendix I* of this guide as well as MOVES documentation for more details.

See documents available on EPA's MOVES website (U.S. EPA 2024a) as well as *NCHRP Web-Only Document 210: Input Guidelines for Motor Vehicle Emissions Simulator Model* (Porter et al. 2015).

Emissions Benefits

Emissions benefits of gap closure projects are driven by mode shifts, and specifically, shifts from driving to active travel induced by the gap closure, which result in reductions in VMT. Mode shifts could be induced by any type of gap closure. Emissions benefits are most likely to be realized when the closure mainly serves utilitarian trips (e.g., commuting, shopping) that might be substituting for car trips, rather than recreational trips that are generally not substituting for driving.

Emissions may be reduced directly along the route of the new facility (e.g., for a new shared-use path or sidewalk that parallels a road where the vehicle trip would have been taken). They may also be reduced on other roads in the vicinity of the facility, on connecting streets or nearby streets from which auto trips are being removed.

Emissions benefits will change over time in proportion to facility usage and changes in vehicle emission rates. As facility usage grows over time, auto travel will be reduced more. However, vehicles are generally getting cleaner on a per-mile basis over time due to evolving federal and state emissions standards, so the benefits per avoided VMT are likely to decline in future years.

Unit of Measure

Emissions benefits can be measured as the monetized value of pollutants avoided by travelers choosing active transportation rather than a typical internal-combustion-engine-powered personal motor vehicle for their trip.

Method Selection

An overview of the methods appears in Figure 10. The Augmented Method replaces some national average values (trip lengths, emission rates) with regional values (see *Appendix H*). Advanced Applications use location-specific modeling to develop data that is much more specific to a given project and context (see *Appendix I*). However, many if not most travel demand models lack the modal detail necessary to apply this method or will not have been run for individual active transportation projects. The impacts of small projects may not be discernible even if the active transportation network and mode choice detail exist in the model.

The emissions analysis process starts with an estimate of mode shift attributable to the gap closure: the number of trips shifted from driving to active transportation. This is calculated as the number of new active transportation trips resulting from the gap closure, multiplied by the share of those trips shifted from driving. The number of trips shifted from driving to active transportation is then multiplied by the length of those trips to get an estimate of VMT reduction. Emission rates specific to each mode and pollutant of interest (expressed in grams per vehicle-mile) can then be applied to changes in VMT. Emission rates may vary by speed; speed-specific rates may be used if the average speed of avoided vehicle trips is known.

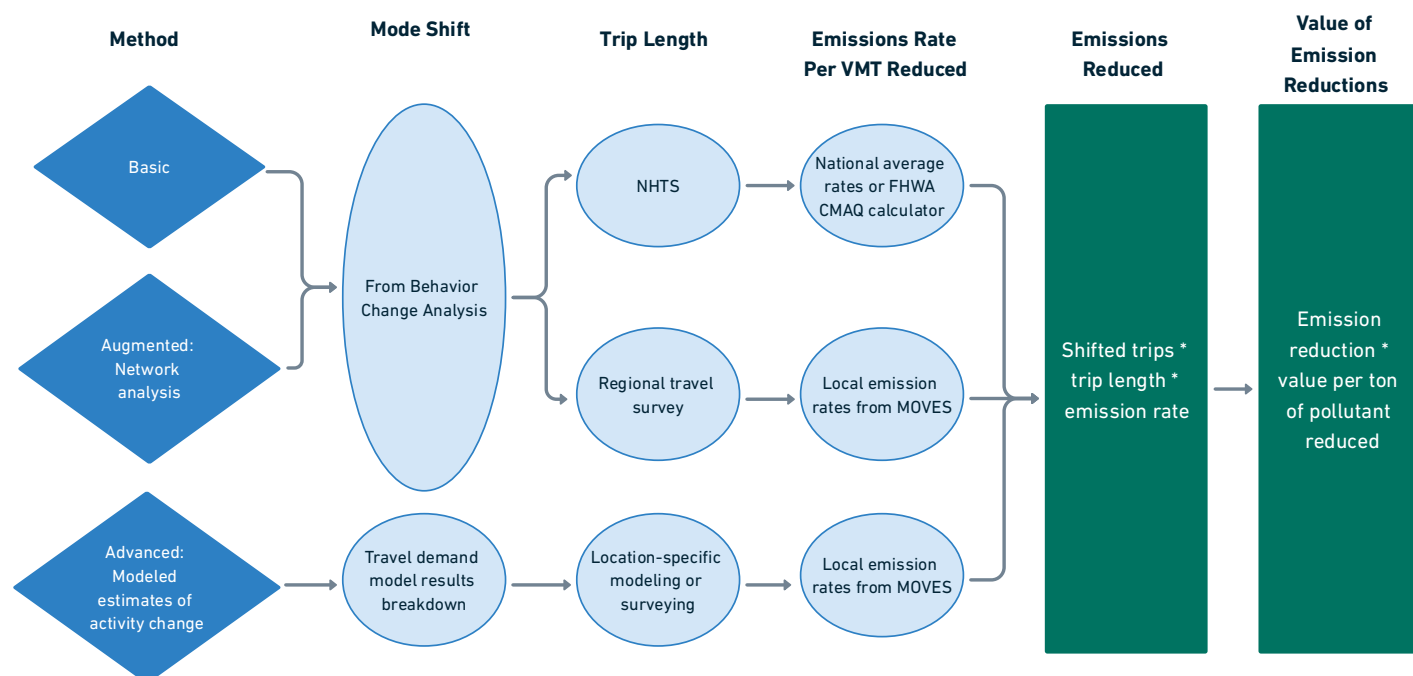


Figure 10. Basic, Augmented, and Advanced Application Methods for calculating emissions benefits.

Data Needs

The data needs for emissions benefit estimation are listed in Table 17.

Other data sources include:

- **ACS.** Data on baseline walk and bike commute mode shares can be used as a benchmark to check reality of mode shift estimates. Data: <https://www.census.gov/programs-surveys/acs/data.html>.
- **State and regional data available from various sources.** Contact your state DOT, MPO or regional planning agency/organization.

Table 17. Data needs for Basic Method emissions benefit estimation.

Key Data Item	Basic Methods
Number of new active trips	Follow procedure outlined in <i>Chapter 3: Step 2. Estimating Behavior Change</i> .
Share who previously traveled by personal motor vehicle	Follow procedure outlined in <i>Chapter 3: Step 2. Estimating Behavior Change</i> .
Average trip length	Average national walk and bicycle trip length from National Household Travel Survey, summary statistics and data explorer: https://nhts.ornl.gov/ . To derive average trip lengths, go to the “Data Explore Tool” and extract person-miles by state and mode of travel as well as person-trips by state and mode of travel. Divide the values to get average miles per trip by mode.
Emission rate per vehicle-mile	National average rates (e.g., from the FHWA Community Multiscale Air Quality Modeling System Emissions Calculator Toolkit [FHWA 2022b]): https://www.fhwa.dot.gov/environment/air_quality/cmaq/toolkit/

Basic Methods

The steps to follow under the Basic Methods approach for estimating emissions benefits are provided in the following list:

1. Estimate number of new active trips.

Estimates of the number of new active trips may be obtained using a variety of methods as described in *Chapter 3: Step 2. Estimating Behavior Change*.

Emissions reductions may be estimated on a daily or annual basis. If daily trip changes are developed and annual emissions benefits are desired, annualize the number of new trips using an appropriate annualization factor. If daily trips represent average annual conditions, multiply by 365. If they represent “active travel season” conditions, a lower annualization factor may be appropriate depending on region-specific climate affecting active travel.

Example: In the example described in step 5 of this section, we use an illustrative value of 200 new trips per day or 73,000 new trips per year.

2. Estimate share who previously traveled by personal motor vehicle.

Estimates of the share who previously traveled by personal motor vehicle (prior drive mode share) may be obtained using a variety of methods as described in *Chapter 3: Step 2. Estimating Behavior Change*. For example, the Caltrans Active Transportation Benefit-Cost Tool default assumption is that 17.6% of new bicycle trips are shifting from cars, and 33.3% of pedestrian trips are shifting from a car trip completely or replacing the car trip by walking plus transit (Volker et al. 2019a, 2019b).

3. Estimate average trip length of shifted personal motor vehicle trips made by active travelers using the gap closure.

The 2017 National Household Travel Survey (NHTS) found that, nationwide, the average length of a bicycle trip is 2.4 miles, and the average length of a walk trip is 0.9 miles. Therefore, the analyst could assume that any shifted personal motor vehicle trips due to the gap closure would be of similar lengths. The NHTS Data Explore Tool allows the user to query this information by state as well as by other categories such as urban vs. rural and population density.

4. Determine emission(s) rate per VMT.

The state-of-practice tool recommended in 49 states for emissions analysis, required for regulatory analysis and recommended for other analysis, is the U.S. Environmental Protection Agency (U.S. EPA) Motor Vehicle Emission Simulator (MOVES) model (U.S. EPA 2024a). (California has a different model, known as EMFAC, that is used for regulatory purposes in that state [CARB, n.d.].)

The simplest way to use MOVES emission factors is to use a tool or tables based on national average data. FHWA’s Congestion Mitigation and Air Quality (CMAQ) Calculator Toolkit—Bicycle and Pedestrian Improvements incorporates pre-loaded national average emission rates (grams per vehicle-mile) from MOVES (FHWA 2022b). The FHWA tool can also be used to translate changes in active travel into changes in emissions. The user needs to provide assumptions about new bike trips, displaced vehicle trips, and average trip lengths. Table 18 provides emission rates in grams per mile (g/mi) based on national MOVES runs. These rates were inferred from the FHWA CMAQ Emissions Calculator Toolkit, “Bicycle and Pedestrian Improvements” module, version 6/2022, by running the model for a reduction of 1,000 daily vehicle trips with average 1-mile trip length. Note that emission rates for a given year (e.g., 2030) will only change when MOVES (or the MOVES rates in the CMAQ tool) are updated.

For greenhouse gas emissions, an average CO₂ emission factor per mile can also be developed based on the average fuel economy of light-duty vehicles (as reported in the U.S. Department of Energy Annual Energy Outlook [AEO], U.S. EIA 2023) and fuel-based emission factors

Table 18. Sample light-duty vehicle emission rates (g/mi).

Pollutant	2025	2030	2035	2040
Carbon Monoxide (CO)	3.648	2.641	1.989	1.664
Particulate Matter <2.5 µm (PM _{2.5})	0.012	0.011	0.010	0.010
Particulate Matter <10 µm (PM ₁₀)	0.047	0.046	0.045	0.045
Nitrogen Oxide (NO _x)	0.191	0.097	0.057	0.045
Volatile Organic Compounds (VOC)	0.156	0.107	0.086	0.078
Carbon Dioxide (CO ₂)	371	336	N/A	N/A
Carbon Dioxide Equivalent (CO ₂ e)	374	339	N/A	N/A
Total Energy Consumption (MMBTU/day)	5.02	4.55	4.25	4.08

Source: FHWA CMAQ Emissions Calculator Toolkit, based on national MOVES runs, 2022 (FHWA 2022b). The tool did not provide CO₂ outputs for 2035 and 2040. Rates are provided as examples for this guide. Documentation of emission rates is from FHWA (2023).

(CO₂ per gallon, CH₄ and N₂O per mile) from a source such as the U.S. EPA (2024b) or World Resources Institute Greenhouse Gas Protocol (World Resources Institute, n.d.).

For example, the 2023 AEO projects a fleet average of 25.8 miles per gallon for light-duty vehicles in year 2025. The CO₂ content of gasoline is approximately 8.9 kg per gallon. This gives an emission rate of 345 g/mi (8,900 g/gal / 25.8 mi/gal). Non-CO₂ greenhouse gases add 2% to 3% to this value.

5. Calculate emissions changes.

To estimate the emissions changes as a result of the project, multiply the number of new trips each day by the prior drive mode share (determined in step 2) by the trip length (determined in step 3) by the emission rate (determined in step 4).

$$\text{Avoided Emissions} = \text{new trips per day} \times \text{prior drive mode share} \times \text{trip length} \\ \times \text{emission rate}$$

Illustrated for CO₂ emissions displaced by a gap closure using national data, including NHTS average mode share and trip length, 2025 emission rate derived from AEO.

Calculate CO₂ emissions per day:

$$\begin{aligned} \text{Avoided Emissions} &= 200 \text{ new bicycle trips per day} \\ &\times 49.5\% \text{ of new bicycle trips had been driving trips previously} \\ &\times 2.4 \text{ miles average bike trip length} \\ &\times 345 \text{ g CO}_2/\text{mi driving in a personal motor vehicle} \end{aligned}$$

$$\text{Avoided Emissions} = 82,000 \text{ g/day} / 1,000 \text{ g/kg} = 82 \text{ kg/day}$$

Transform to per year

$$\text{Avoided Emissions} = 82 \text{ kg/day} \times 365 \text{ days/year} = 30,000 \text{ kg/year}$$

Simplify amount per year (if needed)

$$\text{Avoided Emissions} = 30,000 \text{ kg/year} / 1,000 \text{ kg/metric ton} = 30 \text{ metric tons per year}$$

6. Report emissions benefits in a simple table showing emissions benefits by pollutant.

The table and/or accompanying text should specify the units (e.g., kg or tons), time frame (e.g., daily summer, daily winter, or annual), and evaluation year. The text should also describe the key assumptions used. An example table is shown in Table 19.

7. Monetize benefits by applying damage costs.

Table 20 displays the damage costs per metric ton for three types of emissions. To calculate the emissions benefits for your project, multiply the number of metric tons of reduced emissions by the corresponding damage costs. If your emissions benefits are in kilograms, you will need to convert to metric tons by dividing the number of kilograms by 1,000.

The Augmented Method follows the same steps outlined in the Basic Method, but with different data inputs. The Basic Method uses national data sources, whereas the Augmented Method may use data obtained from local or regional agencies. See Appendix H for Augmented and Appendix I for Advanced Application Methods.

Enhanced Travel Options Benefits

Enhanced travel options benefits measure a reduction in generalized travel cost due to improvements in one or more travel mode options. Generalized travel costs typically include out-of-pocket expenditures and time costs but may also capture quality improvements experienced by the user, such as reduced crowding on transit or the installation of lower-stress active transportation facilities like protected bike lanes or off-street paths.

Unit of Measure

Enhanced travel options benefits stem from behavior change and are often measured in terms of monetized value of benefit per walk-mile or bicycle-mile-traveled. While there may be some overlap with health and safety benefits, many of the quality improvements captured here are unique, and otherwise would not be considered.

Method Selection

The Basic and Augmented Methods both rely on the current U.S. DOT BCA Guidance (U.S. DOT 2023a) to convert increases in biking or walking to estimates of enhanced travel option

Table 19. Sample emissions benefits.

Year	CO ₂ equivalent (metric tons per year)	NO _x (kg/year)	PM _{2.5} (kg/year)	NO _x (kg per summer day)	VOC (kg per summer day)
2020	44.4	22.6	0.6	0.060	0.037

Table 20. Values for monetizing emissions benefits.

Year	CO ₂ (value per metric ton)	NO _x (value per metric ton)	PM _{2.5} (value per metric ton)
2025	\$237	\$20,300	\$975,500

Source: U.S. DOT (2023a).
Note:
These multipliers are typically updated annually and values are provided for the next 30 years. This example is from the current year in 2022 dollars. Users are encouraged to visit the U.S. DOT website for the most up-to-date guidance and values:
<https://www.transportation.gov/mission/office-secretary/office-policy/transportation-policy/benefit-cost-analysis-guidance>

Advanced Application Case Study Enhanced Travel Options Benefits—Portland, OR

Metro—Portland, Oregon’s MPO—applied their regional travel demand model to estimate enhanced travel options benefits generated by previously completed gap-filling bicycle infrastructure projects. The application was conducted as part of a larger research effort, the Active Transportation Return on Investment (ATROI) study.

The ATROI study leveraged the regional bicycle travel model to calculate additional satisfaction created with the implementation of the Eastbank Esplanade, SE 17th Avenue Path, and the Northeast Cully Cycle Track projects. To understand the benefits, researchers changed the existing scenario networks used in the regional travel model to represent “next best” routes without the projects and compared the new model outputs with outputs using post-project networks. The study examined the difference in total expected maximum utility for riders between scenarios. *Utility* in this case refers to additional satisfaction or positive value in the ride experience after implementation of the bicycle projects. Marginal utility of income was used to monetize the utility differences, which were then expanded into future years, annualized, and discounted.

Combining the results of all three projects, the study estimated (per day) about 1,000 new bike trips drawn from other modes (just over 70% from motor vehicles) and over 14,000 existing bike trips that shifted routes to use the new facilities, totaling an additional 7,700 daily miles of travel conducted by bicycle. The total travel options benefits were estimated at more than \$210,000 per year (2010\$). Results from the larger ATROI analysis agreed with limited existing research that suggests travel option benefits may be substantial, potentially similar in magnitude to health benefits from increased physical activity.

Additional details are available in *Appendix K* and at the ATROI website: <https://www.oregonmetro.gov/active-transportation-return-investment-study>.

benefits. The Augmented Method includes modifications which may better capture network effects of gap closures (see *Appendix J*). A summary of an Advanced Applications case study using a regional travel demand model is provided in the above Portland Metro case study. Figure 11 shows a summary of the methods.

Data Needs

The Basic Method relies on existing multipliers to estimate benefits directly from changes in active miles traveled on the upgraded facility itself. Table 21 presents the data items needed to estimate enhanced travel options benefits. The Augmented Method relies on network analysis to add additional benefits related to route shift and is included in *Appendix J*. Network analysis here refers to consideration of how a gap closure might directly connect existing bike or walk facilities which currently are separated by some barrier. This may be especially useful for a short but impactful network gap closure such as a new bridge.

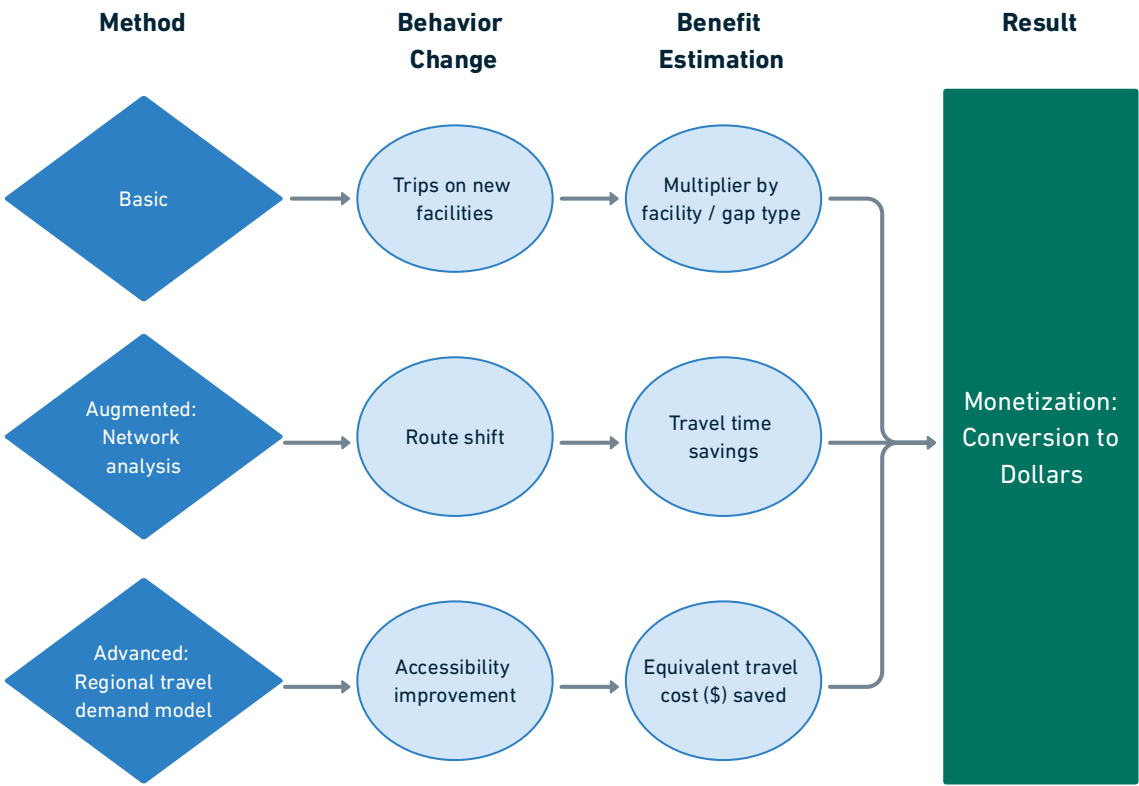


Figure 11. Basic, Augmented, and Advanced Application Methods for calculating enhanced travel options benefits.

Table 21. Data needs for Basic Method enhanced travel options benefit estimation.

Key Data Item	How to Determine
Estimated number of bike or walk trips on new or upgraded gap facility	Follow procedure outlined in <i>Chapter 3: Step 2. Estimating Behavior Change</i> .
Value of enhanced travel options by improvement type	Sourced from U.S. DOT BCA Guidance https://www.transportation.gov/mission/office-secretary/office-policy/transportation-policy/benefit-cost-analysis-guidance

Basic Methods

Basic Methods estimation is described here:

- 1. **Estimate number of bicycle or walk trips on gap closure facility.**
Estimates of the total number of active trips may be obtained using a variety of methods as described in *Chapter 3: Step 2. Estimating Behavior Change*. Note that these benefits are calculated based on all users, not just new users making new trips. However, the estimates should be broken out by existing versus new users.
- 2. **Estimate facility usage.**
Multiply the number of trips by the length of the improved facility (gap closure project). Only include those portions representing quality enhancement relative to existing alternatives. This calculates person-miles walked or person-miles biked that directly experience the upgraded infrastructure. Note that certain pedestrian crossing enhancements apply per use instead of per mile walked.

3. Calculate enhanced amenity benefit value.

Find the most appropriate facility factor in the U.S. DOT BCA Guidance tables, and multiply this by the estimated facility usage.

U.S. DOT's BCA Guidance provides a method that attempts to isolate the value of improved facility amenities to users. Simple multipliers on walking or cycling (miles or trips, depending on feature type) are provided that directly estimate dollar value benefits for certain improvement types. Note that if new facility users (shifted from other routes or for newly generated trips) are included in activity estimates, U.S. DOT BCA Guidance recommends applying a “rule of half” adjustment for those trips (refer to Vadali et al. 2017 for an explanation of the rule of half method and justification). To apply the “rule of half” multiply the number of new users by 0.5 in the enhanced amenity benefit calculation.

U.S. DOT's method considers only travel along the length of the gap closure facility itself. Additional network connections made possible by the project are not considered, and no distinction is made between existing and new bike and walk trips. The U.S. DOT BCA Guidance rates are based on revealed (observed) or stated preference studies that provide marginal rates of substitution—willingness to tradeoff one aspect for another—between facility types.

Suggested values from U.S. DOT when estimating enhanced travel options benefits are shown in Table 22 and Table 23. Users are encouraged to check the U.S. DOT website for updated guidance and recommended values.

For projects where benefits are measured per person-mile walked or bicycled the Estimated Enhanced Travel Options Benefits from Walking or Bicycling are calculated using the following equation:

$$\begin{array}{ccccccc} \text{Estimated} & & \text{Existing trips per} & & \text{Value per} & & \text{Length of} \\ \text{enhanced travel} & = & \text{day} + 0.5 \times \text{new} & \times & \text{person-mile walked} & \times & \text{facility} \times 365 \text{ days} \\ \text{options benefits} & & \text{trips per day} & & \text{or bicycled} & & \text{in miles} \end{array}$$

Note that, the BCA Guidance recommends that the length in miles should be capped at 0.86 miles for pedestrian trips and 2.38 miles for bicycle trips, which are the average lengths of these trips based on 2017 national survey data. So, if a sidewalk project is 2 miles long, the length for the calculation would only be 0.86 miles, unless you have documentation justifying a

Table 22. Pedestrian facility improvements revealed preference values, U.S. DOT BCA Guidance 2023.

Improvement Type	Recommended Value per Person-Mile Walked or per Use (2022 \$) ¹
Expand Sidewalk (per foot of added width, up to 31ft) ²	\$0.11 per foot added, per mile
Install Marked Crosswalk on Roadway with Volumes ≥10,000 Vehicles per Day	\$0.19 per use
Install Signal for Pedestrian Crossing on Roadway with Volumes ≥13,000 Vehicles per Day	\$0.51 per use

Source: Adapted from U.S. DOT (2023a).

U.S. DOT BCA Guidance Notes for Table 22:

¹These values assume an average walking trip speed of 3.2 mph. For the mile-based benefits, the estimated value per user should be capped at 0.86 miles, the average length of a walking trip in the 2017 NHTS, unless the applicant has specific documentation suggesting longer trips or that a trip shorter than 0.86 miles is not feasible on the facility in question. In other words, applicants should not assume all pedestrians travel the full distance of a proposed facility if the facility is longer than 0.86 miles without a clear justification for doing so.

²Value for sidewalk width expansion applicable for sidewalks up to approximately 31 feet, benefits for expansions beyond this width should be described qualitatively.

Table 23. Cycling facility improvements revealed preference values, U.S. DOT BCA Guidance 2023.

Facility Type	Recommended Value per Cycling Mile (2022 \$) ¹
Cycling path with at-grade crossings	\$1.57
Cycling path with no at-grade crossings ²	\$1.97
Dedicated cycling lane	\$1.86
Cycling boulevard / “sharrow”	\$0.29
Separated cycle track	\$1.86

Source: U.S. DOT (2023a).
U.S. DOT BCA Guidance Notes for Table 23:
¹Values should only be applied over sections for which a comparable parallel facility is not available, and only applies to miles cycled on the project facility. These values assume an average cycling trip speed of 9.8 mph or, in the case of off-street paths with no at-grade crossings, a free-flow cycling speed of 12.1 mph. The estimated value per cyclist should be capped at 2.38 miles, the average length of a cycling trip in the 2017 NHTS, unless the applicant has specific documentation suggesting longer trips or that a trip shorter than 2.38 miles is not feasible on the facility in question. In other words, applicants should not assume all cyclists travel the full distance of a proposed facility if the facility is longer than 2.38 miles without a clear justification for doing so.
²The value for a cycling path with no at-grade intersections is higher due to an assumption of higher average speed of 12.1 mph, resulting in less time on the facility, which lowers journey quality benefits but increases travel time savings.

longer assumption. Applying these caps assumes that all the users are not walking or bicycling the entire length of the new facility.

For pedestrian crossing projects where benefits are measured per use, use the following equation:

Estimated enhanced travel options benefits

=

Existing trips per day + 0.5 × new trips per day

×

Value per use

×

365 days

Example: In the example provided here, we use an illustrative sidewalk expanded by 4 feet over 0.5 mile, which is expected to experience 200 walking trips per day (150 existing and 50 new users). The value per use of expanded sidewalks is calculated per foot of added width, so in this example the value per use is \$0.44 (\$0.11 × 4 feet).

150 + (0.5 × 50 trips per day)

×

\$0.11 (2022 \$)

per mile walked

×

4 feet width

× 0.5 miles

×

365 days

=

Estimated enhanced travel options benefit of \$14,053

(2022 \$) per year

Estimated Enhanced Travel Options Benefits

=

(150 existing trips per day + 0.5 × 50 new trips per day)

×

\$0.11 (2022 \$)

per mile walked

×

4 feet

× 0.5 miles

×

365 days

= \$14,053 (2022 \$) per year

The higher value of cycling enhancements compared to pedestrian enhancement is likely due to more developed research literature for cycling enhancements, with different underlying research methodologies, as well as the lower willingness of pedestrians to travel out of their way. For users choosing between a bike boulevard and stand-alone “sharrows,” the facility type described as “sharrows” in the BCA (see Table 23) primarily includes low-volume roadways with shared-lane markings that do not necessarily have other vertical or horizontal traffic calming installations.

Local Economic Benefits

Closing gaps in active travel networks can provide benefits from additional local economic activity. Overall, the research on these benefits is limited, and not typically generalizable across contexts. This section focuses on providing approaches for examining the local economic activity generated by closing active transportation network gaps. The economic activity mechanisms considered for this guide include:

- Increasing local business sales, employment, and/or wages (e.g., if there is more pass-by walking or biking traffic to certain businesses).
- Bringing additional visitors to a community, thus generating additional business activity beyond what is generated by local residents.
- Increasing property values, if people place a premium on direct access to facilities or on neighborhoods with better active travel conditions.

Different types of gap closures are likely to have different effects on these impact metrics:

- Local business activity from pass-by traffic is most likely to be impacted by gap closures along streets with commercial activity as well as nearby parallel facilities.
- New visitors are most likely to be attracted by gap closures to off-street shared-use paths such as a rail trail that is valued for its primarily recreational purpose.
- Property values are most likely to be impacted for residential areas in which general network connectivity is improved or where new facilities create a local access amenity, recreational opportunity, or streetscape improvement.

A recent study of street improvements for active transportation considered elements of economic activity such as wages, employment, and sales tax receipts from nearby businesses in six U.S. cities (PeopleForBikes et al. 2020). This study may be a useful resource for practitioners seeking additional insights into the local economic benefits that can result from closing active transportation network gaps.

Unit of Measure

There have been many studies of the economic impacts of active transportation facilities, examining impacts of (1) increased spending by active transportation tourists; (2) increased business sales or spending based on mode or destination shifts of local travelers; and (3) impacts of facilities on property values.

These studies have used a variety of methods: visitor surveys; time-series analysis comparing business activity before and after an improvement and/or in improved vs. non-improved areas;

Economic Impact Analysis vs. Benefit-Cost Analysis

Economic impact analysis, which measures changes in economic activity, is distinct from BCA, which places a monetary value on the various benefits and costs of an action.

Economic impacts may or may not have net social benefits that can be monetized, depending upon how they are distributed over time and space. Also, some benefits that can be monetized will not necessarily result in economic impacts.

The “property value” economic measure included in this section may reflect monetization of certain benefits of a gap closure (e.g., access or proximity value). Care should be taken not to double-count any benefits that are included in a BCA.

Method Selection

Figure 12 displays the proposed methods for understanding local economic benefits from closing gaps in the active transportation network.

An additional caution with the use of economic impact analysis is that increases in local economic activity may simply be shifted from elsewhere within the region or state, rather than representing net increases in economic activity. For example, people may spend more money in one neighborhood or community that is now more attractive, but less in another neighborhood or community that they now visit less frequently. There may also be some overlap among different categories of local economic benefits; for example, commercial property values may reflect and capitalize other benefits such as business activity, or increases in residential and

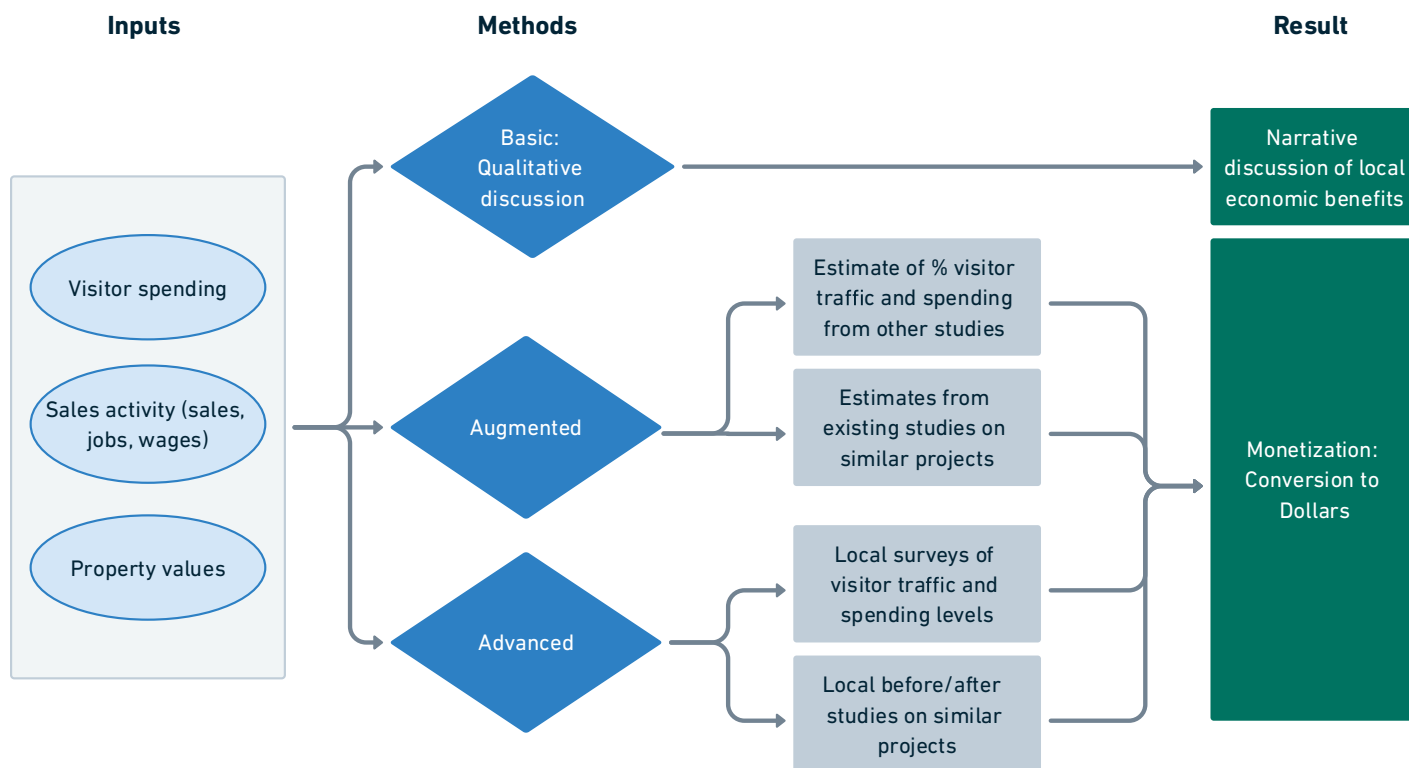


Figure 12. Basic, Augmented, and Advanced Application Methods for calculating local economic benefits.

commercial property values may reflect other benefits such as time savings or lower-stress access routes (which can be separately monetized as enhanced travel options benefits). Finally, the metric chosen to represent economic activity must be carefully chosen because some projects are accompanied by policies meant to combat the negative effects of gentrification that affect some metrics of economic activity.

Data Needs

Table 24 presents the key data items needed for estimating economic activity benefits.

Basic Methods

The Basic Method described in this section relies on summarizing findings from other studies to qualitatively assess the potential benefits. These instructions list the key questions that will impact the economic benefits of closing a gap.

1. **Using the following questions as prompts, describe the type of facility/gap closure, its land-use context, and its expected user characteristics.**
 - Is the increase in active travel from the project expected to be mainly local traffic, visitors, or a mix of both?
 - To what degree are there businesses present adjacent to the project that could benefit from pass-by walking and bicycling traffic?
 - To what degree are there businesses in the vicinity of the project (not necessarily adjacent) that might benefit from increased visitor activity (e.g., restaurants or lodging)?
 - To what degree are there residential properties directly served by the project? Is the project likely to be viewed as an amenity that improves property values?
2. **Qualitatively describe the potential economic benefits in the following categories.**
 - Increases in visitor travel and associated spending.
 - Increases in business activity due to pass-by walking and bicycling traffic.
 - Increases in property values as a result of the project.

Other Benefits

This guide focuses on providing actionable steps for estimating the benefits of closing gaps in active transportation networks, with straightforward methods approachable for many planners. We also wish to highlight other types of benefits that arose through the research and engagement

Table 24. Data needs for Basic Method economic benefit estimation.

Key Data Item	How to Determine
Number of new active trips	Follow procedure outlined in <i>Chapter 3: Step 2. Estimating Behavior Change</i> .
Visitor spending	Type of facility/gap closure and expected user characteristics (local vs. visitors).
Business activity—sales, jobs, wages	Type of facility/gap closure and land-use context—degree to which there is commercial activity along or near the facility that benefits from walk and bicycle pass-by traffic.
Property values	Type of facility/gap closure and land-use context—degree to which there is residential property adjacent to or served by the facility.

efforts for this project, but are lacking sufficient existing research for estimation, and/or are more complex and locally specific than the benefits discussed earlier in this chapter. Such benefits that are particularly important to planners, leaders, and decision-makers when considering transportation infrastructure projects include ecosystem services benefits, resiliency benefits, and noise/water improvement benefits. Research in this area is ongoing, but we recommend you at least consider how the gap closure project might provide ecosystem services or resiliency benefits using these resources.

Ecosystem Services Benefits

Ecosystem services are direct and indirect benefits to humans from the natural places near where we live: specifically, things like clean air and water, fertile soil for food production, pollination, or flood control. This is an ongoing area of research and types of benefits currently being studied include improved health, stronger social ties, and reduced costs of healthcare (U.S. EPA 2023b).

Active gap closure investments can often include complementary investments in conservation that preserve natural environments while improving access to natural spaces such as parks. Trail corridors, for example, can often serve as both transportation corridors and can provide important buffering spaces for conservation areas by including land protection as part of their right-of-way acquisition. For example, the planned Great Springs Trail in Texas between San Antonio and Austin seeks to ensure that 50,000 acres of sensitive aquifer recharge area are preserved to protect drinking water for future generations while providing an important sink for carbon dioxide (Alta 2022).

The U.S. EPA maintains a listing of ecosystem decision support tools which support varying levels of ecosystem services estimation (U.S. EPA 2023b). For example, the U.S. EPA has developed the Final Ecosystem Goods and Services (FEGS) Scoping Tool (U.S. EPA 2023a). This tool is intended to be used at the early scoping stage of a project to identify key persons and ecosystem services that may be relevant to the project. A recent U.S. EPA report FEGS Scoping Tool Analysis of Beneficiaries and Environmental Attributes for the Tillamook River Wetlands Project provides an example of the use of this tool.

In addition, there are tools provided by the Federal Emergency Management Agency (FEMA) that are specifically oriented toward providing a rough estimate of benefits from the protection of natural lands by tying land cover categories to the monetary value of ecosystem services they provide on a dollars per acre per year basis. FEMA provides an integrated BCA Toolkit that helps simplify the estimation of benefits by grant applicants who would like to understand the value of ecosystem services associated with nature-based solutions or environmental mitigation projects (FEMA 2023).

Resiliency Benefits

Some locations may expect more frequent natural disasters in the future, such as tornados or hurricanes, disrupting utilities and supply chains. Active modes are often the most useful immediately after a natural disaster, as they require human power and can be maneuvered in and out of boats, or around obstacles such as downed trees in the roadway.

Specifically for trails, the FHWA has sponsored a guide, “Trails as Resilient Infrastructure” which demonstrates how these facilities are inherently resilient, can be planned to be more resilient and sustainable, and have a place in responding to natural hazards and disasters (FHWA 2023).

Noise and Water Pollution Benefits

Noise and water pollution are additional environmental costs from motor vehicle travel. Research is sparse, but noise reduction benefits from active travel mode shift have been included in two applications (Li and Faghri 2014; Decisio and Transaction Management Centre 2012). Each of these relied on existing local estimates of noise per VMT and assumed a proportionate reduction. In both cases the share of total benefits was small ($<1\%$). These issues are very important to the public, however, so it is recommended that the plan for assessment as part of a future environmental process be communicated during project engagement.



CHAPTER 5

Step 4. Communicating the Results

This chapter focuses on conveying data findings and the results of complex gap closure analyses to different audiences and considers the typical communication methods. The chapter first discusses different audiences and provides guidance on how best to communicate with each. We discuss commonly used communication tools with a focus on the development of infographics, charts, and tables; example tools and materials are linked throughout for reference. Finally, we refer readers to additional resources including NCHRP reports about communicating data and developing high-quality graphic materials.



Addressing Your Audience

Communicating the benefit and the need to invest in a gap closure project can look different depending on the intended audience. Identifying the right audience will help ensure that your message reaches the right people using the right level of detail and conveys the relevant information. Each audience has different interests and objectives and levels of expertise when interacting with data and often has limited time to fully focus on and understand a concept or idea. Translating data into easy-to-understand tables, charts and infographics which resonate with the audience's values and underlying concerns can help agencies and practitioners maximize valuable communication dollars.

Who Is Your Audience?

Prior to developing a communication campaign or infographic communicating specific benefits, it is important to consider and define your target audience or audiences and what is important to them. Identifying their key values will help you convey your information in more targeted ways that connect with issues they care deeply about. Providing overly technical data or findings to the general public will not translate to strong support for building sidewalk compared to a story about the current hardships experienced by elderly residents walking along a high-speed arterial roadway with no sidewalk to reach the grocery store, along with an estimate of the number of other seniors who would benefit from the improvement. By identifying the issues that resonate most with audiences and using effective communication methods, practitioners will be able to use data to help tell the story of today's need and paint a vision of a better tomorrow.

Typical Audiences

The following groups are likely to consume information about the benefits of bicycle and pedestrian gap closure:

- Federal and state elected officials.
- County/city elected officials and advisory committees (appointed, quasi-elected).
- Engineers, technical staff, grant application reviewers.
- The news media.
- The general public.

Federal and State Elected Officials

The focus of gap closure benefits is usually local or regional, but the funding for closing gaps often comes from major federal grant programs such as the RAISE (Rebuilding American Infrastructure with Sustainability and Equity) program. Federal and state elected officials are often one step removed from local transportation decisions, but their support is invaluable when seeking grant funding. When developing grant applications, agencies will often seek letters of support from their federal and state elected officials to highlight the political support and general need for the gap closure (examples of grant application letters of support and their connections with highlighting benefits data are available on the York County Economic Alliance’s “Codorus Creek Beautification” webpage [YCEA 2023]). Communication with these officials may be infrequent but is invaluable to garnering support and accessing funding. Time with these officials is highly limited so communicating directly and concisely is of the utmost importance. When presenting the benefits of a gap closure, highlight large-scale macro-level benefits and direct benefits to the people in the elected official’s district; their constituents are often their primary concern. If an elected official has run on a platform focused on particular issues that could be addressed by the gap closure (such as traffic safety or revitalizing particular neighborhoods), highlight these in communications. While these officials are often supportive of closing gaps within the active transportation network, their limited time impacts the level of original content that they are typically able to develop. This means that providing template letters of support to these officials is a good strategy to make support easy and to ensure they have some talking points to build off when customizing the letter.

Key Communication Objectives

- Communicate benefits to their constituency.
- Connect benefits with their platform issues.

Top Questions

- How does this benefit the people in my district?
- Are there any issues of gentrification or displacement with this project?
- How does this project help bring more jobs and economic vitality to my district?

County/City Elected Officials and Advisory Committees (Appointed, Quasi-Elected)

Elected officials at the county and local level are typically aware of and involved in local public works projects from a decision-making standpoint. Elected city and county officials make decisions that directly impact the local transportation network by adopting local plans, allocating funding to projects, and providing support for public works projects when seeking additional grant funding. It is important to provide these elected officials with a strong understanding of what will be gained for their local community when allocating often limited public funds toward a specific improvement project. These officials are typically concerned with being good stewards of scarce public dollars and staff resources and want to be sure that their decisions to invest time and money into specific projects will have a tangible benefit. Additionally, these

elected officials are often concerned with needing to defend the investment decision within their community during and after construction based on the impact that can come to local businesses with prolonged construction projects; therefore, a key focus of these officials is often on the short- and long-term economic benefits from gap closures. Communicating with these officials is typically more frequent for practitioners at public agencies and may include opportunities for more in-depth discussions such as a board workshop or public hearing for delving into nuances of benefits analysis.

Members of advisory boards may be appointed or quasi-elected and are typically either advocates or industry professionals. These members have a narrower focus than the larger public bodies (city council/county board of supervisors) and will typically delve deeper into technical questions and considerations. Despite the narrower focus, the general information presented to these bodies will typically mirror that presented to members of larger public bodies.

Key Communication Objectives

- Communicate benefits to their constituency (or advisory board issue).
- Connect benefits with their platform issues and local needs.
- Showcase the return on investment of scarce local funds.

Top Questions

- How does closing this gap directly benefit the people in my district?
- Will this project cause issues of gentrification or displacement for people in my district?
- How will this project improve safety for school children in my district?

Engineers, Technical Staff, Grant Application Reviewers

This audience typically has a greater understanding of public works projects and are industry professionals familiar with technical jargon and heavy technical writing. These technical staff members have many responsibilities which may include reviewing grants or maintaining and implementing public works improvements. While reviewing grant applications for funding, members of this audience are typically most concerned with assessing the level of need and identifying how the proposed project will specifically address that identified need and to what level. It is important to highlight the direct connection between the existing need and the specific improvement identified to address that need. Using overly flowery or non-precise language when communicating with this audience can result in a project receiving a lower number of points or a project not being awarded funding on a grant application because the need was not accurately represented.

Key Communication Objectives

- Communicate the needs and benefits of the project.
- Clearly showcase the connection between need and proposed project.

Top Questions

- Does the proposed gap closure project meet engineering guidelines?
- What are the specific benefits for low-income and minority populations from this project?
- How involved was the public through this planning process?

The News Media

The news media are valuable partners for public works staff to promote the benefits of specific projects and initiatives within the community, including gap closure projects to the general

public. The news media can be instrumental in increasing exposure for initiatives and specific projects by putting them on the public's radar and artfully connecting the project with real issues for the public. Transportation agencies and practitioners can benefit from involving the news media early on upcoming projects to highlight key information and foster a strong working relationship to meet the typically tight deadlines. To satisfy typically tight content limits, the practitioner must provide a concise summary of the issue, need, and benefit of the specific gap closure project in a way that is approachable. The news media are also highly skilled at reformatting complex concepts into language that is approachable and understandable by the typical member of the public; news media staff may help offer suggestions for reformatting messaging that is too technical or difficult to understand. There are a wide variety of news media platforms that can be used to disseminate information and all of which are helpful for reaching a large audience; practitioners are encouraged to use multiple platforms for distributing a message to reach a broad audience.

Key Communication Objectives

- Clearly showcase the connection between need and proposed project.
- Highlight the key issues impacting the general public.

Top Questions

- Is this news important to our readers or viewers?
- Can we present this information concisely in a way that would be understood by our readers or viewers?
- Is this an ongoing issue/project or a one-time announcement?

The General Public

This group is the largest and most diverse audience of all and requires communication that is simple and suitable for all educational, technical, and interest levels. Because people have many concerns in their daily lives, the general public is typically uninterested in public works projects or plans until the project has a direct and immediate personal impact or benefit. People are often too busy to thoroughly read detailed methodologies and findings from a dense analysis on why a specific project will benefit them; this audience needs simple and concise messaging which directly ties the specific project with their personal needs and concerns. Communicating with the general public will typically involve either working with the news media or releasing content directly from the agency, such as a press release or social media post.

Clearly highlighting the benefit that will come to the community or directly to the individual living in the community helps to make the project benefits resonate with this audience and feel more tangible. It is often beneficial for public agencies to pre-test their communication materials for the general public in order to ensure that the message is digestible and understandable by the general public; this may be done internally or in collaboration with partners in the news media.

Key Communication Objectives

- Highlight how the project will benefit the individual directly.
- Showcase the indirect benefits from active transportation projects for those who do not bike or walk.

Top Questions

- How does this project benefit me?
- I'm not a walker or biker—does this project still benefit me?

- Why are we building one type of facility compared to another type (e.g., bike lane vs. shared-use path)?

Audiences and their associated communication objectives and example questions are included for reference in Table 25.

The Communication Process

When communicating the results of complex data analytics and the long-term macro-level benefits of specific improvements, it is important for the practitioner to convey a message that resonates with their audience. Practitioners are encouraged to consider the following

Table 25. Key communication objectives and top questions for target audiences.

Audience	Key Communication Objective	Top Questions
Federal and state elected officials	• Communicate benefits to their constituency. • Connect benefits with their platform issues.	• How does this benefit the people in my district? • Are there any issues of gentrification or displacement with this project? • How does this project help bring more jobs and economic vitality to my district?
County/city elected officials and advisory committees	• Communicate benefits to their constituency (or advisory board issue). • Connect benefits with their platform issues and local needs. • Showcase the return on investment of scarce local funds.	• How does closing this gap benefit the people in my district? • Will this project cause issues of gentrification for people in my district? • How will this project improve safety for school children in my district?
Engineers, technical staff, grant application reviewers	• Communicate the needs and benefits of the project. • Clearly showcase the connection between need and proposed project.	• Does the proposed gap closure project meet engineering guidelines? • What are the specific benefits for low-income and minority populations from this project? • How involved was the public through this planning process?
The news media	• Clearly showcase the connection between need and proposed project. • Highlight key issues impacting the local community.	• Is this news important to our readers or viewers? • Can we present this information concisely in a way that would be understood by our readers/viewers? • Is this an ongoing issue/project or a one-time announcement?
General public	• Highlight how the project will benefit the individual directly. • Showcase the indirect benefits from active transportation projects for those who do not bike or walk.	• How does this project benefit me? • I'm not a walker or biker, does this project still benefit me? • Why are we building one type of facility compared to another type?

five elements of the communication process when crafting messaging and outreach campaigns. More information regarding the communication process is provided by Zmud et al. (2009):

- **Context.** The setting or background within which the audience will understand and assess the messaging (i.e., larger trends in the world, nation, state such as a budget crisis or rise in pedestrian deaths). The context that the messaging is delivered in will shape the strategy, content, channel, and style of communication going forward.
- **Strategy.** The approach to influence decision-makers and the larger target audience. To identify a communication strategy, practitioners should identify the decision-makers, assess what is valuable from the audience's perspective, consider messaging that will resonate with core values of the audience, and then determine the best messenger.
- **Content.** The information itself that is presented to the audience, which is determined by the context and strategy. When crafting content, the practitioner should use clear, direct, and active language; minimize jargon; use humor where appropriate; and focus on telling the story of the data rather than pure data itself.
- **Channels.** The medium through which the content is presented to the audience, including printed or published materials, oral channels (word of mouth), broadcast (radio/television), and internet/computer based.
- **Style.** This represents the packaging that communication is in when presented to the audience inclusive of the design, layout, color, and fonts used to display the message. ADA accessibility requirements should be a key consideration as these elements are formulated.

If a strong connection between the data presented and the audience's values or key issues is not established, even the most important findings or data may be quickly forgotten or overlooked by your audience. Apply the following concepts to help establish a better connection with your audience (Mendonca and Miller 2007):

- **Simplicity.** Short and simple messages are easier to remember and repeat.
- **Unexpectedness.** Messages which are common sense or expected will not stick in the reader's mind.
- **Concreteness.** Stating clearly and specifically what, who, and how will help a reader understand a message more easily than an abstract concept of what could occur over a long time period.
- **Credibility.** Audiences are experts at detecting spin in messaging, especially if the underlying message does not reflect their lived experience. For example, shying away from or painting a better picture than the existing realities which pedestrians and bicyclists face every day may undermine an agency's credibility when engaging with the general public.
- **Stories and emotions.** Connecting data with human emotions and stories will help to mentally stimulate the reader/listener and will better resonate with them. Stories that involve people and emotions are more engaging than data-heavy descriptions of analysis and will do a better job resonating with the reader/listener and leave a more lasting impression.

Communicating Uncertainty

Users of this guide will produce estimates of benefits based on the empirical research to date (which is limited), a range of assumptions, and different degrees of simplification. Therefore, it is important to remember that these are estimates, which come with a range of sources of error. Communicating some level of uncertainty when sharing the resulting estimates is necessary.

Some best practices for communicating uncertainty to decision-makers and broader audiences include the following:

- Clearly explain that these are estimates based on data from a range of places and various assumptions, and acknowledging the limitations.

- Present at least two estimates, with one using more conservative assumptions. Explain the key differences in assumptions between the estimates. This could be done, for example, by producing behavior-change estimates for both a “low” and “medium” degree of improvement or by using more conservative assumptions about route shift potential or existing active travel levels at a location.
- Consistently use accurate language. Refer to the numbers as estimates rather than definitively stating, for example, that “the project will result in 656 additional bicycle trips.” Instead, “We estimate that the project will result in 580 to 660 additional bicycle trips.”
- Round off the estimates. Providing a precise number, such as 656 trips or 2.34 tons of emission reductions implies an inappropriate level of certainty.

Frequently Used Tools

Infographics, charts, and tables are all tools that are frequently used to summarize robust technical information. The tools are summarized here. The section concludes with additional reading, which provides more detailed discussions of how to convey robust quantitative information.

The combination of the context, strategy, content, channel, and style for communicating the benefits of gap closure often lends itself to the development of one or more infographic in support of the project. An infographic at its core is simply a clear and concise graphical representation of information or data, which can take many forms. Due to the multitude of benefits that are generated from the closure of gaps in the active transportation network, there are often numerous aspects to highlight and combining them together onto a single infographic can help to visualize the project benefits in aggregate. Infographics can take a wide variety of forms and there is no one size fits all. However, infographics and data visualization should follow a handful of best practices identified in this section to best convey their important information. Furthermore, this section highlights typical infographics used to showcase the various types of benefits and includes different examples from projects around the country. These examples are intended to be used as inspiration for practitioners to craft their own infographics using in-house or third-party software; online infographic generation is available through a variety of websites which may help to streamline the creation of simple infographics for practitioners. For a more highly detailed description of the wide variety of data visualization options a practitioner could consider including as part of their infographic, please refer to *NCHRP Web-Only Document 226: Data Visualization Methods for Transportation Agencies* (Higgins et al. 2017).

Communicating Effectively

When presenting data to any audience, especially those of a non-technical background, it is critical to display it in a way that directly presents the right information in a clear manner that tells a larger story beyond just the data. This means making a connection between the data point and the larger context that it comes from. Data by itself are often unhelpful without making a more direct and clear human connection to the larger community. Extrapolating what an increase in ridership activity would mean in terms of climate benefits, transportation time savings, health improvements, reductions in VMT, and economic benefits to the local economy will resonate more with wider audiences than a simple increase in bicycle usage. Practitioners must use data to tell the larger story to communicate the true benefit of gap closure projects most effectively with their target audience.

Beyond telling the larger story, practitioners must ensure that each graph and chart presented has a clear message and takeaway to keep their audience focused on the key communication objective. The most effective graphics are:

- Quickly comprehended and understood by the reader.
- Relevant to the data and topic or answer one or more fundamental questions.
- Formatted with a sense of balance, proportion, and clarity of design.
- Able to stand on their own if lifted from the page (no accompanying text).
- Aligned between the data, analysis, and scale.

Furthermore, when creating graphs which highlight key data, practitioners should ensure that each graph includes basic elements including a title, scale, units, and simple colors.

Sample Graphics

Sample graphics that highlight the benefits of various aspects of gap closure projects are highlighted in Figures 13 and 14. You are encouraged to use these examples for inspiration when creating your own project-specific graphics.

Wolf River Greenway—Memphis, TN

The *Memphis Gets Moving: Health & Economic Impacts of Building the Wolf River Greenway* document (Alta 2015) presents the estimated benefits to the Memphis, TN, area following the construction of the proposed 36-mile-long Wolf River Greenway linking Memphis with nearby suburban communities. This document presents a clear and concise overview of the total benefits to the community as well as a breakdown of each element included in the overall analysis in a single page spread (e.g., public health, transportation, environmental, and economic). This one-stop shop for all benefits on a single page is presented in approachable and accessible language that is easy for many audiences to understand and support.

Great Springs Trail Project—San Antonio, TX, to Austin, TX

The *Economic Benefits of the Great Springs Trail* report (Alta 2021) highlights the potential benefits from the construction of the proposed Great Springs Trail, which would link San Antonio with Austin, TX, and include nearly 170 miles of trails. The report showcases overall benefits to the area across economic, land and water, transportation, and health metrics and includes a detailed summary of each metric. The brief report is easy to read and clearly highlights the need for the trail as well as the significant benefits in clear and approachable language using graphics that can stand on their own and create impactful messages around the benefits of the trail.

Codorus Greenway—York County, PA

This example (Figure 15) from the Codorus Greenway RAISE Grant Application (YCEA 2022) highlights important takeaways for grant reviewers who are technical in nature and interested in specific metrics regarding benefits.

Further Reading

You are encouraged to review the additional resources provided in the following list for further detailed explanations and example materials before developing communication materials and content. These additional resources provided highly detailed information about developing communication plans and identify the benefits and typical uses of the wide array of graphs and tables that are available to practitioners.

- *NCHRP Web-Only Document 226: Data Visualization Methods for Transportation Agencies* (Higgins et al. 2017).
- *NCHRP Report 610: Communication Matters: Communicating the Value of Transportation Research Guidebook* (Zmud et al. 2009).



Source: Alta (2015).

Figure 13. Wolf River Greenway sample graphic.



Source: Alta (2021).

Figure 14. Example graphic from the Great Springs Trail Benefits Report.

OVER 20 YEARS...



The motor vehicle trip replacement factor for all active mode trips is **0.36**, meaning we can expect new bike and pedestrian facilities to **replace 36% of car trips**.



The Codorus will **replace 1,396,900 total vehicle miles** and provide **\$11,740,000 in health benefits** related to increased physical activity.

Source: YCEA (2022).

Figure 15. Codorus Greenway RAISE grant application graphic showing health benefits.

- *NCHRP Research Report 926: Guidance to Improve Pedestrian and Bicyclist Safety at Intersections*, “Chapter 2: Identify and Collect the Data for Analysis” (Sanders et al. 2020).
- *Bridging the Gap Between Agencies and Citizens* (Bremmer and Bryan 2008).
- “Crafting a Message that Sticks: An Interview with Chip Heath,” *The McKinsey Quarterly* (Mendonca and Miller 2007).
- *Making Bikes Count: Effective Data Collection, Metrics, & Storytelling* (NACTO 2022).
- *Metro Active Transportation: Return on Investment Study*, pages 26–33 (Metro 2022).



CHAPTER 6

Steps in Action (Analysis Walkthroughs)

Introduction

This chapter illustrates how to calculate benefits for two hypothetical gap closure projects. The two walk-through examples cover the benefits described in the guide and two of the most common project types: installation of a new facility and upgrade of an existing facility. Table 26 shows a summary of the two examples and the benefit categories that were calculated for each.

Southwest Case Study

Name: Shared-Use Path.

User Type: Bicycle/Pedestrian.

Context of Study Area: Suburban.

Scale of Study Area: Neighborhood.

Barrier Form: Missing Facility.

Deficiency:

- Safety issue: one fatality (Pedestrian), four serious injuries (two Pedestrian, two Bicycle), two nonserious injuries (Pedestrian) and one possible injury (Bicycle) on the corridor in last 5 years.
- Lack of connectivity along the corridor for active transport.

Project Length: Approximately 3 Miles.

Proposed Gap Closure Method: Construction of a 3-mile, shared-use path linking a suburban neighborhood to a shopping center.

Route Shift Expected: Yes.

Tools and Models Used: GIS, Python based scripting tools (not necessary).

Data Sources

- OpenStreetMap (OSM) network
- Count databases for comparable facilities with existing counts
- U.S. Census 5-year ACS summary estimates
- Collision data

This analysis employs the use of U.S. Census 5-year ACS summary estimates for Means of Transportation to Work (ID: B08301). As the facility does not currently exist, all trips along the facility will be considered new trips. However, it is important to note that trips along the facility could simply be shifted from alternative routes. In place of ACS data, demand data such as local, national, or regional activity-based model results could be used.

Table 26. Characteristics of analysis walk-through examples.

General Location of the Example	Southwest, USA	Northeast, USA
User Type Impacted	General	Bicyclists
Type of Project	Shared-use path (new facility)	Bike lane upgrade
Behavior Change	✓	✓
Health Benefits	✓	✓
Safety Benefits	✓	✓
Emissions Benefits	✓	Not in this example
Enhanced Travel Option Benefits	Not in this example	✓
Local Economic Activity Benefits	Not in this example	Not in this example

Behavior Change

The first step to estimate behavior change is to determine if there should be a baseline estimate. Since this is a new facility, there is no existing baseline and we proceed to the estimating change in activity using the Context Transfer Process. The proposed project is over 2 miles long, so the project will be split into three 1-mile segments to conduct this analysis (see *Appendix B*).

To conduct the Context Transfer Process, we used three comparable facilities with existing counts to create a robust estimate for our project. Comparable facilities were selected based on the relative similarity to the project location in terms of population density, topography, land-use characteristics, facility characteristics, and nearby relative averaged daily traffic volumes.

The strength of a context transfer is highly dependent on the amount of data available and on how comparable the available data are to the proposed project. We recommend using at least three to five count locations at comparable facilities to create a more normalized comparison than simply using one or two comparable facility count locations. Existing counts for the comparable facilities used in this example are found in Table 27. In our project, the counts for the comparable facilities were annualized. If using shorter periodic counts, you either need to annualize before this step or apply an annual adjustment factor.

The next step is to identify demand metrics available at the project area and the comparable facilities. For this example, we are using outputs from U.S. Census 5-year ACS summary estimates for Means of Transportation to Work (Table ID: B08301) at the Census Block Group level as the demand data to relate the comparable facilities to our project. Active travel demand is calculated by looking at all bicycle and pedestrian trips within each modes' influence area (0.5 mile for walking and 1.25 miles biking).

Table 27. Details of comparable facilities chosen to estimate bicycling and walking trips for the project.

Count Location	Annual Average Daily Pedestrian Counts	Annual Average Daily Bike Counts
Count Location #1	213	97
Count Location #2	407	211
Count Location #3	317	174
Average Existing Counts at Comparable Facility Count Locations	312	161

Ideally, the demand data is proportionally allocated to each buffer of the 1-mile segments for the respective bicycle and pedestrian areas of influence (see *Proportional Allocation* in *Chapter 3: Step 2. Estimating Behavior Change*). If you do not have access to GIS software and are unable to proportionally allocate the data, you can total the number of bicycle and pedestrian trips in CBGs that are in or partially in the influence areas.

Next, average the demand data at the comparable facilities as shown in Table 28.

Next, sample the same demand data [all active trips in the U.S. Census 5-year ACS summary estimates for Means of Transportation to Work (ID: B08301)] that occur in the project area of influence, as shown in Table 29.

Calculate the Context Transfer Estimates:

To calculate the context transfer estimates on the project segments, create context adjustment factors by dividing the demand rate of each 1-mile segment along the proposed trail by the average demand rate for the comparable facilities. Then multiply that context adjustment factor by the average of existing counts along comparable facilities:

$$\text{Estimated Trips Per 1-Mile Project Segment} = (X/Y) \times Z$$

Where:

X = Pedestrian or bicycle demand rate near proposed project 1-mile segment

Y = Pedestrian or bicycle average of demand rates near the comparable facility count locations

Z = Pedestrian or bicycle average of existing counts along comparable facilities

Reminder: This calculation uses demand from within the buffer of the respective mode's influence area of each 1-mile segment of the proposed trail.

Table 28. Demand at comparable facilities, by mode influence area.

Count Location	All Active Trips in Pedestrian Influence Area (0.5 Mile Buffer)	All Active Trips in Bicycle Influence Area (1.25 Mile Buffer)
Count Location #1	5,482	7,468
Count Location #2	9,954	11,430
Count Location #3	7,774	9,836
Average Active Trips at comparable facility count locations	7,737	9,578

*If desired, users could use other statistical methods such as using a minimum and maximum to communicate uncertainty.

Table 29. Demand for the representative count locations along the project, by mode influence areas.

Count Location	All Active Trips in Pedestrian Influence Area (0.5 Mile Buffer)	All Active Trips in Bicycle Influence Area (1.25 Mile Buffer)
Project 1-mile Segment #1	7,428	8,874
Project 1-mile Segment #2	7,185	8,245
Project 1-mile Segment #3	7,860	9,023

Next, since the proposed project is longer than 2 miles, you transform estimated trips per mile-long segment to total estimated person-trips along a proposed project, by summing the respective estimates and then dividing by the appropriate NHTS Trip Length Factor. For walking, the trip length factor is 0.86 miles, and for bikes, the trip length factor is 2.38 (see *Appendix B*).

$$\text{Estimated Trips} = \text{SUM}(\text{Estimated Trips Per Mile Segment}) / F$$

Where:

F = Pedestrian or Bicycle NHTS Trip Length Factor

Using the example project the estimates would look like this:

Walk:

$$\text{Estimated Trips Walk Mile Segment 1} = (7,428 / 7,737) \times 312 = 300$$

$$\text{Estimated Trips Walk Mile Segment 2} = (7,185 / 7,737) \times 312 = 290$$

$$\text{Estimated Trips Walk Mile Segment 3} = (7,860 / 7,737) \times 312 = 317$$

Bike:

$$\text{Estimated Trips Bike Mile Segment 1} = (8,874 / 9,578) \times 161 = 149$$

$$\text{Estimated Trips Bike Mile Segment 2} = (8,245 / 9,578) \times 161 = 139$$

$$\text{Estimated Trips Bike Mile Segment 3} = (9,023 / 9,578) \times 161 = 152$$

Total After Adjusting for Trip Length:

$$\text{Estimated Walk Trips} = (300 + 290 + 317) / 0.86 = 1,054$$

$$\text{Estimated Bike Trips} = (149 + 139 + 152) / 2.38 = 185$$

The demand estimates for the new trail are 1,054 pedestrians and 185 bicyclists per day.

Finally, since different benefit analyses need information regarding what type of new trip the proposed facility will attract (e.g., mode shift, route shift), we need to calculate the change stratification. Route shift is expected for this project, as there is an existing parallel corridor 0.5 mile from the facility that people walking and biking are expected to detour from onto the new proposed shared-use path. As a result, we use the values in Table 6 with route shift to look up travel shift values. Table 30 includes the operation and results of calculating travel shifts.

Health Benefits

To quantify the benefits of increased physical activity related with the proposed facility, we will use U.S. DOT multipliers in conjunction with the estimated number of trips on the proposed facility. The results of this analysis are dollar figures.

Table 30. Change stratification calculations for the Southwest walk-through example.

Number of New Active Travelers	Mode Shift from Motor Vehicles	Activity Shift/Trip Generation/New Travel	Route Shift	Other Mode Shift
Pedestrians	$1,054 \times 0.18 = 190$	$1,054 \times 0.06 = 63$	$1,054 \times 0.50 = 527$	$1,054 \times 0.26 = 274$
Bicyclists	$185 \times 0.17 = 31$	$185 \times 0.11 = 20$	$185 \times 0.60 = 111$	$185 \times 0.12 = 22$

To estimate the health benefits of the project, apply U.S. DOT mortality reduction multipliers to the induced and vehicle shift new trip counts. U.S. DOT mortality reduction factors for a pedestrian trip are \$7.63 and for a bicyclist trip \$6.80. Furthermore, U.S. DOT multipliers only apply to the ages of 20 to 74 (68% of all trips) for walking and 20 to 64 (59% of all trips) for biking (note: all monetary estimates in 2022 \$). The following equation is used to calculate the daily dollar value of health benefits:

$$\text{Daily Dollar Figure} = Z \times Y \times X$$

Where:

X = Percentage of users within relevant age range

Y = Mortality reduction factor bicyclist or pedestrian

Z = Sum of vehicle and induced demand shifts of new trips

For this case study, the calculations are as follows:

$$\text{Walking Daily Dollar Figure} = (190 + 63) \times \$7.63 \times 0.68 = \$1,312.67$$

$$\text{Bicycling Daily Dollar Figure} = (31 + 20) \times \$6.80 \times 0.59 = \$204.61$$

$$\text{Annual Total} = (\$1,312.67 + \$204.61) \times 365 = \$553,806.18$$

Emission Benefits

To quantify the benefits of shifting from driving to active travel induced by the proposed facility, we will use FHWA emission rates combined with U.S. DOT monetized values per metric ton to come to dollar sums. The emissions rates are found in Table 18 and the U.S. DOT monetized values per metric ton come from Table 20. Emission rates and U.S. DOT monetized values used are both for the year 2025. For this example we will use the average length of bicycle and pedestrian trips from the NHTS.

To estimate the emissions changes because of the project, multiply the number of new trips each day by the prior drive mode share by the trip length by the emission rate. Similar to the health benefits, the new trips used in this analysis stem only from the trips attributed to the induced and vehicle shift.

$$\text{Avoided Annual Emissions} = \text{new trips per day} \times \text{prior drive mode share}$$

$$\times \text{trip length in miles} \times \text{emission rate in g/mi}$$

$$\times 365 \text{ days/year}$$

72 Estimating Benefits of Closing Gaps in Active Transportation Networks: A Guide**Walk:**Avoided Emissions (CO₂)

$$= \left[\frac{(190 + 63) \text{ trips per day} \times 0.71}{\times 0.86 \text{ mi} \times 371 \text{ g/mi} \times 365 \text{ days/year}} \right] / 1,000 \text{ g/kg} = \mathbf{20,919.2 \text{ kg/year}}$$

Avoided Emissions (NO_x)

$$= \left[\frac{(190 + 63) \text{ trips per day} \times 0.71}{\times 0.86 \text{ mi} \times 0.191 \text{ g/mi} \times 365 \text{ days/year}} \right] / 1,000 \text{ g/kg} = \mathbf{10.8 \text{ kg/year}}$$

Avoided Emissions (PM_{2.5})

$$= \left[\frac{(190 + 63) \text{ trips per day} \times 0.71}{\times 0.86 \text{ mi} \times 0.012 \text{ g/mi} \times 365 \text{ days/year}} \right] / 1,000 \text{ g/kg} = \mathbf{0.7 \text{ kg/year}}$$

Bike:Avoided Emissions (CO₂)

$$= \left[\frac{(31 + 20) \text{ trips per day} \times 0.71}{\times 2.38 \text{ mi} \times 371 \text{ g/mi} \times 365 \text{ days/year}} \right] / 1,000 \text{ g/kg} = \mathbf{11,670.0 \text{ kg/year}}$$

Avoided Emissions (NO_x)

$$= \left[\frac{(31 + 20) \text{ trips per day} \times 0.71}{\times 2.38 \text{ mi} \times 0.191 \text{ g/mi} \times 365 \text{ days/year}} \right] / 1,000 \text{ g/kg} = \mathbf{6.0 \text{ kg/year}}$$

Avoided Emissions (PM_{2.5})

$$= \left[\frac{(31 + 20) \text{ trips per day} \times 0.71}{\times 2.38 \text{ mi} \times 0.012 \text{ g/mi} \times 365 \text{ days/year}} \right] / 1,000 \text{ g/kg} = \mathbf{0.4 \text{ kg/year}}$$

Combined Total Avoided EmissionsTotal Annual Avoided CO₂ Emissions = 20,919.2 kg + 11,670.0 kg = 32,589.2 kg**= 32.59 metric tons**Total Annual Avoided NO_x Emissions = 10.8 kg + 6.0 kg = 16.8 kg = **0.0168 metric tons**Total Annual Avoided PM_{2.5} Emissions = 0.7 kg + 0.4 kg = 1.1 kg = **0.0011 metric tons**

Monetized Avoided Emissions

Monetized CO₂ Avoided = 32.59 metric tons CO₂ × \$237 per metric ton CO₂ = **\$7,723.64**

Monetized NO_x Avoided = 0.0168 metric tons NO_x × \$20,300 per metric ton NO_x = **\$340.59**

Monetized PM_{2.5} Avoided = 0.0011 metric tons PM_{2.5} × \$975,500 per metric ton PM_{2.5}
= **\$1,028.27**

Total Annual Benefits from Emissions Reduction = **\$9,092.50**

Safety Benefits

To quantify the benefits that our project will bring to enhanced safety for active transportation trips, we will use CMFs to estimate the relative reduction in crashes we will expect on the facility. We will use the crash-history approach in this example. After conducting this step, we will monetize the reduction in crashes.

The first step in the crash-history approach is to identify the appropriate crashes that have occurred in the project area. The following crashes are relevant for the crash-history approach: (1) crashes that have occurred directly on the facility, and (2) crashes that have occurred on facilities near the proposed facility from which pedestrians or bicyclists may be expected to detour if an improved facility is installed to close the gap. In this example, there is an existing parallel corridor 0.5 miles from the facility that people walking and biking are expected to detour from onto the new proposed shared-use path. The crash history was collected from the parallel facility and summarized in Table 31.

CMFs are then used to estimate the reduction in crashes that building the proposed facility will have, as shown in Table 32. The CMFs used in this section are the following:

- Adding Sidewalk (CMF: 11246) Crash Reduction Value: 0.402
- Install Shared Path (CMF: 9250) Crash Reduction Value: 0.25

Table 31. Summary of crashes for the Southwest walk-through example.

Crash Severity	Crashes with 1+ Pedestrians	Crashes with 1+ Bicyclists (and 0 Pedestrians)
Fatal Injury (K)	1	0
Suspected Serious Injury (A)	2	2
Suspected Minor Injury (B)	2	0
Possible Injury (C)	0	1
No Injury (O)	0	0

Table 32. Applying CMFs in Southwest walk-through example.

Row ID	Crash Severity	Number of Pedestrian Crashes Avoided by Gap Closure	Number of Bicycle Crashes Avoided by Gap Closure
1	Fatal Injury (K)	1 × 0.402 = 0.402	0
2	Suspected Serious Injury (A)	2 × 0.402 = 0.804	2 × 0.25 = 0.5
3	Suspected Minor Injury (B)	2 × 0.402 = 0.804	0
4	Possible Injury (C)	0	1 × 0.25 = 0.25
5	No Injury (O)	0	0

After calculating the relative number of bicycle crashes that could be avoided by a gap closure, these values are then monetized according to U.S. DOT KABCO values. The formula for calculating the monetary values is specified in Table 33.

Total Benefits

The total benefits from the proposed facility are found from summing up the relative monetary benefits identified for this case study. The monetary benefits of each respective benefit type are identified and summed as follows:

Estimated Pedestrian and Bicycle Counts: 1,054 pedestrians and 185 bicyclists	
Health Benefits:	\$553,806
Emission Benefits:	\$9,092
Safety Benefits:	\$6,790,313
Total Annual Benefits:	\$7,353,211

Northeast Case Study

Name: Striped Bicycle Lane to a Protected Bike Lane on Urban Arterial.

User Type: Bicycle.

Context of Study Area: General Urban.

Scale of Study Area: Neighborhood.

Facility Type: Corridor Design and Operation.

Deficiency:

- Safety issue: LTS 4 facility, although no crashes on specific segment in last 5 years.
- Lack of connectivity along the corridor for active transport.

Project Length: 1.5 Miles

Proposed Gap Closure Method: Construction of a 1.5-mile, Protected Bike Lane upgrading an existing Striped Bicycle Lane, on an urban corridor with businesses on both sides of the street.

Route Shift Expected: Yes.

Tools and Models Used: GIS, Python based scripting tools (not necessary).

This example uses existing baseline bicycle counts and the Percent Change Process to estimate behavior change due to the gap closure project.

Table 33. Example calculations for monetizing safety benefits.

Row ID	Severity	Monetary Value of Avoided Pedestrian Crashes	Monetary Value of Avoided Bicycle Crashes
1	Fatal Crashes (K)	$0.402 \times \$12,500,000 = \$5,025,000$	0
2	Suspected Serious Injury Crashes (A)	$0.804 \times \$1,188,200 = \$955,313$	$0.5 \times \$1,188,200 = \$594,100$
3	Suspected Minor Injury Crashes (B)	$0.804 \times \$233,800 = \$187,975$	0
4	Possible Injury or Complaint of Pain Crashes (C)	0	$0.25 \times \$111,700 = \$27,925$
5	Property Damage Only Crashes (O)	0	0
6	SUBTOTAL Monetary Benefit by Mode	\$6,168,288	\$622,025
7	TOTAL Monetary Benefits of Gap Closure	\$6,790,313	

Data Sources

- Crash data
- OpenStreetMap (OSM) network
- Bicycle and pedestrian count data on the existing facility

Behavior Change

The first step of any gap analysis is calculating the number of estimated users on the new proposed facility. This specific project would upgrade an existing striped bicycle lane to a protected bike lane. This proposed project will only include benefits for bicyclists, as the pedestrian experience is assumed to be unchanged.

Since there is current biking on the facility (an existing baseline), this example uses the Percent Change Process to estimate behavior change on the proposed facility. The proposed facility provides a lower-stress option and better access to destinations than existing parallel facilities in the area, with either a more direct route or a reasonable detour, so it is expected to induce a route shift.

The first necessary type of data for the Percent Change Process is counts on the existing facility. The percent change factors are applied to these existing counts along the facility. Existing counts for this case study are found in Table 34, where the counts for the comparable facilities were an annual average. If using shorter periodic counts, you either need to annualize before this step or apply an annual adjustment factor.

Next, gather or calculate LTS scores on the existing network. Once calculated or gathered, analyze the relative change in LTS scores expected from implementing the project. In this example the original facility was scored LTS 4, and the project improved the score to LTS 2. As a result, the relative degree of improvement number to determine the expected behavior change is 2: Medium (see *Chapter 3: Step 2. Estimating Behavior Change*). This calculation is summarized in Table 35.

After calculating the LTS and the relative average daily count along the facility, apply the Percent Change Process to calculate estimated counts along the facility. This involves looking up the relative percent change factor in either Table 1 or Table 2. This project is expected to have a significant route shift, because although there is an existing bike lane, it is a high-stress corridor and is uncomfortable for the average cyclist to use the facility given its LTS score of 4. The proposed facility will transform this segment of the high-stress corridor into one where most cyclists will feel comfortable on it, opening access to bicyclists to all of the businesses that are on both sides of the corridor. Because of this expected significant route shift, we will be using Table 2 for our percent change factors.

The calculation to create the Percent Change Process estimates is as follows:

To estimate the Percent Change Process estimations because of the project, apply the relevant percentage increase in bicycle travel to the existing counts:

$$\text{Estimated Trips Per Mile Segment} = Y + (Y \times X)$$

Where:

X = Percentage increase in bicycle travel

Y = Average daily count

Table 34. Existing counts of bicycle trips along the Class II facility.

Count Location	Annual Average Bike Counts
Count Location #1	240

Table 35. Calculations for LTS improvement after implementing the project.

Scenario	LTS Score
Existing	4
Proposed	2
Change	2

Using the example project, the estimates would look like this:

$$\text{Estimated Trips} = 240 + (240 \times 1.74) = 658$$

The estimate of active travel behavior for the new facility is **658 bicycle trips per day**.

Finally, since different benefit analyses need information regarding what type of new trip the proposed facility will attract (e.g., mode shift, route shift) a change stratification calculation must be conducted. Use the values for travel shifts from Table 6 to calculate the change stratification. Table 36 includes the operation and the change stratification calculation results.

$$\text{Estimated New Trips} = 658 - 240 = 418$$

Health Benefits

To quantify the benefits of increased physical activity related with the proposed facility, we will use U.S. DOT multipliers in conjunction with the estimated new trips attributable to vehicle shift and induced trip new trips. The results of this analysis are dollar figures.

To estimate the health benefits of the project, apply U.S. DOT mortality reduction multipliers to the induced and vehicle shift new trip counts. The U.S. DOT mortality reduction factor for a bicyclist trip is \$6.80. Furthermore, U.S. DOT multipliers only apply to the ages of 20 to 64 (59% of all trips) for biking:

$$\text{Daily Dollar Figure} = Z \times Y \times X$$

Where:

- X = Percentage of population within 20–64 age range
- Y = Mortality reduction factor for bicyclists
- Z = Sum of vehicle and induced demand shifts of new trips

$$\text{Bicycle Daily Dollar Figure} = (71 + 46) \times \$6.80 \times 0.59 = \$469$$

$$\text{Annual Total} = ((71 + 46) \times \$6.80 \times 0.59) \times 365 = \$171,332$$

Enhanced Travel Options Benefits

To quantify the benefits of enhanced travel options associated with the project, we will use U.S. DOT rates regarding specific infrastructure upgrades, in conjunction with existing and new trips along the proposed facility.

Table 36. Change stratification calculations for Northeast walk-through example.

Number of New Active Travelers	Mode Shift from Motor Vehicles	Activity Shift/Trip Generation/New Travel	Route Shift	Other Mode Shift
Bicyclists	$418 \times 0.17 = 71$	$418 \times 0.11 = 46$	$418 \times 0.60 = 251$	$418 \times 0.12 = 50$

To estimate the enhanced travel options benefits of this project, apply U.S. DOT bicycling facility improvement values in Table 23 to the estimated users along the facility, using the “rule of half” to adjust benefits for new users. In this example, it is an upgrade from a striped bicycle lane to a protected bike lane and the following equation is used:

Daily Dollar Figure = (W + 0.5 × Y) × Z × X

Where:

- X = Recommended benefit value of improved enhanced travel options (the recommended value per cycling mile is \$1.86 for this example)
- Y = Estimated new trips along the facility
- W = Estimated existing trips along the facility
- Z = Miles of proposed facility

Bicycle Daily Dollar Figure = (240 + 0.5 × 418) × 1.5 × \$1.86 = \$1,253

Annual Total = \$1,253 × 365 = \$457,345

Safety Benefits

To quantify the benefits that the proposed facility will bring to enhanced safety for active transportation trips, we will use CMFs to estimate the relative reduction in crashes we will expect on the facility. We will use the crash multiplier approach for this example (assuming there is no historic crash data available for the facility). After conducting this step, we will monetize the reduction in crashes.

The first step in the crash multiplier approach is identifying the relative length of the segment and intersections affected that will be used in the calculations. In this example the relative distance of the proposed facility is 1.5 miles and there are two intersections in the project area, and these numbers are reflected in Table 37.

The next step is to use the crash multiplier tables in Appendix G to estimate the relative number of crashes that would be expected to be seen on the facility. The crash multipliers used were from Table G-4, the table listing crash multipliers for bicycles for a minor arterial facility type. A sample calculation follows. All other calculations and results are seen in Table 38.

Crash Estimate = distance of proposed facility × crash multiplier

Once the relative estimates of crashes along the facility are calculated, CMFs are then used to estimate the reduction in crashes that building the proposed facility will have.

The CMF used in this section is as follows:

- Convert Traditional or Flush Buffered Bike Lane to Separated Bike Lane with a Blend of Flexi-Post and Other Vertical Elements (CMF: 11303), Crash Reduction Value: 0.398

Table 37. Relative mileage of project.

Functional Class (Midblock/Segment) or Maximum Functional Class (Intersection)	Miles in Project Area	Intersections in Project Area
Minor Arterial	1.5	2

Table 38. Example calculations of expected crashes for Northeast walk-through example.

Row ID	Severity	Bicyclists Use ESTIMATE unless HISTORY is available, and Subtotal in Table 12 is greater than 0: Crashes in One Year
1	Fatal Crashes (K)	$1.5 \times 0.005323610 / 5 = 0.0016$
2	Suspected Serious Injury Crashes (A)	$1.5 \times 0.051905198 / 5 = 0.0156$
3	Suspected Minor Injury Crashes (B)	$1.5 \times 0.308325750 / 5 = 0.0925$
4	Possible Injury or Complaint of Pain Crashes (C)	$1.5 \times 0.217380745 / 5 = 0.0652$
5	Property Damage Only Crashes (O)	$1.5 \times 0.048799759 / 5 = 0.0146$
6	TOTAL: K + A + B + C + O	0.1895

The formula for calculating the reduction in crashes is as follows, with example calculations provided in Table 39:

Estimated number of bicycle crashes avoided = crash estimate $\times$ crash reduction value

After calculating the relative number of bicycle crashes that could be avoided by a gap closure, these values are then monetized according to U.S. DOT KABCO values. The formula for calculating the monetary values is specified in the following formula, and example calculations are provided in Table 40.

Monetary value of reduced crashes = estimated number of bicycle crashes avoided

$\times$ U.S. DOT KABCO value of crash type

Table 39. Bicycle crashes avoided in Northeast walk-through example.

Row ID	Severity	Number of Bicycle Crashes Avoided by Gap Closure
1	Fatal Crashes (K)	$0.0016 \times 0.398 = 0.0006$
2	Suspected Serious Injury Crashes (A)	$0.0156 \times 0.398 = 0.0062$
3	Suspected Minor Injury Crashes (B)	$0.0925 \times 0.398 = 0.0368$
4	Possible Injury or Complaint of Pain Crashes (C)	$0.0652 \times 0.398 = 0.0259$
5	Property Damage Only Crashes (O)	$0.0146 \times 0.398 = 0.0058$

Table 40. Monetized value of reduced crashes in Northeast walk-through example.

Row ID	Severity	Monetary Value of Avoided Bicycle Crashes
1	Fatal Crashes (K)	$0.0006 \times \$12,500,000 = \$7,500$
2	Suspected Serious Injury Crashes (A)	$0.0062 \times \$1,188,200 = \$7,367$
3	Suspected Minor Injury Crashes (B)	$0.0368 \times \$233,800 = \$8,604$
4	Possible Injury or Complaint of Pain Crashes (C)	$0.0259 \times \$111,700 = \$2,893$
5	Property Damage Only Crashes (O)	$0.0058 \times \$5,000 = \29
6	TOTAL Monetary Benefits of Gap Closure	\$26,393

Total Benefits

The total benefits from the proposed facility are found from summing up the relative monetary benefits identified for this case study. The monetary benefits of each respective benefit type are identified and summed as follows:

Estimated Bicycle Count: 658 bicyclists

Health Benefits: \$171,332

Enhanced Travel Options Benefits: \$457,345

Safety Benefits: \$26,393

Total Annual Benefits: \$655,070

APPENDIX A

Behavior Change Technical Details

To support behavior-change estimation, we developed a range of general factors that blended various existing tools—particularly the Caltrans Tool, which provided the initial tables and estimates—and adapted research findings. We labeled these “Caltrans” and “Research” estimates. Because evidence for responses to many pedestrian and bicycle infrastructure types has been sparse to date, we tried to make use of all available evidence, rather than pick a single “best” source. The wide range of supporting studies and methodologies made construction of confidence intervals or precision estimates impossible. Users are encouraged to test the sensitivity of their analyses to different input assumptions and to consider results at different response levels (low, medium, high) to better understand behavior-change uncertainty. This appendix documents the construction of mode and route shift percentage change estimates (low-capacity behavior change) and network impedance factors (medium-capacity behavior change).

Percentage Change Estimates

We relied on five primary research sources to augment the existing Caltrans estimates for bicycle behavior change:

1. Broach et al. 2012—revealed preference (RP) bicyclist route choice model
2. Broach and Dill 2016—RP mode choice model, including walking and biking
3. McNeil et al. 2015—stated preference (SP) study of bike lane treatments
4. Broach and Dill 2015—RP pedestrian route choice model
5. Sevtsuk et al. 2021—RP pedestrian route choice model

From the RP studies, we calculated elasticities from the published models that roughly corresponded to the various combinations of facility type and degree of improvement over existing conditions (low, medium, high). Choice elasticities were calculated as arc elasticities from the multinomial logit (MNL) models presented using the usual formula:

$$E_{P,i|dX} = B_{X,i} X_i (1 - P_i) dX$$

Where:

$E_{P,i|dX}$ is the elasticity of the probability (P) of choosing alternative i given a change in attribute X
 $B_{X,i}$ is the MNL parameter estimate of attribute X for alternative i .

With certain exceptions as noted in subsequent tables and text that follows, general assumptions were:

- Set X equal to 0.5 (the midpoint of the arc between not having the facility and having it).
- Set initial probability of selection P at 0.5 in all cases (equivalent to assuming, e.g., current option captures half the potential bike volume or half the relevant trips) for convenience and

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to provide generally conservative estimates (since setting low selection probabilities inflates the elasticity calculation).

- Assumed no detour required to use new facility.
- Structured Basic and Advanced methods such that each step was roughly equivalent to a single “step” in quality (e.g., for an off-street path, “low” would reference an on-street bike lane, “medium” moderate traffic minor arterial without a bike lane, and “high” a major arterial without bike lane).
- Where separate parameters were provided for commute and non-commute cycling, weight by approximate share of bike trips in most recent NHTS (2017) for each purpose, rounded to 20% commute and 80% non-commute.
- Research-based mode shift estimates did not account for new trips. We applied the Caltrans Tool’s estimate of new demand to inflate the calculated elasticities (1.3x mode shift for bike, 1.1x mode shift for walk). E.g., if calculated mode shift for bike was 50%, we used $50\% \times 1.3 = 65\%$ as the research-based estimate.

Resulting elasticities under different scenarios were used directly as the research-based estimates for percent change (e.g., a calculated elasticity of 0.3 was interpreted as an expected 30% increase in cycling at a location). Where available, route and mode shift elasticities were added together to produce a total expected percent change. Table A-1 and Table A-2 describe the sources and calculation results for estimated impacts of each gap-closing facility type. Where both Caltrans and research estimates were available, results were averaged to produce a final percentage change estimate.

The Augmented Method for percent change estimation (*Appendix C*) requires an additional assumption that original research relating network quality to shifts in bicycling will apply similarly to shifts in walking from pedestrian network improvement. The research team consulted

Table A-1. Sources and calculated estimates for percent change in bicycle travel.

Facility added	Research-based estimates			Caltrans
	Sources (see list provided earlier in Appendix A for reference)	Low/Med/High (R = route shift, M = mode shift, including new trip adjustment)	Low/Med/High Total increase (route + mode + new)	Caltrans estimate (route + mode + new)
Trails/Off-street paths	1,2	R: 52%/181%/297% M: 40%/72%/114%	92%/253%/411%	N/A
Conventional bike lane	1,2	R: 13%/100%/258% M: 9%/37%/75%	22%/137%/332%	-21% ^b /124%/268%
Buffered/Protected bike lane	1,2,3 ^a	R: 31%/130%/297% M: 22%/44%/88%	53%/174%/385%	77%/174%/271% ^c
Bicycle Boulevard	1,2	R: 39%/168%/283% M: 29%/61%/103%	67%/229%/387%	N/A
Road diet ^d	N/A	N/A	N/A	7%/12%/25%

Notes:

^a SP study (3) used to create preference factors that could be applied to conventional bike lanes in 1 & 2, since buffered and protected bike lanes were not included in those studies.

^b Non-intuitive negative value replaced with research-based estimate.

^c Caltrans estimates for protected bike lanes were counterintuitively lower than those for conventional bike lanes, so we used the estimates from buffered bike lane studies only.

^d Based on studies reviewed, this estimate is for impacts from primary road diet elements (narrowing, lane and speed reduction) and not reflective of adding specific bicycle infrastructure.

Table A-2. Sources and calculated estimates for percent change in pedestrian travel.

Facility added	Research-based estimates			Caltrans
	Sources (see list provided earlier in Appendix A for reference)	Low/Med/High (R=route shift, M=mode shift, including new trip adjustment)	Low/Med/High Total increase (route + mode + new)	Caltrans estimate (route + mode + new)
Crossing island	N/A	N/A	N/A	5%/10%/15%
Midblock/collector marked crossing	4	R: 5%/11%/26% M: N/A ^a	11%/24%/57%	N/A
Signalized arterial crossing	2,4	R: 14%/27%/41% M: 7%/14%/21%	21%/41%/62%	N/A
Road diet	N/A	N/A	N/A	8%/15%/30%
Sidewalk	2,5	R: 10%/15%/20% ^b M: 12%/23%/35%	21%/38%/54%	12%/23%/33%
Trail/off-street path	4	R: N/A ^d M: 69%/87%/138% ^c	129%/162%/255% ^c	N/A

Notes:

a. Mode + new trip impacts imputed based on Caltrans shift proportion multipliers.

b. Instead of presence/absence, measured as sidewalk width (5). We tested 5 feet, 7.5 feet, and 10 feet versus no sidewalk for the scenarios.

c. Scenarios tested were: (L=sidewalk on residential street, M=residential street missing sidewalk, H=major commercial street w/ sidewalk).

d. Route shift impacts imputed based on Caltrans shift proportion multipliers.

data from original research to explore whether pedestrian sensitivities to route quality potentially should be less than this elasticity, based on findings that people walking are less willing to shift routes modes (Broach et al. 2012; Broach and Dill 2015, 2016). Instead, we found it could be justified to be higher, in relative terms. To remain conservative, we use an assumed elasticity of 2.37 for both modes.

Change Stratification Estimates

The Caltrans Tool includes suggested shift change proportions attributable to different sources for both pedestrian and bike projects: route, shift from vehicle modes, shift from other modes, new travel. We adjusted these proportions in two ways from the original:

- Where no significant route shift effect expected, re-allocate among remaining shift categories total 100%. This avoids underestimating the impact of projects with primarily mode shift potential (e.g., a new bridge without existing alternatives).
- Where research estimates were available for both route and mode shift:
 - Calculate the implied route shift allocation as $\text{route shift} / (\text{total shift})$ and the remaining categories as $100\% - \text{route shift}$.
 - Allocate among remaining based on Caltrans shift proportions.
 - Average these new estimates with those of other facility types to arrive at a general average shift allocation.

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Impedance Adjustment Factors

LTS provides a method for rating segments (and, more recently, intersections) into discrete categories thought to correspond to perceived stress or network quality by different cycling sub-populations. The discrete categories lead to deterministic, all-or-nothing networks that do not map easily to conventional network impedances (time or perceived cost of travel), creating issues for traditional network analysis. To create a more flexible network definition that better aligns with the low-capacity, percentage change option, impedance adjustment factors were created by mapping existing RP route choice research models to LTS categories.

LTS categories were mapped to research model variables based on a mix of overlapping definitions along with some qualitative judgment about the intended meaning of a given LTS category (e.g., “comfortable for people of all ages and abilities”). There are currently a limited range of facility attributes that have been modeled, and LTS definitions include multiple factors, making one-to-one matching infeasible. Users are encouraged to consider their unique local context, and to seek out updated research that might update the mappings or impedances provided here.

Once a mapping from LTS to research models was established, impedances were established based on the marginal rate of substitution (MRS) between route distance and a given segment or intersection improvement. Distance could then be converted to travel time based on assumed base speeds for a given travel mode. The resulting MRS-based distance/time values represent the combined effect of delay and perceived cost relative to a reference case. The reference case is typically an off-street or very low-volume and low speed facility with adequate bicycle or pedestrian infrastructure. An example impedance example is provided:

- **Context.** Biking on an off-street path vs. an on-street, conventional striped bike lane.
- **Relevant research model MRS.** 0.24 (i.e., cyclist willing to ride 24% farther to use the better facility).
- **Impedance factor.** 1.24 (i.e., 1 mile on a striped bike lane is equivalent to 1.24 miles on an off-street path); note that it does not matter whether this is applied to travel speed/time or distance, since travel speed/time is fixed for each mode.

Table A-3 and Table A-4 provide the specific mappings used to generate the impedance values from various research model MRS estimates. The mappings should be used as a guide to customize factors as needed to suit a specific project context. For example, a protected bike lane might use the LTS 1 value (1.0) or an interpolated value between LTS 1 and LTS 2 (1.0–1.4), based on a qualitative assessment of the specific facility relative to the mappings shown.

Table A-3. Segment level LTS scores and impedance values—Pedestrian and bicycle multiplier LTS score to research mappings.

LTS Score	Bicycle Mapping ^{1,2}	Pedestrian Mapping ³
LTS 1	Off-street multiuse path	5-foot sidewalk, no traffic, low speed traffic (15 mph) ⁴
LTS 2	Average with/without bike lane, minor arterial	5-foot sidewalk, 500 cars/hour, 25 mph traffic
LTS 3	No bike lane, major arterial	5-foot sidewalk, 1,000 cars/hour, 30 mph traffic
LTS 4	No bike lane, highway	No sidewalk, 2,000 cars/hour, 40 mph traffic

Notes:

¹ Bicycle segment mapping to Broach et al. 2012 and Broach and Dill 2016.

² Weighted average of commute (20%) and non-commute (80%) values based on rounded NHTS 2017 bike trip splits.

³ Pedestrian segment mapping to Sevtsuk et al. 2021.

⁴ Model did not include fixed reference values, only differences, so these should be considered approximate or hypothetical base values.

Table A-4. Intersection level LTS scores and impedance values—Pedestrian and bicycle multiplier LTS score to research mappings.

LTS Score	Bicycle Mapping ^{1,2}	Pedestrian Mapping ³
LTS 1	Average of turn, signal, and stop sign delay without significant motor vehicle traffic	Local street crossing or collector crossing <u>with</u> marked crosswalk
LTS 2	Unsignalized crossing, collector	Collector crossing <u>without</u> marked crosswalk
LTS 3	Unsignalized crossing, minor arterial	Interpolated between LTS 2 and 4 ⁴
LTS 4	Unsignalized crossing, primary arterial or greater	Unsignalized minor arterial or greater crossing

Notes:

¹ Bicycle intersection mapping to Broach et al. 2012; Broach and Dill 2015.

² Weighted average of commute (20%) and non-commute (80%) values based on rounded NHTS 2017 bike trip splits.

³ Pedestrian intersection mapping to Broach et al. 2012; Broach and Dill 2016.

⁴ No additional intersection crossing types were available in reviewed research models, so the simple average was calculated between categories.



APPENDIX B

When to Adjust for Trip Distance

For facilities longer than 2 miles, it is important to adjust estimates for trip distance of the modes whose behavior is being modeled. This need arises from two interrelated issues:

- Changes in bicycling and walking activity are estimated using real or synthetic counts and become less representative of the area estimated with increasing facility length. This observation on spatial representation is well documented in spatial statistical phenomena and is often the basis of distance decay methodologies in accessibility modeling (Committee of the Transport Access Manual 2020).
- When creating trip estimates for facilities above 2 miles in length, you should adjust for user trip length. Otherwise, you risk double counting trips and over-estimating bicycling and walking activity on the facility. If you have multiple segments with real or synthetic counts, it is possible that the count could capture the same trip twice if observed within the average trip distance. For example, imagine there is a trail that is 10 miles long. Context transfer can be used to estimate synthetic counts on 10 different 1-mile-long segments. If we added up all these counts, we are likely double counting multiple bicyclists whose average trip length is approximately 3 miles on this trail. This is conceptually illustrated in Figure B-1.

The following steps outline a procedure to address this possibility of double counting trips when using synthetic counts to estimate bicycling and walking activity on longer facilities:

1. Create a person-miles-traveled estimate.

Multiply the facility's total length by the total number of trips estimated by context transfer for all 1-mile synthetic counters along that facility. If you have segmented your project substantially different from 1 mile, you should use a length-weighted sum of the number of trips where the weight is the ratio of the segment length to 1 mile. So, if a synthetic count segment is 0.5 miles long, you would multiply the synthetic count by 0.5 before summation.

2. Convert person-miles-traveled to unique trips.

Divide the person-miles-traveled estimate by the appropriate average trip distance in miles for bicycling and walking, respectively. This calculates unique trip estimates that adjust for the length of the proposed facility and can be used for benefit calculations.



Figure B-1. Illustrated possible double counting of synthetic counts.

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Where possible, the average trip distance by mode can be a locally-derived trip distance average from either project-specific intercept surveys or relatively recent local travel surveys. If such surveys are not available, the analyst can use national average trip distances by mode: based on the 2017 National Travel Survey, the average trip distance is approximately 0.86 miles for walking and the average trip distance is 2.38 miles for bicycling.

When adjusting the overall estimate of behavior change, if the only count estimate available was calculated for all active travelers (bicycling, walking and other active modes combined), use a mode share weighted average trip distance to adjust for trip distance.

The formula to adjust estimates of behavior change for trip distance to calculate an estimate of unique trips is:

$$Unique\ Trips = \frac{\sum_{i=1}^{segments} (Sampled\ Facility\ Length \times Synthetic\ Count)}{Mode\ Average\ Trip\ Distance}$$

Where:

Unique Trips = the unique number of trips for a specific modal estimate

Sampled Facility Length = the total facility length under study (typically in miles)

Synthetic Count = the total number of trips estimated by summing all synthetic counters estimates generated using the context transfer process (assuming a 1-mile spacing)

Mode Average Trip Distance = the average trip distance for the mode being evaluated (typically in miles)

We consider this adjustment to be optional for facilities shorter than 2 miles, but highly recommend this procedure or a similar adjustment be made on facilities longer than 2 miles.

APPENDIX C

Augmented Methods for Estimating Behavior Change

Network Analysis Procedures

Both the Percent Change and Context Transfer Augmented Methods rely on network analysis to evaluate changes in low-stress accessibility. These procedures require a user to have advanced familiarity with GIS and working familiarity with network analysis with more advanced software tools such as ArcGIS Network Analyst, QGIS, or programming languages like R and Python. This section first describes the network analysis approach and then the relevant calculation associated with each approach. The network analysis approaches outlined here attempt to identify how to evaluate accessibility to jobs (or some other destination type) using best practice adjustments for network quality (Committee of the Transport Access Manual 2020; Sundquist et al. 2021; Litman 2022).

Within our gap closure framework, network connectivity depends not only on available walkable or bikeable travel segments but also on whether they offer a comfortable experience for active travelers. Travelsheds are common conceptual tools used to understand how connectivity influences a traveler's options and experience. A travelshed represents the area a traveler can traverse using a given mode of travel up to a specified maximum amount of time or perceived cost (time plus comfort/discomfort along a route). Conceptual diagrams of walking and cycling travelsheds are shown in Figure C-1, where orange and blue show the travel area possible within 10 minutes walking and biking, respectively. The more constrained red portions show the impact of user sensitivity to travel conditions, defined by metrics such as LTS. While LTS is often implemented as an “all-or-nothing” cost (i.e., an individual will travel only on facilities below some threshold) we instead implement increased stress as an additional cost, or impedance, that adds to perceived travel time. This approach has several advantages, including more robust research support, lower sensitivity to minor network coding errors, and a formal, statistical relationship between perceived and actual time or distance.



Figure C-1. What is low-stress network connectivity and why is it important?

C-2 Estimating Benefits of Closing Gaps in Active Transportation Networks: A Guide**Augmented Methods for Calculating Metrics with Network Analysis**

This process calculates change in low-stress network accessibility to jobs (or some other destination) via the transportation metric. The amount of change observed is used to further refine estimated percent change.

Use the following process:

1. Gather required data.

Identify an evaluation extent based on a buffer around the project area, with the buffer radius being approximately as large as a 30-minute travel distance for each target mode as the crow flies. This will ensure sufficient data for the actual influence areas around the project. For bicycling, we recommend 5 miles around the project site, and for walking, 1.5 miles. We recommend 7.5 miles if e-micromobility is considered explicitly as part of the project. The extent of the network should be documented as part of the analysis. This assumes a walk speed of 3 mph, a bicycling speed of 10 mph, and an e-micromobility speed of 15 mph.

Develop a routable network dataset that has LTS scores associated with it for the appropriate modes. This can be built using commercial software such as ArcGIS Network Analyst or open-source alternatives such as R5, QNEAT, Pandana, and OSMnx.

While access to any sort of destination can be used (e.g., schools and parks), this guide focuses on access to jobs, which is commonly used in planning documents and, at the present time, the only measure with research support that quantifies network improvement impacts on active travel. Job densities are also likely highly correlated with a range of non-work opportunities, and we feel job access is a reasonable current proxy impacting a broad range of trip types. We recommend using U.S. Census longitudinal employer-household dynamics (LEHD) for jobs, typically choosing to include all jobs in all sectors.

In addition to employment data, population data should be gathered. Typically, results are reported in the same geographic areas used to gather population data. For the purposes of this analysis, we refer to these geographies as summary zones, understanding they could be anything from TAZs to CBGs.

2. Develop comparison networks.

Develop two LTS networks representing two scenarios: one before implementation and one after implementation of the gap closure project. Changes from the before to after network should reflect only those changes directly part of the gap closure project scope.

These LTS networks should have higher attention for accuracy closer to the project, and defer to a locally available LTS network if available. Document the specific LTS scoring methodology used for the evaluation, and ensure it represents at least four levels of comfort.

Particularly for intersections, scores may be calculated and modeled through a variety of means. In ArcGIS Pro (Esri), for example, these could be modeled as barriers with a scaled cost for traversal. Overall, the project team has found that the increased complexity of modeling intersection LTS typically has a modest effect on the modeled analysis results.

For scoring methods, we recommend using the Mineta Transportation Institute's Bicycle LTS and the ODOT's Pedestrian Level of Stress Methods (as applicable) as minimum baselines for the rigor expected (Mekuria et al. 2012; ODOT 2023).

For the scenario after implementation of the gap closure project, represent the project by updating existing facilities or modeling a new facility. For upgrades to an existing facility: update the appropriate modal LTS score and update the associated impedance. For modeling a new facility: add a new link to the network, input the appropriate LTS score, and calculate the associated impedance. If the network analysis includes intersections, additional modifications to data used to represent intersections may be necessary to appropriately model the gap closure. Gap closures will often be part of a larger package of improvements in the vicinity. You should only update the network with changes covered by the gap project itself. However,

you may choose to modify the base network to include additional planned projects that might be leveraged by the gap closure. The ability to capture these complementary changes is a key advantage of using the Augmented Methods versus the Basic Methods.

3. **Adjust network impedances.**

Within the chosen network analyst software, adjust the impedances to travel along the network using values from Table C-1 or C-2. Intersection representation is optional. Travel times for each segment are scaled based on the following base travel speeds. While different speeds may be used to model different populations (e.g., older adults or people who use mobility devices), the following speeds are commonly used industry wide for a variety of purposes:

- Pedestrians: 3 mph
- Bicyclists: 10 mph
- E-Micromobility: 11 mph to 15 mph (NYSERDA 2015)

4. **Calculate access under both scenarios: before and after implementation of gap closure project.**

Access is defined as the stress-adjusted travelshed around the gap closure project for each mode. A measure of time is used to allow comparison between the access stress-adjusted travelsheds before and after implementation of the gap closure project. Based on the original research underlying this method, a bicycle travel time of 30 minutes (5 miles at 10 miles per hour) is recommended for analysis purposes (Broach and Dill 2017). For walking, we recommend an analysis travel time of 20 minutes (1 mile at 3 miles per hour), reflecting the shorter observed duration of walking trips (NHTS 2017). Justification could be made for reducing these extents to 20 and 15 minutes, respectively, to better match observed behavior (NHTS 2017). For certain destination types, such as transit stops, this distance would warrant being reduced even further to 5 to 10 minutes.

5. **Compare the access travelsheds in the two scenarios and summarize the difference in jobs and population.**

Summarize the difference in jobs and population between the access travelsheds for each mode. Calculate the population-weighted average of the number of jobs within the access

Table C-1. Segment level LTS scores and impedance values—Pedestrian and bicycle multipliers.

LTS Score	Bicycle Impedance Multiplier	Pedestrian Impedance Multiplier
LTS 1	1	1
LTS 2	1.5	2
LTS 3	3	3
LTS 4	8	6

Note:

Additional detail and documentation are provided in *Appendix A: Impedance Adjustment Factors* section.

Table C-2. Intersection level LTS scores and impedance values—Pedestrian and bicycle in minutes.

LTS Score	Bicycle Impedance (Minutes)	Pedestrian Impedance (Minutes)
LTS 1	0.02	0
LTS 2	0.1	0.3
LTS 3	0.2	0.6
LTS 4	0.5	0.9

Note:

Additional detail and documentation are provided in *Appendix A: Impedance Adjustment Factors* section.

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travelshed before and after given the change in the networks, and ideally there is some work done to address the sampling effects (potential bias from defining the network starting point(s) that represent the center of each travelshed).

There are two common approaches to this type of analysis:

- Develop an origin-destination matrix that summarizes the number of jobs within the travelshed based on travel times imputed to job locations.
- Develop travelshed geometries of both the baseline and change travelsheds, and summarize the number of jobs accessed. Using proportional allocation tools (*Chapter 4: Step 3. Estimating Benefits*) will increase the accuracy of the percent change in low-stress accessibility.

A best practice accessibility analysis would account for multiple types of factors to determine access improvement ratios. At minimum, we recommend accounting for sampling bias, edge effects, and diminishing utility with distance to destinations.

- **Sampling bias.** Sampling biases occur in network analysis as a result of network location having sometimes non-linear impacts on accessibility results. A visual example of this is the starting point for an accessibility analysis being a dead-end road vs. in the middle of a dense grid. This is best countered with sampling multiple locations within a geography that results are being summarized into or conducting a disaggregated access analysis.
- **Edge effects.** This is another type of network analysis artifact that comes from the extent of the access analysis not being large enough to capture the extent of possible travel from the analysis zones of the study area. Where possible, this is best countered by ensuring network and other data are gathered within an extent covering the largest anticipated analysis area.
- **Diminishing utility with distance.** Accessibility analysis has often originated with the understanding that as destinations increase in distance from a person's start location, the utility decreases by some function. Often this can be modeled accurately with user surveys of a transportation service identifying the changes in user concentration with proximity, but it is often modeled using linear, exponential, or similar distance decay functions. This guide recommends implementation of some type of distance decay function as a closer approximation of available behavior-change literature, even though the concept is often hard to explain to lay audiences. Committee of the Transport Access Manual (2020) and Sundquist et al. (2021) provide good starting points on this topic. However, utilization of a distance decay function is optional.

Other considerations are documented in multiple resources that discuss the best practice approaches for accessibility analysis, including:

- *Transport Access Manual: A Guide for Measuring Connection between People and Places* (Committee of the Transport Access Manual 2020).
- State Smart Transportation Initiative's *Measuring Accessibility: A Guide for Transportation and Land Use Practitioners* (Sundquist et al. 2021).
- FHWA's *Guidebook for Measuring Multimodal Connectivity* (FHWA 2018a).

The equation for computing a population-weighted average measure of job accessibility for all summary zones that touch the access travelshed is the following:

$$a = \sum_{i=1}^n \left(j_n \frac{(p_n)}{\sum_{i=1}^n p_n} \right)$$

Where:

a = the population-weighted measure of accessibility: in this example, the population-weighted average number of jobs

p_n = the population of summary zone n

j_n = a metric for accessibility, in this case jobs in summary area n

n = the geographic summary area, typically census geography or regionally modeled transportation analysis zone.

6. Calculate a percent change of access.

Based on the population-weighted accessibility gains, calculate a percent change to be used in the behavior-change assessments. This percentage can be used to either assess a percent change in expected active travelers on an existing facility or to evaluate the context transfer of one facility to another. The equation for this procedure would be the following, using the equation from step 5:

$$\text{Percent Change} = \frac{\text{Job Access After} - \text{Job Access Before}}{\text{Job Access Before}}$$

Where:

Percent Change = the percent change in low-stress job accessibility within the influence area

Job Access After = the population-weighted average total number jobs accessible on a low-stress network within the influence area after a gap closure is implemented

Job Access Before = the population-weighted average total number jobs accessible on a low-stress network within the influence area during the baseline condition

7. Prepare and document key outputs from the analysis.

The following outputs might be used for future analysis, visualization, or public communications:

- **Summary zones access gains.** The summary zones used to evaluate changes in low-stress accessibility should be set aside for visualization and further analysis.
- **Percentage change in accessibility.** The percent change in low-stress job accessibility within the influence area should be summarized for the projects.

Percent Change with Network Analysis

Basic Methods rely on expected behavior change based on gap facility improvement type, using average responses from relevant research across a range of local contexts. The Augmented Method measures changes in low-stress network accessibility due to a specific gap closure, and then calculates an expected behavior shift directly. This method should better capture the network effects resulting from a gap closure. The intermediate result of “percent change in low-stress (walk or bike) accessibility” will also be helpful in communicated results.

The process is as follows:

1. Following the network analysis procedures, determine the percent change in population-weighted average change in low-stress job accessibility within the influence area.
2. The percent change in low-stress accessibility is used to evaluate the degree of improvement that results from the gap closure directly. This guide uses elasticities and impedances derived from route preference and mode shift literature on bicyclists and pedestrians.
 - a. The percent change in low-stress accessibility is interpreted to have an equivalent elasticity to the similar route quality index method from Broach and Dill (2017), which found that for every 10% change in a population-weighted route quality index to job locations, bicycle commuting rates were expected to increase 23.7% (i.e., elasticity of 2.37).

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For estimation purposes, this guidance is applied directly to percent change to estimate all bicycle and pedestrian mode shift (see *Appendix A: Behavior Change Technical Details* for additional detail and justification).

- b. Calculate mode shift share of all trips, and create an inflation factor relative to the percentage of all mode shift trips to other types.
 - c. If using our mode shift tables to make this determination (*Chapter 3: Step 2. Estimating Behavior Change*, Table 6), the inflation multiplication factors to apply are shown in Table C-3 (e.g., for a pedestrian gap closure with route shift, the calculated change would be multiplied by 2.27 to account for new travel and route shifted trips, in addition to walk trips shifted from other modes).
 - d. Some benefit categories require shifts from motor vehicle modes only. In those cases, deflation factors are provided to remove mode shift from non-vehicle modes (transit, bike, etc.).
3. Calculate percentage change as a function of the percent change in low-stress job accessibility, of elasticity and the commute share adjustment factor. This estimate will be bounded by the highest behavior-change estimate in the behavioral response tables (*Chapter 3: Step 2. Estimating Behavior Change*, Table 1 to Table 4) provided in this guide + 25%. We included the 25% increment to allow for additional headroom to account for gap closure network effects, which were often excluded in the research estimates underlying our shift percent estimates. The formula for this calculation is as follows:

$$\text{Percentage Change} = \text{MIN}\left(\left(\text{Cap} \times 1.25\right), \left(\text{PCLSJA} \times \text{RQE} \times \text{CAF}\right)\right)$$

Where:

Percent change = the percent change in active travelers to expect

Cap = the highest activity change identified in our behavioral response tables

PCLSJA = the percent change in low-stress job accessibility of CBGs within the influence area

RQE = the route quality index elasticity derived from literature

CAF = the commute share adjustment factor that adjusts the expected share of active travelers to include new trips and route shift (determined based on whether or not there is route shift expected for the facility)

4. Apply percentage change to baseline. The percent change is applied to determine the change in active travel on the facility post-implementation. This volume estimate is directly integrated into best estimates. In some cases, the Augmented Method might produce lower estimates than the Basic table lookups. Usually this will reflect local conditions with relatively low demand in the project's influence area, perhaps due to low surrounding population or

Table C-3. Inflation factors to apply to Augmented Method percent change estimates.

Adjustment Multiplier	Bicycle Shifts and New Travel	Bicycle Shifts and New Travel	Pedestrian Shifts and New Travel	Pedestrian Shifts and New Travel
	With Route Shift	No Route Shift	With Route Shift	No Route Shift
Apply to mode shift to get total change	3.45x	1.39x	2.27x	1.14x
Apply to mode shift to get vehicle mode shift	0.59x	0.60x	0.41x	0.41x

few opportunities to access. You may use conflicting results like this to either re-assess the Basic analysis assumptions and results or present the conflicting results as a way of communicating uncertainty (see *Communicating Uncertainty* in Chapter 5: Step 4. *Communicating the Results*).

5. For facilities shorter than 2 miles, this synthetic count can serve as a trip estimate for the project. For facilities longer than 2 miles, consult *Appendix B: When to Adjust for Trip Distance*.

Context Transfer Based on Increases in Low-Stress Accessibility

Context transfer with network analysis functions very similarly to conventional context transfer, except the demand metric that is used to create an adjusted rate of trips between sites changes in low-stress job accessibility. The steps for this process are as follows:

1. Identify a location of a comparable facility that is in a similar context and new facility type.
 - a. Comparable facilities are facilities that have similar characteristics with respect to functional classification, land-use, and built context, and geographic proximity (similar geographic region).
 - b. If a set of facilities with similar contexts are available with counts, they can be used to create an average rate.
 - c. Comparability for network analysis needs to be also considered. The facilities should be of similar lengths in areas with similar network characteristics. This can be determined by looking at various network evaluating metrics such as node to link ratios, intersection densities, or similar metrics.
2. Conduct a network analysis for before-and-after conditions for the comparable facility and the proposed project. Follow the network analysis procedures outlined in this guide for both locations, and the summary zones around the project will be directly used for context transfer.
3. Compute before-and-after percentages of changes in accessibility for each mode within each summary zone.

$$\text{Percent Change}_{\text{zonal}} = \frac{\text{Job Access After} - \text{Job Access Before}}{\text{Job Access Before}}$$

Where:

Percent Change = the percent change in low-stress job accessibility within the influence area

Job Access After = average total number jobs accessible on a low-stress network within the summary zone after a gap closure is implemented

Job Access Before = average total number jobs accessible on a low-stress network within the summary zone during the baseline condition

4. Sample the percent change low-stress job accessibility around the count location and the new project using proportional allocations with the influence area of the facilities. The influence buffer size is based on the mode whose volumes are being estimated. The maximum length of a sampling segment should be 1 mile. If the facilities are longer than this, the segmentation should be split to be approximately 1-mile segments for repeated sampling.
 - a. The sampling of percent changes in access should use proportional averages where the area weights an average proportion sampled from different zonal geographies, while the result should be equivalent when computing a percent change in access.

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5. Develop a metric rate based on the low-stress accessibility gain at the count location relative to the demand low-stress accessibility gain at the new location. For example, if an existing facility with a count of 50 daily pedestrians had a sampled pedestrian access gain of 10%, its demand rate would be 10 pedestrians per 1% of pedestrian access gain.
 - a. If multiple count locations are available, use them to develop an average rate of bicycle and pedestrian demand per unit of the demand metric.
6. Apply the metric rate to the sampled segments that make up the facility. If multiple segments make up the project, then it is assumed that by sampling approximately 1-mile segments we are generating a synthetic count for that segment that is representative behavior along 1 mile of the facility.
7. For facilities shorter than 2 miles, this synthetic count can serve as a trip estimate for the project. For facilities longer than 2 miles, this guide recommends consulting *Appendix B: When to Adjust for Trip Distance*.

APPENDIX D

Advanced Application Case Study: Behavior Change – State of Utah

In 2018, the Utah Legislature passed Senate Bill 136, resulting in a directive for the Utah Department of Transportation (UDOT) and relevant partners to “develop statewide strategic initiatives across all modes of transportation.” This directive placed renewed focus on the need for accurate link-level modal travel estimates across the state’s entire transportation network.

Funded through a variety of initiatives, the result is a fit-to-purpose bicycle demand model—a customized model that is designed and tailored to meet the project parameters including desired outputs, data inputs, staff capacity and available software. The primary purpose of the bike model is to better account for existing bike travel patterns and future opportunities by leveraging a model informed by the latest behavior-change research. This model builds upon the same behavior-change research informing the methods built into this guide (Broach and Dill 2017) to be sensitive to different facility types, slopes, and traffic characteristics with traffic costs derived from UDOT’s model implementation as shown in Table D-1. Better information on these patterns supports prioritizing and decision-making for bike infrastructure through demand analysis/forecasts, first mile/last mile transit analysis and better understanding of network and land-use related impacts.

The model follows a familiar modeling process: trip generation estimates bike trips per household based on demographics and access to destinations; trip distribution allocates these trips based on purpose and origin-destination types; destination choice assigns trips using a gravity model approach; and trip assignment calculates the best paths with the lowest perceived travel costs between origin-destination pairs. The results of the model are O-D trip table at the microzone level and daily link-level bicycle volume estimates. Successful deployment of these types of tools will require substantial investments in development of data and methods as well as ongoing maintenance to keep the system viable.

Table D-1(a). Bike network cost constants and coefficients used in Utah bike network travel demand model.

Variable	Additional Distance (miles)	Notes
Turn	0.034	Presence of a turn
Stop sign	0.0037	Presence of a stop sign
Traffic signal	0.017	Presence of a traffic signal
Parallel traffic L heavy	0.18	Left turn parallel to heavy traffic; 20k + AADT
Cross traffic LS medium	0.05	Left turn or straight across medium traffic; 10–20k AADT

AADT: Annual average daily traffic

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Table D-1(b). Bike network cost constants and coefficients used in Utah bike network travel demand model.

Variable	Length Multiplier	Notes
Distance	1.0	Default
Bike blvd	-0.108	Bike class = 3b, 3c
Bike path	-0.16	Bike class = 1a, 1b, 1c, 1
Bike lane light	0	Bike class = 2a, 2b, 2, 3a and light traffic
Bike lane heavy	1.65	Bike class = 2a, 2b, 2, 3a and heavy traffic
No bike lane heavy	7.157	Heavy traffic
Slope 2–4 percent	0.371	Percent slope (moderate hill)
Slope > 6 percent	3.239	Percent slope (steep hill)



APPENDIX E

Augmented Methods for Safety Benefit Analysis

Projects with more technical expertise can utilize these Augmented Methods to develop more robust results that align with current best practices. Building on the previously described methods, Augmented Methods will typically include GIS-based network analysis to refine the results obtained using Basic Methods. Augmented Methods also assume project resources are spent on acquiring, verifying, and/or enhancing data specific to the project study area, rather than making broader assumptions that would allow the use of national statistics or state-level information. For example, the Augmented Method may use an existing high-injury network (HIN) derived from crash data in addition to crash data.

Unit of Measure

Safety benefits in the Augmented Method are reported in two ways: a total number of avoided crashes and/or a total monetary value of those avoided crashes. Avoided crashes can be reported by travel mode and severity.

Data Needs

The Augmented Method uses GIS data about the road network to calculate local crash multipliers based on crash profiles. Road network data could include segments and/or intersections classified by HIN status, functional class, and/or roadway attributes. See Table E-1 for more information on data types needed for the Augmented Method.

There are particular analysis assumptions and limitations that should be considered by the project team when beginning an Augmented safety benefits analysis.

First, the overall project scale and land-use context should be carefully defined.

- For rural, small town, and other low crash concentration contexts for the Augmented Method, consider calculating multipliers from a larger geographic scale, e.g., consider calculating for all comparable land-use or intensity contexts in an entire state instead of a single city, for which the total number of crashes may be very low.
- In addition, parallel facilities in the project area should be included to account for safety benefits of people shifting routes to the new, safer gap closure facility.

Second, the particular type of gap closure can affect the types and complexity of multipliers and CMFs included in the analysis.

- Consider calculating multipliers to be applied per mile or per intersection as best fits the gap closure project and CMF definition. Multipliers for the Basic Method have been provided at the intersection and midblock/segment scale to allow for this flexibility.

E-2 Estimating Benefits of Closing Gaps in Active Transportation Networks: A Guide**Table E-1. Data needs for Augmented Method safety benefit estimation.**

Data Type	Crash Rate Method (Augmented Method)
Historic crash data	For gap closure project site and any relevant parallel facilities – number of crashes by mode and severity. Number of crashes by mode and severity for larger study area (e.g., city or state) is also needed.
Crash estimation data	Historic crash data for relevant jurisdiction. Road network with relevant attributes of interest (e.g., crash risk factors, HIN, etc.).
Project area data	Centerline miles AND/OR count of intersections within gap closure project and any relevant parallel facilities, grouped by same attributes used in crash estimation.
Gap closure improvement type(s) – to the extent known	Type of facility or countermeasures being used to close gap (e.g., adding a separated bike facility, reducing motor vehicle travel lanes, etc.).
CMFs from CMF Clearinghouse (preferred) and other sources (if needed)	CMFs matched to gap closure improvement type(s) as closely as practical.

- For gap closures that consist of a single intersection or a series of intersections along a corridor, multipliers per intersection are most appropriate.
- For gap closures that consist of linear facilities, multipliers per mile are most appropriate.
- For mixed treatments at the segment and intersection scale, consider calculating midblock crash rates per mile and intersection crash rates per intersection separately.
- For gap closures that forge an entirely new network connection (e.g., a new bridge with no nearby existing bridges), there may not be parallel facilities available. Consider using an influence area around the gap closure project instead.

Data Aggregation and Unit of Analysis

- This analysis can be done at the segment or intersection level, or both.
- The simplest version is to use a single unit type (segments or intersections, not both) that best matches the gap type and the project elements required to close it.
- For linear gaps, where closing the gap would require primarily linear/segment facilities (e.g., bike lanes, shared-use paths, sidewalks) and where intersection treatments along those facilities may not yet be known, use segments as the unit of analysis.
- For crossing gaps, where closing the gap would primarily require intersection or midblock crossing treatments at spot locations (e.g., crosswalk markings, crosswalk enhancements), use intersections as the unit of analysis.
- If a gap closure project would include a mix of linear/segment facilities and intersection facilities, and if the intersection facilities are known, a mix of intersection and segment units can be used. The data input requirements are the same, but calculations are more complex than using a single unit of analysis. Using separate intersection and segment/midblock units of analysis allows one to separately analyze intersection crashes with intersection treatment CMFs and segment or midblock crashes with segment treatment CMFs.

Methods

The Augmented Method follows the same steps as the previously described, Basic Method, except for **Step 3a: Choosing a set of crash rate multipliers**. End users will calculate their own baseline multipliers using a crash profile approach and refer to these custom-calculated

multipliers instead of published multipliers. The following outline describes the steps necessary to calculate custom multipliers:

- Acquire ideally 5 and a minimum of 3 years of crash data for an area either containing the gap/project area or a peer area to the gap/project area (e.g., the entire city, county, region, or state that contains the gap).
- Develop crash profiles for the entire area.
- Measure the concentration of crashes per intersection and/or per mile within those crash profiles.
- Use those concentrations or rates as baseline multipliers.

This method is based on systemic analysis, which proactively screens the network for known safety risk factors. We use the rate or concentration of crashes per intersection and/or per mile in various crash profiles as the way to estimate risk. Refer to existing resources on systemic analysis for Vulnerable Road Users for additional guidance (Thomas et al. 2018).

1. Choose the desired scale and unit of analysis for multipliers. Table E-2 presents three options with common use cases and considerations.
2. Choose the desired crash profile type(s) that will be developed. This will depend on data availability and quality at the scale(s) of interest. Table E-3 lists three approaches that may be used when developing crash profiles and multipliers.
3. If the gap closure project will use CMFs for specific types of pedestrian or bicycle crashes (e.g., pedestrian in crosswalk with motorist turning left), these approaches can be repeated across multiple crash types to maximize alignment with CMFs.
 - a. However, note that sparse or zero-inflated data may become a problem if too many crash types and too many profiles/risk factors are used. While there is no universal threshold, a crash profile analysis probably needs at least 100 crashes total and at least 10 crashes per variable of interest, whichever is greater. Refer to other guidance on systemic analysis (Thomas et al.; U.S. DOT 2013).
4. Use the template in Table E-4 for calculating multipliers from crash profiles.
 - a. List any appropriate contextual factors in Row 1. Repeat this analysis if necessary to address different contexts (e.g., urban vs. rural, different crash types, etc.).

Table E-2. Scale of analysis for baseline crash rate multipliers.

Scale	Decision Factors (e.g., Gap Type)	Application Notes
Intersection Crashes per Intersection	Best for gap closure projects that include intersection-specific treatments and CMFs for one or more intersections (e.g., series of intersection treatments along a high-stress gap). If linear facilities are also planned, combine with Midblock Crashes per Mile.	Can be used alone or in combination with midblock crashes per mile. Do not combine with All Crashes per Mile.
Midblock/Segment (non-intersection) Crashes per Mile	Best for gap closure projects that include CMFs for linear facilities (e.g., adding sidewalks, bike lanes, or shared-use paths). If intersection treatments are also planned, combine with Intersection Crashes per Intersection.	Can be used alone or in combination with intersection crashes per intersection.
All Crashes (intersection + midblock/segment) per Mile	Best for gap closure projects that include linear facilities (e.g., adding sidewalks, bike lanes, or shared-use paths), where specific intersection CMFs are not yet known.	Do not use in combination with intersection crashes per intersection. This option is best if intersection treatments are unknown, and no intersection CMFs are being used.

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Table E-3. Approaches for crash profiles that can be used to calculate baseline crash rate multipliers.

Example Approaches for Crash Profiles	Additional Data Inputs Required	Appropriate Scale(s)	Appropriate Geographies
Systemic Risk Factor Combinations	Risk factors of interest (e.g., motorist AADT, number of lanes, signal phasing, etc.) joined to segments and intersections areawide	Intersection, midblock/segment, or all crashes	Depends on data availability and coverage of desired risk factor variables. Common geographies might be an entire city or the state highway network of an entire state.
Functional Class or Facility Type	Functional class or facility type classification joined to segments and intersections areawide	Intersection, midblock/segment, or all crashes	Functional class is widely available at most scales; this profile type can be done at virtually any geography (city, county, region, state).
Existing HIN	HIN status of segments citywide in lieu of functional class or facility type method. Profiles categorize data as 'On HIN' vs 'Off HIN'	All crashes	Depends on the HIN's scale (e.g., citywide, regional, state). City is the most common geography, though regional and statewide HIN are increasingly common.

Table E-4. TEMPLATE for calculating multipliers for a set of crash profiles for a specific mode and context. Repeat for all modes and contexts appropriate to the gap closure project.

Context:	E.g., urban statewide, rural statewide, citywide, specific crash types, etc.										
Mode:	E.g., pedestrians										
		# Crashes Matching Profile, by Severity					BASELINE MULTIPLIERS: Rate or Concentration of Crashes Per Mile or Intersection Matching Profile, by Severity <i>(Divide # Crashes by Miles or Intersections to calculate)</i>				
Column Number											
1	2	3	4	5	6	7	8	9	10	11	12
Crash Profile Definitions	Miles or # Intersections Matching Profile	Fatal (K)	Serious Injury (A)	Minor Injury (B)	Possible Injury (C)	Property Damage Only (O)	Fatal (K)	Serious Injury (A)	Minor Injury (B)	Possible Injury (C)	Property Damage Only (O)

- Specify the mode being analyzed in Row 2. Repeat this analysis for each active mode (pedestrians, bicyclists).
- List the crash profiles being analyzed in Column 1.
- For each crash profile, measure the number of miles and/or intersections matching the profile and context in the area of interest. List these in Column 2.
- For each profile, measure the number of crashes by severity matching the profile, context, and mode in the area of interest. List these in Columns 3 to 7.
- Divide the number of crashes by the number of intersections or miles. These are the baseline multipliers to use in Step 3a. List these in Columns 8 to 12.

Advanced Application Case Study: Safety Benefits – Madison, WI

Introduction

Our methods for estimating safety benefits of active transportation gap closure projects rely on systemic estimates of crashes per mile or per intersection either published in this guide or calculated from local data. Agencies with existing SPFs can employ those rather than using multipliers based on crash profiles. The general process is the same as the Augmented and Basic Methods; the only difference is that agencies would use baseline rates derived from SPFs instead of from the multiplier tables provided for the Basic Method or from custom-created multipliers from the Augmented Method. Table E-3 summarizes different potential approaches to estimating baseline crashes, showing how SPFs fit into the approach.

To illustrate this approach, we used safety data from the Madison-area MPO. Researchers from the Wisconsin Traffic Operations and Safety Laboratory (TOPS) at the University of Wisconsin – Madison (UWM) estimated SPFs for arterial and collector segments and intersections in the Madison-area MPO. The models used severity and mode distribution factors to expand the models into estimates by mode and severity (TOPS Laboratory, n.d.). Empirical Bayes (EB) adjustments were used to blend model predictions with observed crash history. Finally, the project team calculated an equivalent property damage only (EPDO) score using the EB-adjusted expected number of crashes by mode and severity at each location. Details about the estimation and application of the SPFs have been published by the original researchers. These data were used to define the region's HIN. Figure F-1 shows a map of the regional HIN.

The TOPS researchers provided access to these data to test the Advanced Application Method to safety benefits estimation. Note that the study data span 2017–2020 and conditions may have changed since the study completed.

Madison Case Study

Downtown Madison is located on an isthmus between Lake Mendota and Lake Monona. U.S. Highway 151/East Washington Avenue is the primary route across this isthmus. The project team identified a section of U.S. Highway 151 that extends from the Yahara River on the east side of the isthmus to the state capitol on the west side as a “gap” in the network based on coverage of the HIN in this area as well as the constrained topography. Figure F-2 shows the HIN in this area.

This stretch of Highway 151 is a six-lane divided arterial with parking lanes and sidewalks on both sides. The speed limit was set to 35 mph for most of the gap corridor at the time the safety data were collected, though speed limits have since been lowered to 25 mph. Motorist annual average daily traffic (AADT) for most of the corridor range from 18,000 to 24,000 in each direction, with lower values closer to the capitol. Bicycle LTS ranges from 3 to 4, making this corridor

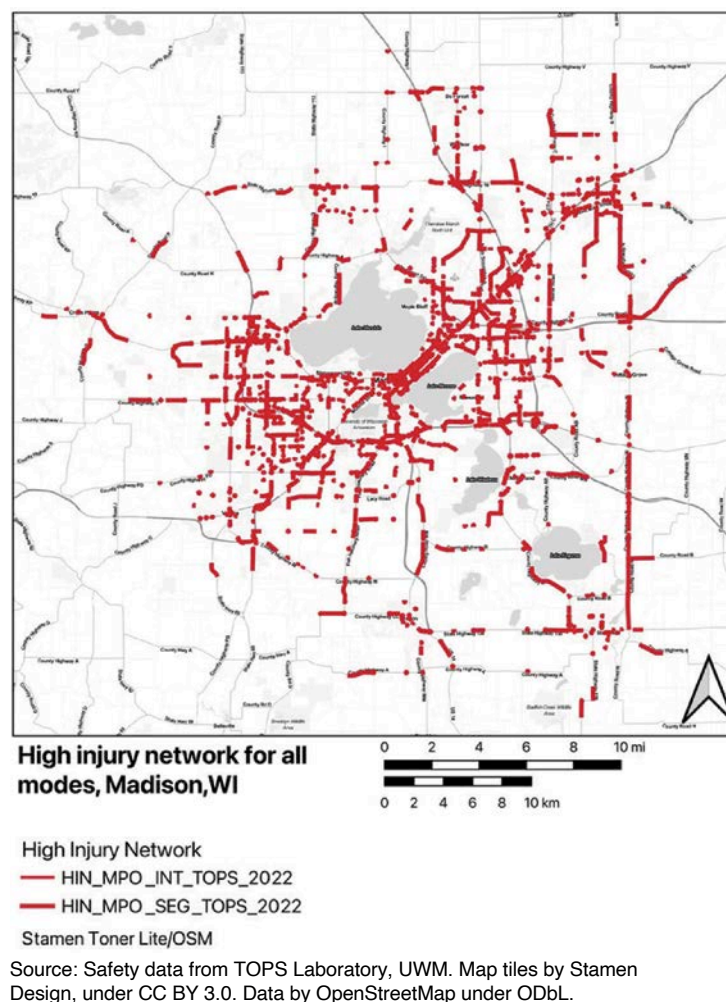
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Figure F-1. Map of Madison, WI MPO regional high-injury network.

impassible for most bicyclists. Given the constrained topography and network and the high LTS along the three main arterials crossing the isthmus, this section constitutes an active transportation network gap.

Closing a gap may attract travelers from other parallel facilities, conferring a safety benefit to people presently using other facilities. Per our safety methodology approach, we recommend using parallel facilities from which pedestrians or bicyclists may realistically detour once a gap is closed, and incorporating crash baselines from these parallel facilities into the benefits estimation.

For this case study, we identified two parallel facilities for East Washington that are roughly 0.3 mi (0.5 km) in either direction: Williamson Street to the south, and Johnson Street to the north. Both corridors cross the Yahara River and provide continuous connectivity to downtown Madison. Williamson is a 4-lane undivided arterial, with the outer two lanes functioning as street parking outside of rush hour. Johnson starts as a 4-lane divided arterial on the eastern end, and then splits into paired one-way streets with Gorham Avenue. Gorham has two lanes plus on-street parking. The one-way portion of Johnson has two lanes, a left side striped (non-buffered) bike lane, and on-street parking. Both streets have sidewalks, 25 mph speed limits, and bicycle

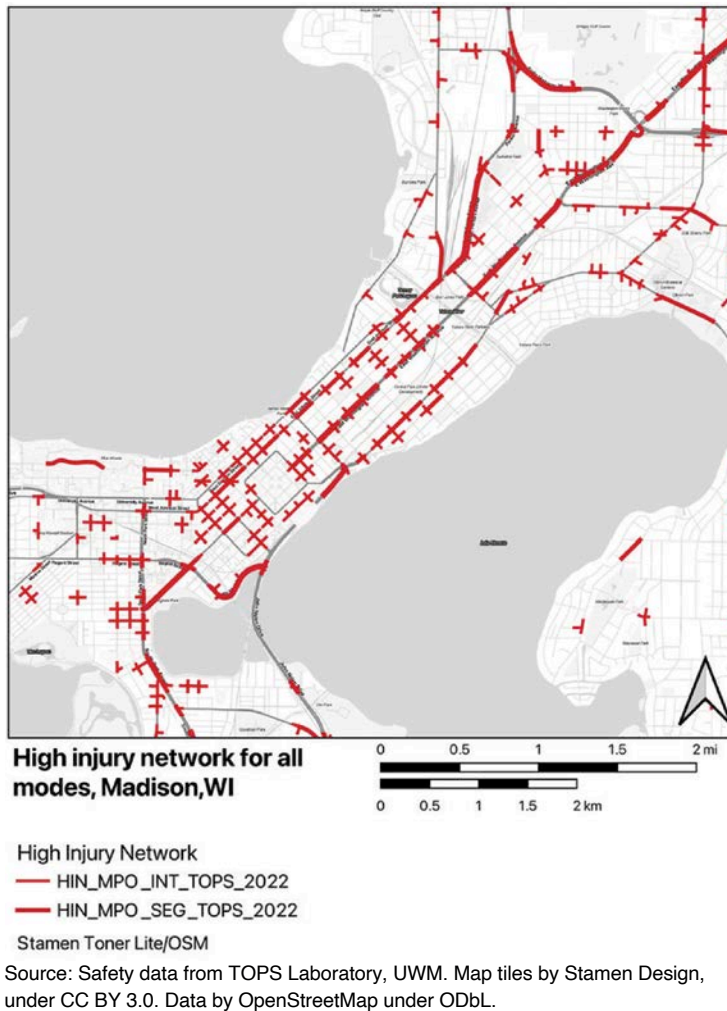


Figure F-2. High-injury network near Downtown Madison, WI.

LTS ranging from 3 to 4. Figure F-3 shows both the gap corridor (East Washington) and the parallel corridors (Williamson and Johnson/Gorham) overlaid with the HIN.

Both the gap corridor and the two identified parallel corridors have seen a high number of fatal and injury crashes for all modes. Table F-1 summarizes observed crash history by mode and severity for the study period (2017–2020). These totals are achieved by summing all segment and intersection observed crashes across the study area’s extent (gap corridor and parallel corridors). The same process was used to calculate totals of expected crashes from EB-adjusted SPF outputs as shown later in Table F-5.

The general workflow for safety benefits estimation consists of:

1. Estimating baseline crash totals for the gap corridor and parallel facilities
2. Applying CMFs to these totals
3. Calculating the reduction and/or final crash totals
4. Aggregating these totals using crash costs, equivalent property damage only (EPDO) weights, or another weighting scheme

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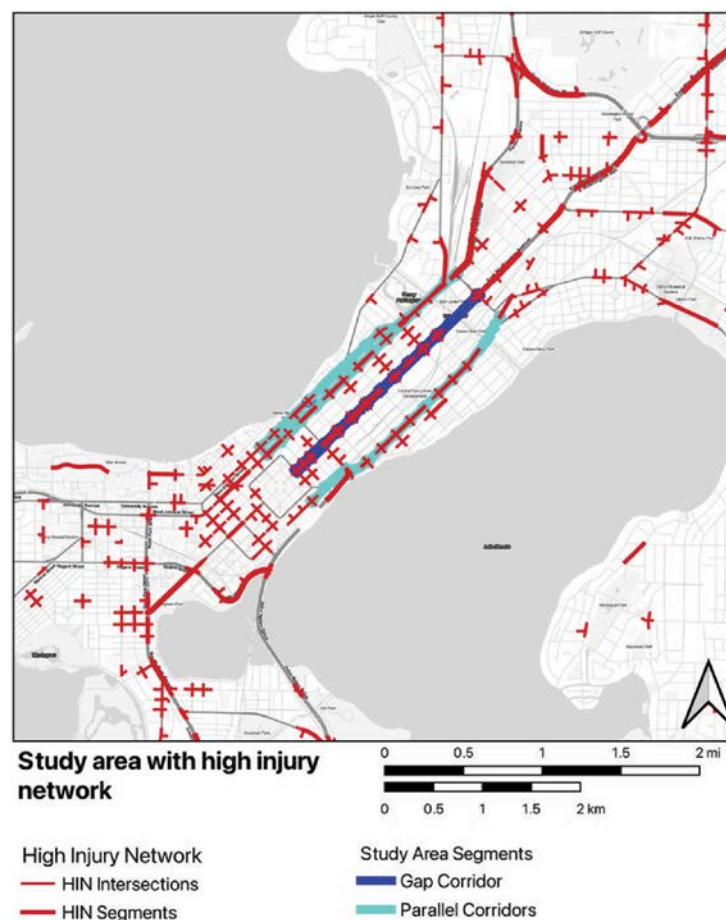


Figure F-3. Gap corridor and parallel corridors overlaid with high-injury network near Downtown Madison, WI.

Table F-1. Observed crash history in study area (gap corridor and parallel corridors).

Severity	Gap Corridor			Parallel Corridors		
	Pedestrian Crashes	Bicyclist Crashes	Motorist Crashes	Pedestrian Crashes	Bicyclist Crashes	Motorist Crashes
Fatal (K)	1	0	1	0	0	1
Serious Injury (A)	3	1	11	2	3	10
Minor Injury (B)	7	4	46	9	16	55
Possible Injury (C)	4	2	106	4	6	73
Property Damage Only (O)	2	0	435	0	8	535

When CMFs apply uniformly to all severities, these steps can be simplified, with CMFs being applied directly to aggregate weighted totals (crash costs, EPDO, or similar). The Madison data from the TOPS lab contains pre-calculated EPDO totals based on crash valuations for Wisconsin by mode and severity (published in Claros et al. 2020). We illustrate these calculations both ways – first, using the simplified approach and Wisconsin-specific EPDO values, and second, using individual calculations by severity and U.S. DOT crash cost valuations from the

2023 U.S. DOT BCA Guidance (the most recent guidance available at the time the case study was completed). The Wisconsin and U.S. DOT costs are summarized in Table F-2 and EPDO weights are summarized in Table F-3. Note that Wisconsin uses the motorist EPDO crash cost as the denominator for EPDO weights for all modes. The distribution of costs from Wisconsin is much flatter than U.S. DOT’s crash cost distribution; one fatal crash using Wisconsin’s rates is equivalent to 129.4 to 155.5 property damage only crashes (depending on the mode), whereas one fatal crash using U.S. DOT’s rates is equivalent to 2,974.4 property damage only crashes. The U.S. DOT rates hew much more closely to empirical research on the value of a statistical life, which inevitably leads to a much stronger emphasis on life-altering fatal and serious injury crashes than property damage crashes. Updated 2024 U.S. DOT BCA crash costs are also presented in Table G-17.

The next part of the safety benefits estimation workflow requires choosing CMFs that best represent the types of gap closure treatments for the study area. A project to close this gap along East Washington might include the following elements:

- Roadway reallocation, reducing through lanes from 6 to 4 or 2
- Addition of shared-use path or separated bicycle lanes
- Pedestrian crossing enhancements like leading pedestrian intervals (LPIs).

The FHWA CMF Clearinghouse does not have a CMF for reallocating a 6-lane urban arterial like East Washington; the only “road diet” CMFs available are for 4-to-3 conversions, resulting

Table F-2. Crash costs by mode and severity from Wisconsin TOPS lab and U.S. DOT.

Severity	Wisconsin Cost: Pedestrian	Wisconsin Cost: Bicyclist	Wisconsin Cost: Motorist	U.S. DOT Cost: All Modes
Fatal (K)	\$3,305,922	\$3,147,627	\$3,782,512	\$11,600,000
Serious Injury (A)	\$433,383	\$362,759	\$389,169	\$554,800
Minor Injury (B)	\$113,100	\$90,303	\$107,674	\$151,100
Possible Injury (C)	\$73,539	\$60,060	\$56,365	\$77,200
Property Damage Only (O)	\$35,692	\$49,042	\$24,322	\$3,900

Notes:

Crash costs presented in this table are disaggregated by mode, which is consistent with Wisconsin’s reporting standards. Aggregation of costs across all modes, as shown for U.S. DOT is also common.

Table F-3. EPDO weights derived from Wisconsin TOPS lab and U.S. DOT crash costs.

Severity	Wisconsin EPDO: Pedestrian	Wisconsin EPDO: Bicyclist	Wisconsin EPDO: Motorist	U.S. DOT EPDO: All Modes
Fatal (K)	135.9	129.4	155.5	2,974.4
Serious Injury (A)	17.8	14.9	16.0	142.3
Minor Injury (B)	4.7	3.7	4.4	38.7
Possible Injury (C)	3.0	2.5	2.3	19.8
Property Damage Only (O)	1.5	2.0	1.0	1.0

Notes:

EPDO weights presented in this table are disaggregated by mode, which is consistent with Wisconsin’s reporting standards. Aggregation of weights across all modes, as shown for U.S. DOT is also common.

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in one through lane in each direction and a two-way center left-turn lane, plus bike lanes. The most recent high-quality bike lane CMFs are available for 4-lane roads, but do not capture the safety benefit of the lane reduction. Regardless, these two CMFs are very close to one another (0.53 for 4-to-3 conversion and 0.54 for adding bike lanes). Given these imperfect options and the closeness of the two CMFs, we will test the safety calculations using the more conservative of the two CMFs (adding bike lanes). We will also include the CMF for adding LPIs. These CMFs are shown in Table F-4 and in Table G-16, the larger CMF reference table in this document.

The CMF for LPIs applies to pedestrian crashes only, all severities, per Table G-16. Because it is unclear how many pedestrians would detour from parallel corridors to the gap corridor, we will apply that CMF to pedestrian crashes in the gap corridor only, not the parallel corridors.

The CMF for adding bicycle lanes applies to crashes for all modes and all severities, per Table G-16. The CMF for bicycle lanes is intended for 4-lane roads and assumes a 20% increase in bicyclist traffic. Since we are assuming that bicyclist traffic would detour from parallel corridors, assuming a small reduction is appropriate, though CMF Clearinghouse also has bike lane CMFs for 0% and 40% increases in bicycle traffic. Because of this assumed diversion, we will apply the CMF to bicyclist crashes on both the gap corridor and the parallel corridors. We will apply the bike lane CMF to motorist and pedestrian crashes from the gap corridor only. Table F-4 summarizes which CMFs are applied to which modes and locations.

The results of applying the CMFs to each mode and severity or to each mode's EPDO scores are summarized in the following series of tables:

- Table F-5 shows the baseline conditions in the gap corridor and the parallel corridors by mode and severity, along with each mode's EPDO and crash costs calculated using both Wisconsin rates and U.S. DOT rates. Note that these baselines are different from the observed crash history from Table F-1 because the baseline comes from EB-adjusted SPF estimates of crashes.
- Table F-6 shows the final conditions that result from multiplying baseline crash estimates by the CMFs listed in Table G-16.
- Table F-7 summarizes the difference in EPDO and crash cost totals between final and baseline conditions.

As evident from the tables, the applied CMFs imply a nearly 24% decrease in EPDO or crash costs between baseline and final conditions. Most of this difference comes from the reduction in motorist crashes on the gap corridor—many pedestrian and bicyclist safety enhancements also benefit drivers. The reductions shown using the U.S. DOT crash costs are larger than those calculated using the Wisconsin rates due to the higher value placed on fatal and serious injury crashes.

Table F-4. CMFs applied to each mode and location. Standard practice when applying multiple CMFs is to multiply them.

Crash Type	Gap Corridor	Parallel Corridors
Pedestrian Crashes	LPI CMF (0.81) Bike Lane CMF (0.54)	None
Bicyclist Crashes	Bike Lane CMF (0.54)	Bike Lane CMF (0.54)
Motorist Crashes	Bike Lane CMF (0.54)	None

Table F-5. Baseline conditions—gap corridor and parallel corridors before CMF application.

Outcome	Gap Corridor				Parallel Corridors				Gap + Parallel
Mode	Pedestrian	Bicycle	Motorist	SUBTOTAL	Pedestrian	Bicycle	Motorist	SUBTOTAL	TOTAL
K Crashes	0.3	0.1	0.8	1.2	0.3	0.1	1.0	1.4	2.5
A Crashes	2.0	0.8	8.4	11.2	2.4	1.1	9.2	12.7	24.0
B Crashes	4.2	5.3	49.7	59.3	5.2	7.2	60.3	72.6	131.9
C Crashes	2.0	1.6	86.3	89.8	2.4	2.2	75.0	79.6	169.4
O Crashes	14.1	13.4	418.6	446.1	22.7	23.8	495.8	542.4	988.5
WI Cost	\$2,893,837	\$1,716,502	\$26,857,984	\$31,468,323	\$3,727,736	\$2,542,389	\$29,982,881	\$36,253,006	\$67,721,328
WI EPDO	119.0	70.6	1,104.3	1,293.8	153.3	104.5	1,232.7	1,490.5	2,784.4
U.S. DOT Cost	\$5,089,939	\$2,112,186	\$30,258,078	\$37,460,203	\$6,254,762	\$2,653,950	\$33,070,984	\$41,979,696	\$79,439,899
U.S. DOT EPDO	1,305.1	541.6	7,758.5	9,605.2	1,603.8	680.5	8,479.7	10,764.0	20,369.2

Table F-6. Final conditions—gap corridor and parallel corridors after CMF application.

Outcome	Gap Corridor				Parallel Corridors				Gap + Parallel
Mode	Pedestrian	Bicycle	Motorist	SUBTOTAL	Pedestrian	Bicycle	Motorist	SUBTOTAL	TOTAL
K Crashes	0.1	0.0	0.5	0.6	0.3	0.0	1.0	1.3	1.9
A Crashes	0.9	0.5	4.5	5.9	2.4	0.6	9.2	12.2	18.1
B Crashes	1.9	2.9	26.9	31.6	5.2	3.9	60.3	69.3	100.9
C Crashes	0.9	0.8	46.6	48.3	2.4	1.2	75.0	78.6	126.9
O Crashes	6.2	7.2	226.1	239.5	22.7	12.9	495.8	531.4	770.9
WI Cost	\$1,265,764	\$926,911	\$14,503,311	\$16,695,987	\$3,727,736	\$1,372,890	\$29,982,881	\$35,083,507	\$51,779,493
WI EPDO	52.0	38.1	596.3	686.5	153.3	56.4	1,232.7	1,442.5	2,128.9
U.S. DOT Cost	\$2,226,339	\$1,140,580	\$16,339,362	\$19,706,282	\$6,254,762	\$1,433,133	\$33,070,984	\$40,758,879	\$60,465,161
U.S. DOT EPDO	570.9	292.5	4,189.6	5,052.9	1,603.8	367.5	8,479.7	10,451.0	15,503.9

Table F-7. Reduction in estimated crashes from baseline to final conditions resulting from CMF application.

Outcome	Gap Corridor				Parallel Corridors				Gap + Parallel
Mode	Pedestrian	Bicycle	Motorist	SUBTOTAL	Pedestrian	Bicycle	Motorist	SUBTOTAL	TOTAL
WI Cost	-\$1,628,073	-\$789,591	-\$12,354,673	-\$14,772,336	\$0	-\$1,169,499	\$0	-\$1,169,499	-\$15,941,835
WI EPDO	-66.9	-32.5	-508.0	-607.4	0.0	-48.1	0.0	-48.1	-655.4
U.S. DOT Cost	-\$2,863,600	-\$971,606	-\$13,918,716	-\$17,753,921	\$0	-\$1,220,817	\$0	-\$1,220,817	-\$18,974,738
U.S. DOT EPDO	-734.3	-249.1	-3,568.9	-4,552.3	0.0	-313.0	0.0	-313.0	-4,865.3

APPENDIX G

Safety Benefit Analysis Reference Tables

Crash Multiplier Set 1: All (Intersection + Midblock/Segment) Crash Rates per Mile for All Areas (Urban + Rural) from Six Counties within CA State Database

The crash multipliers in Table G-1 and G-2 are based on crash data from six counties within California: Alameda, Contra Costa, San Diego, San Francisco, San Mateo, and Santa Clara. These represent the rate of crashes over 5 years per mile on the specified functional class. For use over a longer or shorter time horizon, divide by 5 to convert to an annual rate. Rates are calculated on all crashes and roads within the six counties—urban and rural combined. Rates are calculated from both intersection and midblock crashes combined. These should not be combined with intersection-specific or midblock-specific crash rate multipliers, since both intersection and midblock/segment crashes are reflected in these combined “All Crashes” rates. Functional class is defined as the functional class of the segment closest to the crash for all crashes.

Table G-1. All pedestrian crash multipliers per mile by functional class and severity derived from UC Berkeley SafeTREC TIMS data 2018–2022.

Functional Class	Fatal (K) per Mile	Suspected Serious Injury (A) per Mile	Suspected Minor Injury (B) per Mile	Possible Injury (C) per Mile
Interstate	0.05145867	0.02471637	0.01985413	0.02188007
Other Freeway or Expressway	0.03834261	0.02350031	0.01978973	0.01175016
Other Principal Arterial	0.09370425	0.07467057	0.1954612	0.13762811
Minor Arterial	0.05413422	0.08898776	0.22432332	0.18464961
Major Collector	0.01337223	0.03428155	0.11111111	0.08460977
Minor Collector	0	0.00529101	0.01587302	0.01058201
Local	0.0005493	0.0011492	0.00398245	0.00388126

Notes:

Calculated using data retrieved from UC Berkeley SafeTREC TIMS crash data and the Caltrans Linear Referencing Services dataset. Crash rates are approximate and subject to data limitations.

G-2 Estimating Benefits of Closing Gaps in Active Transportation Networks: A Guide**Table G-2. All bicyclist crash multipliers per mile by functional class and severity derived from UC Berkeley SafeTREC TIMS data 2018–2022.**

Functional Class	Fatal (K) per Mile	Suspected Serious Injury (A) per Mile	Suspected Minor Injury (B) per Mile	Possible Injury (C) per Mile
Interstate	0.04092383	0.02471637	0.01985413	0.02188007
Other Freeway or Expressway	0.0296846	0.02350031	0.01978973	0.01175016
Other Principal Arterial	0.04612006	0.07467057	0.1954612	0.13762811
Minor Arterial	0.04857249	0.08898776	0.22432332	0.18464961
Major Collector	0.01167031	0.03428155	0.11111111	0.08460977
Minor Collector	0	0.00529101	0.01587302	0.01058201
Local	0.00041921	0.0011492	0.00398245	0.00388126

Notes:

Calculated using data retrieved from UC Berkeley SafeTREC TIMS crash data and the Caltrans Linear Referencing Services dataset. Crash rates are approximate and subject to data limitations.

Crash Multiplier Set 2: All (Intersection + Midblock/Segment) Crash Rates per Mile for Urban Areas from MN State Database

The crash multipliers in Table G-3 and G-4 are based on crash data from MN, in Census-designated urbanized areas only. These represent the rate of crashes over 5 years per mile on the specified functional class. For use over a longer or shorter time horizon, divide by 5 to convert to an annual rate. Rates are calculated from both intersection and midblock crashes combined. These should not be combined with intersection-specific or midblock-specific crash rate multipliers, since both intersection and midblock/segment crashes are reflected in these combined “All Crashes” rates. Functional class is defined as either the highest functional class present at the intersection, for intersection crashes, or as the functional class of the segment closest to the crash, for midblock (non-intersection) crashes.

Table G-3. Urban area all pedestrian crash multipliers per mile by functional class and severity derived from Minnesota DOT data 2017–2021.

Functional Class	Fatal (K) per Mile	Suspected Serious Injury (A) per Mile	Suspected Minor Injury (B) per Mile	Possible Injury (C) per Mile	Property Damage Only (O) per Mile
Principal Arterial – Freeway/ Expressway	0.025505164	0.035069600	0.105208801	0.082891783	0.015940727
Principal/Major Arterial (non-Freeway)	0.026978829	0.052731348	0.096878524	0.072352315	0.008584173
Minor Arterial	0.034603466	0.140632033	0.371765437	0.300783969	0.036821636
Major/Minor Collector	0.005696470	0.032583811	0.070408374	0.055369693	0.005240753
Local Road	0.001394127	0.007634507	0.018986687	0.011684115	0.001593288

Notes:

Calculated using data retrieved from Minnesota Department of Transportation and OpenStreetMap. Crash rates are approximate and subject to data limitations.

Table G-4. Urban area all bicyclist crash multipliers per mile by functional class and severity derived from Minnesota DOT data 2017–2021.

Functional Class	Fatal (K) per Mile	Suspected Serious Injury (A) per Mile	Suspected Minor Injury (B) per Mile	Possible Injury (C) per Mile	Property Damage Only (O) per Mile
Principal Arterial – Freeway/ Expressway	0.006376291	0.007970364	0.070139201	0.081297710	0.012752582
Principal/Major Arterial (non-Freeway)	0.007357863	0.018394656	0.122631043	0.077257557	0.019620967
Minor Arterial	0.005323610	0.051905198	0.308325750	0.217380745	0.048799759
Major/Minor Collector	0.000683576	0.009797929	0.065395481	0.039419575	0.009114353
Local Road	0.000199161	0.002323546	0.013808500	0.007435346	0.001925224

Notes:

Calculated using data retrieved from Minnesota Department of Transportation and OpenStreetMap. Crash rates are approximate and subject to data limitations.

Crash Multiplier Set 3: All (Intersection + Midblock/Segment) Crash Rates per Mile for Rural Areas from MN State Database

The crash multipliers in Table G-5 and G-6 are based on crash data from MN, in Census-designated rural areas only. These represent the rate of crashes over 5 years per mile on the specified functional class. For use over a longer or shorter time horizon, divide by 5 to convert to an annual rate. Rates are calculated from both intersection and midblock crashes combined. These should not be combined with intersection-specific or midblock-specific crash rate multipliers. Functional class is defined as either the highest functional class present at the intersection, for intersection crashes, or as the functional class of the segment closest to the crash, for midblock (non-intersection) crashes.

Table G-5. Rural area all pedestrian crash multipliers per mile by functional class and severity derived from Minnesota DOT data 2017–2021.

Functional Class	Fatal (K) per Mile	Suspected Serious Injury (A) per Mile	Suspected Minor Injury (B) per Mile	Possible Injury (C) per Mile	Property Damage Only (O) per Mile
Principal Arterial – Freeway/ Expressway	0.000000000	0.001851975	0.005555925	0.003703950	0.000000000
Principal/Major Arterial (non-Freeway)	0.001530363	0.001530363	0.005815381	0.001530363	0.000000000
Minor Arterial	0.000291930	0.002481402	0.003503156	0.001605613	0.000583859
Major/Minor Collector	0.000182052	0.000655385	0.001383592	0.000291282	0.000109231
Local Road	0.000012203	0.000073220	0.000207457	0.000085423	0.000048813

Notes:

Calculated using data retrieved from Minnesota Department of Transportation and OpenStreetMap. Crash rates are approximate and subject to data limitations.

G-4 Estimating Benefits of Closing Gaps in Active Transportation Networks: A Guide**Table G-6. Rural area all bicycles crash multipliers per mile by functional class and severity derived from Minnesota DOT data 2017–2021.**

Functional Class	Fatal (K) per Mile	Suspected Serious Injury (A) per Mile	Suspected Minor Injury (B) per Mile	Possible Injury (C) per Mile	Property Damage Only (O) per Mile
Principal Arterial – Freeway/ Expressway	0.011111851	0.005555925	0.007407900	0.001851975	0.001851975
Principal/Major Arterial (non-Freeway)	0.003060727	0.004591090	0.007345744	0.002448581	0.001224291
Minor Arterial	0.002043508	0.001751578	0.002627367	0.001167719	0.000437894
Major/Minor Collector	0.000655385	0.001092309	0.001310771	0.000436924	0.000072821
Local Road	0.000097627	0.000353897	0.000341694	0.000158644	0.000024407

Notes:

Calculated using data retrieved from Minnesota Department of Transportation and OpenStreetMap. Crash rates are approximate and subject to data limitations.

Crash Multiplier Set 4: Fatal Crash Rates from FARS (Fatal Crashes Only; Not Preferred)

The crash multipliers in Table G-7 are based on FARS crash data from 2015 to 2019. These multipliers are for fatal crashes only. These represent the rate of crashes over 5 years per mile on the specified functional class. For use over a longer or shorter time horizon, divide by 5 to convert to an annual rate.

Table G-7. Pedestrian and bicycle fatal crash multipliers derived from FARS 2015–2019.

Functional Class	Fatal Pedestrian Crashes per Mile	Fatal Bicyclist Crashes per Mile
Major Arterial	0.017649889	0.002085138
Minor Arterial	0.041508085	0.005536307
Major Collector	0.010394488	0.001646725
Minor Collector	0.007892883	0.001601086
Local/Undesignated Road	0.000516971	0.000105633

Notes:

Calculated using FARS and Highway Performance Monitoring System (HPMS) data, 2015–2019, via Safer Streets Priority Finder. Crash rates are approximate and subject to data limitations.

Crash Multiplier Set 5: Intersection Subset Crash Rates per Intersection for Urban Areas from MN State Database

The crash multipliers in Table G-8 and G-9 are based on intersection crash data from MN, in Census-designated urbanized areas only. These represent the rate of crashes over 5 years per intersection on the specified functional class. For use over a longer or shorter time horizon, divide by 5 to convert to an annual rate. Rates are calculated from intersection crashes, defined as occurring within 200 feet of an intersection. These can be used alone or alongside midblock crash rates per mile (Crash Multiplier Set 7: Midblock/Segment Crash Rates per Mile for Urban Areas from MN State Database). Functional class is defined as the highest functional class present at the intersection.

Table G-8. Urban area intersection subset pedestrian crash multipliers per intersection by functional class and severity derived from Minnesota DOT data 2017–2021.

Functional Class	Fatal (K) per Intersection	Suspected Serious Injury (A) per Intersection	Suspected Minor Injury (B) per Intersection	Possible Injury (C) per Intersection	Property Damage Only (O) per Intersection
Principal Arterial – Freeway/ Expressway	0.001331203	0.003194888	0.014376997	0.011980831	0.001597444
Principal/Major Arterial (non-Freeway)	0.004587156	0.008027523	0.015137615	0.011467890	0.001605505
Minor Arterial	0.003204086	0.013420014	0.037148827	0.030369166	0.003436267
Major/Minor Collector	0.000776438	0.005797402	0.012578291	0.009576065	0.000879963
Local Road	0.000200872	0.001090450	0.002467860	0.001707415	0.000258264

Notes:

Calculated using data retrieved from Minnesota Department of Transportation and OpenStreetMap. Crash rates are approximate and subject to data limitations.

Table G-9. Urban area intersection subset bicycle crash multipliers per intersection by functional class and severity derived from Minnesota DOT data 2017–2021.

Functional Class	Fatal (K) per Intersection	Suspected Serious Injury (A) per Intersection	Suspected Minor Injury (B) per Intersection	Possible Injury (C) per Intersection	Property Damage Only (O) per Intersection
Principal Arterial – Freeway/Expressway	0.000798722	0.001331203	0.010915868	0.013578275	0.002129925
Principal/Major Arterial (non-Freeway)	0.001146789	0.003440367	0.021559633	0.013532110	0.003440367
Minor Arterial	0.000557232	0.005247272	0.031390759	0.022196424	0.004922220
Major/Minor Collector	0.000051763	0.001759925	0.012112428	0.007816140	0.001759925
Local Road	0.000028696	0.000416093	0.002410468	0.001248278	0.000330005

Notes:

Calculated using data retrieved from Minnesota Department of Transportation and OpenStreetMap. Crash rates are approximate and subject to data limitations.

Crash Multiplier Set 6: Intersection Subset Crash Rates per Intersection for Rural Areas from MN State Database

The crash multipliers in Table G-10 and G-11 are based on intersection crash data from MN, in Census-designated rural areas only. These represent the rate of crashes over 5 years per intersection on the specified functional class. For use over a longer or shorter time horizon, divide by 5 to convert to an annual rate. Rates are calculated from intersection crashes, defined as occurring within 200 feet of an intersection. These can be used alone or alongside midblock crash rates per mile (Crash Multiplier Set 8: Midblock/Segment Crash Rates per Mile for Rural Areas from MN State Database). Functional class is defined as the highest functional class present at the intersection.

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Table G-10. Rural area intersection subset pedestrian crash multipliers per intersection by functional class and severity derived from Minnesota DOT data 2017–2021.

Functional Class	Fatal (K) per Intersection	Suspected Serious Injury (A) per Intersection	Suspected Minor Injury (B) per Intersection	Possible Injury (C) per Intersection	Property Damage Only (O) per Intersection
Principal Arterial – Freeway/Expressway	0.001745201	0.000000000	0.000872600	0.000872600	0.000000000
Principal/Major Arterial (non-Freeway)	0.000476554	0.001143729	0.001810904	0.000667175	0.000095311
Minor Arterial	0.000269088	0.000336361	0.000874538	0.000403633	0.000201816
Major/Minor Collector	0.000069188	0.000322878	0.000415129	0.000138376	0.000000000
Local Road	0.000012515	0.000150182	0.000225273	0.000100121	0.000012515

Notes:

Calculated using data retrieved from Minnesota Department of Transportation and OpenStreetMap. Crash rates are approximate and subject to data limitations.

Table G-11. Rural area intersection subset bicycle crash multipliers per intersection by functional class and severity derived from Minnesota DOT data 2017–2021.

Functional Class	Fatal (K) per Intersection	Suspected Serious Injury (A) per Intersection	Suspected Minor Injury (B) per Intersection	Possible Injury (C) per Intersection	Property Damage Only (O) per Intersection
Principal Arterial – Freeway/Expressway	0.000000000	0.000872600	0.001745201	0.000872600	0.000000000
Principal/Major Arterial (non-Freeway)	0.000285932	0.000476554	0.001524971	0.000381243	0.000000000
Minor Arterial	0.000134544	0.000605449	0.001479987	0.000605449	0.000134544
Major/Minor Collector	0.000023063	0.000184502	0.000576568	0.000092251	0.000046125
Local Road	0.000000000	0.000037546	0.000150182	0.000050061	0.000037546

Notes:

Calculated using data retrieved from Minnesota Department of Transportation and OpenStreetMap. Crash rates are approximate and subject to data limitations.

Crash Multiplier Set 7: Midblock/Segment Subset Crash Rates per Mile for Urban Areas from MN State Database

The crash multipliers in Table G-12 and G-13 are based on midblock (non-intersection) crash data from MN, in Census-designated urbanized areas only. These represent the rate of crashes over 5 years per mile on the specified functional class. For use over a longer or shorter time horizon, divide by 5 to convert to an annual rate. Rates are calculated from midblock (non-intersection) crashes, defined as occurring more than 200 feet from an intersection. These can be used alone or alongside intersection crash rates per intersection (Crash Multiplier Set 5: Intersection Crash Rates per Intersection for Urban Areas from MN State Database). Functional class is defined as the functional class of the segment closest to the midblock (non-intersection) crash.

Table G-12. Urban area midblock subset pedestrian crash multipliers per mile by functional class and severity derived from Minnesota DOT data 2017–2021.

Functional Class	Fatal (K) per Mile	Suspected Serious Injury (A) per Mile	Suspected Minor Injury (B) per Mile	Possible Injury (C) per Mile	Property Damage Only (O) per Mile
Principal Arterial – Freeway/Expressway	0.017534800	0.015940727	0.019128873	0.011158509	0.006376291
Principal/Major Arterial (non-Freeway)	0.002452621	0.009810483	0.015942036	0.011036794	0.000000000
Minor Arterial	0.003992708	0.012421757	0.016858099	0.010647220	0.003992708
Major/Minor Collector	0.002278588	0.007063623	0.015038682	0.013215811	0.001367153
Local Road	0.000464709	0.002589094	0.007568120	0.003784060	0.000398322

Notes:

Calculated using data retrieved from Minnesota Department of Transportation and OpenStreetMap. Crash rates are approximate and subject to data limitations.

Table G-13. Urban area midblock subset bicyclist crash multipliers per mile by functional class and severity derived from Minnesota DOT data 2017–2021.

Functional Class	Fatal (K) per Mile	Suspected Serious Injury (A) per Mile	Suspected Minor Injury (B) per Mile	Possible Injury (C) per Mile	Property Damage Only (O) per Mile
Principal Arterial – Freeway/Expressway	0.001594073	0.000000000	0.004782218	0.000000000	0.000000000
Principal/Major Arterial (non-Freeway)	0.001226310	0.000000000	0.007357863	0.004905242	0.001226310
Minor Arterial	0.000000000	0.001774537	0.008429049	0.005323610	0.001774537
Major/Minor Collector	0.000455718	0.002050729	0.012076517	0.005012894	0.001367153
Local Road	0.000066387	0.000398322	0.002655481	0.001659675	0.000398322

Notes:

Calculated using data retrieved from Minnesota Department of Transportation and OpenStreetMap. Crash rates are approximate and subject to data limitations.

Crash Multiplier Set 8: Midblock/Segment Subset Crash Rates per Mile for Rural Areas from MN State Database

The crash multipliers in Table G-14 and G-15 are based on midblock (non-intersection) crash data from MN, in Census-designated rural areas only. These represent the rate of crashes over 5 years per mile on the specified functional class. For use over a longer or shorter time horizon, divide by 5 to convert to an annual rate. Rates are calculated from midblock (non-intersection) crashes, defined as occurring more than 200 feet from an intersection. These can be used alone or alongside intersection crash rates per intersection (Crash Multiplier Set 6: Intersection Crash Rates per Intersection for Rural Areas from MN State Database). Functional class is defined as the functional class of the segment closest to the midblock (non-intersection) crash.

G-8 Estimating Benefits of Closing Gaps in Active Transportation Networks: A Guide**Table G-14. Rural area midblock subset pedestrian crash multipliers per mile by functional class and severity derived from Minnesota DOT data 2017–2021.**

Functional Class	Fatal (K) per Mile	Suspected Serious Injury (A) per Mile	Suspected Minor Injury (B) per Mile	Possible Injury (C) per Mile	Property Damage Only (O) per Mile
Principal Arterial – Freeway/Expressway	0.007407900	0.005555925	0.005555925	0.000000000	0.001851975
Principal/Major Arterial (non-Freeway)	0.001530363	0.000918218	0.001530363	0.000306073	0.000918218
Minor Arterial	0.001459648	0.001021754	0.000729824	0.000291930	0.000000000
Major/Minor Collector	0.000546155	0.000582565	0.000655385	0.000218462	0.000072821
Local Road	0.000085423	0.000207457	0.000122033	0.000061017	0.000012203

Notes:

Calculated using data retrieved from Minnesota Department of Transportation and OpenStreetMap. Crash rates are approximate and subject to data limitations.

Table G-15. Rural area midblock subset bicycle crash multipliers per mile by functional class and severity derived from Minnesota DOT data 2017–2021.

Functional Class	Fatal (K) per Mile	Suspected Serious Injury (A) per Mile	Suspected Minor Injury (B) per Mile	Possible Injury (C) per Mile	Property Damage Only (O) per Mile
Principal Arterial – Freeway/Expressway	0.000000000	0.000000000	0.001851975	0.001851975	0.000000000
Principal/Major Arterial (non-Freeway)	0.000612145	0.000000000	0.000918218	0.000306073	0.000000000
Minor Arterial	0.000000000	0.001167719	0.000291930	0.000291930	0.000291930
Major/Minor Collector	0.000145641	0.000364103	0.000473334	0.000145641	0.000036410
Local Road	0.000012203	0.000036610	0.000061017	0.000036610	0.000012203

Notes:

Calculated using data retrieved from Minnesota Department of Transportation and OpenStreetMap. Crash rates are approximate and subject to data limitations.

Commonly Used CMFs for Pedestrian and Bicycle Countermeasures

Table G-16. Commonly used CMFs for pedestrian and bicycle gap closure treatments.

Treatment Type	Applicable Mode(s)	Applicable Severities	Apply to Segments/Inter-Sections/Both	CMF	CRF	Source	Notes/Considerations
Converting four-lane roadways to three-lane roadways with center turn lane (road diet)	All	All	Any (not specified)	0.53	47%	Persaud et al. 2010, via CMF Clearinghouse	
Modify signal phasing (implement an LPI)	Pedestrian	All	Signalized intersections	0.81	19%	Goughnour et al. 2018, via CMF Clearinghouse	Sites where LPI implemented at all crossings. <i>Other CMFs available for implementing LPI on only major or minor crossings. See Source.</i>
Install bicycle lanes	All	All	Segments	0.54	46%	Avelar et al. 2021, via CMF Clearinghouse ¹	4-lane roads, assuming 20% increase in bicyclist volumes. <i>Other CMFs available for 2-lane roads and varying levels of expected change in bicyclist volumes. See Source.</i>
Upgrade traditional bike lanes or flush buffered bike lanes to separated bike lanes	Bicycle	All	All	0.60	40%	Dixon et al. 2023, via CMF Clearinghouse	Assumes separated bike lane uses a mix of flexi-posts and other vertical elements. <i>Other CMFs available for various specific combinations of before-and-after conditions. See Source.</i>
Install a PHB or high-intensity activated crosswalk (HAWK)	Pedestrian	All	Intersections	0.54	46%	Fitzpatrick et al. 2019, via CMF Clearinghouse	Reference category includes a mix of unsignalized and signalized intersections.

Notes:

¹CMF Clearinghouse's version of this CMF contains a typo. Refer to the original publication (Avelar et al. 2021) for correct CMF values and applicability.

CMF = crash modification factor; CRF = crash reduction factor

Standard U.S. DOT Crash Costs—KABCO Scale

Table G-17. Crash costs.

Crash Severity	U.S. DOT BCA Guidance—2022 Monetized Crash Cost Values
Fatal Injury (K)	\$12,500,000
Serious Injury (A)	\$1,188,200
Minor Injury (B)	\$233,800
Possible Injury (C)	\$111,700
Property Damage Only (O)	\$5,000

Source:

Table A-1, U.S. DOT 2023a.



APPENDIX H

Augmented Methods for Emissions Benefits Analysis

The Augmented Method follows the same steps outlined in the Basic Method, but with different data inputs. The Basic Method uses national data sources, whereas the Augmented Method may use data obtained from local or regional agencies.

Unit of Measure

Emissions benefits can be measured as the monetized value of avoided pollutants. An overview of the methods appears in Figure H-1.

The process starts with an estimate of the mode shift, which is the number of new active transportation trips as a result of the gap closure multiplied by the share of those trips shifted from driving. The number of trips shifted from driving to active transportation is then multiplied by the length of those trips to get an estimate of VMT reduction. Emission rates specific to each mode and pollutant of interest (expressed in grams per vehicle-mile) can then be applied to changes in VMT. Emission rates may vary by speed; speed-specific rates may be used if the average speed of avoided vehicle trips is known.

Data Needs

Table H-1 displays the data needs by capacity level.

The following list presents key data sources for emissions benefits analysis:

- **NHTS.** Summary statistics and data explorer: <https://nhts.ornl.gov/>. To derive average trip lengths, go to the “Data Explore Tool” and extract person-miles by state and mode of travel as well as person-trips by state and mode of travel. Divide the values to get average miles per trip by mode.
- **ACS.** Data on baseline walk and bike commute mode shares can be used as a benchmark to check reality of mode shift estimates. Data: <https://www.census.gov/programs-surveys/acs/data.html>
- **FHWA CMAQ.** Emissions Calculator Toolkit (FHWA 2022b). https://www.fhwa.dot.gov/environment/air_quality/cmaq/toolkit/
- **State and regional data available from various sources.** Contact your state DOT, MPO, or regional planning agency/organization.

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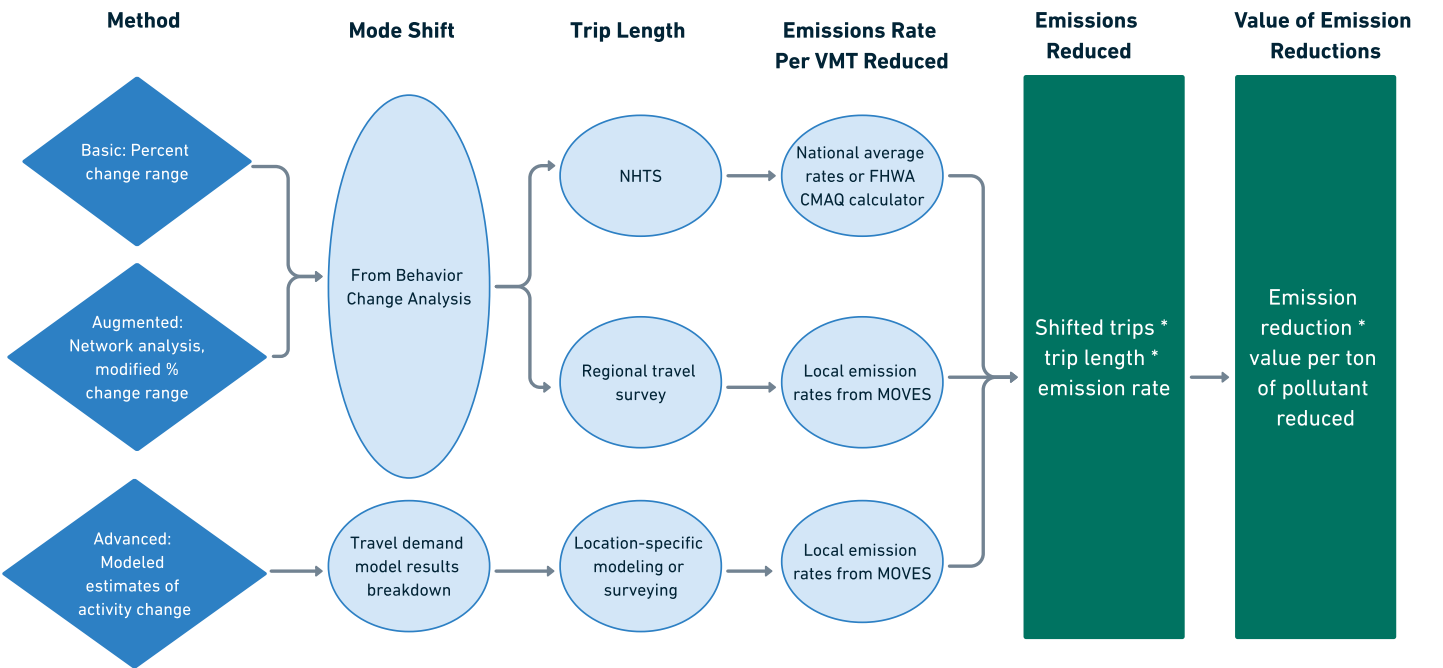


Figure H-1. Basic, Augmented, and Advanced Application Methods for calculating emissions benefits.

Table H-1. Data needs for Basic, Augmented, and Advanced Application Method emissions benefit estimation.

Key Data Item	Basic Method	Augmented Method	Advanced Application
Number of new active trips	Described in <i>Chapter 3: Step 2. Estimating Behavior Change</i>		
Share who previously traveled by personal motor vehicle	Described in <i>Chapter 3: Step 2. Estimating Behavior Change</i>		
Average trip length	Average national walk and bicycle trip length from National Household Travel Survey	Average walk and bicycle trip length from a regional household travel survey	Location-specific trip modeling or surveying
Emission rate per vehicle-mile	National average rates (e.g., from the FHWA CMAQ Calculator Toolkit)	Local emission rates from U.S. EPA's MOVES runs conducted by a regional or state agency for other projects	Project-specific MOVES runs using local data/inputs

Methods

The steps for estimating emissions benefits using Augmented Methods are presented in the following list:

1. Estimate number of new active trips.

Estimates of the number of new active trips may be obtained using a variety of methods as described in *Chapter 3: Step 2. Estimating Behavior Change*.

Emissions reductions may be estimated on a daily or annual basis. If daily trip changes are developed and annual emissions benefits are desired, annualize the number of new trips using an appropriate annualization factor. If daily trips represent average annual conditions, multiply by 365. If they represent “active travel season” conditions, a lower annualization factor may be appropriate depending on region-specific climate affecting active travel.

Example: In the example provided here, we use an illustrative value of 300 new trips per day or 109,500 new trips per year.

2. Estimate share who previously traveled by personal motor vehicle.

Estimates of the share who previously traveled by personal motor vehicle may be obtained using a variety of methods as described in *Chapter 3: Step 2. Estimating Behavior Change*. For example, the Caltrans Active Transportation Benefit-Cost Tool default assumption is that 17.6% of new bicycle trips are shifting from cars, and 33.3% of pedestrian trips are shifting from a car trip completely or replacing the car trip by walking plus transit (Volker et al. 2019a, 2019b).

3. Estimate average trip length of shifted personal motor vehicle trips made by active travelers using the gap closure.

The 2017 NHTS found that, nationwide, the average length of a bicycle trip is 2.38 miles, and the average length of a walk trip is 0.86 miles. Therefore, the analyst could assume that any shifted personal motor vehicle trips due to the gap closure would be of similar lengths. The NHTS Data Explore Tool allows the user to query this information by state as well as by other categories such as urban vs. rural and population density.

4. Determine emission(s) rate per VMT.

The state-of-practice tool recommended in 49 states for emissions analysis, required for regulatory analysis and recommended for other analysis, is the U.S. Environmental Protection Agency (U.S. EPA) MOVES model (U.S. EPA 2024a). (California has a different model, known as EMFAC, that is used for regulatory purposes in that state [CARB, n.d.])

The simplest way to use MOVES emission factors is to use a tool or tables based on national average data. FHWA’s CMAQ Calculator Toolkit – Bicycle and Pedestrian Improvements incorporates pre-loaded national average emission rates (grams per vehicle-mile) from MOVES (FHWA 2022b). The FHWA tool can also be used to translate changes in active travel into changes in emissions. The user needs to provide assumptions about new bike trips, displaced vehicle trips, and average trip lengths.

Table H-2 provides emission rates in g/mi based on national MOVES runs. These rates were inferred from the FHWA CMAQ Emissions Calculator Toolkit, “Bicycle and Pedestrian Improvements” module, version 6/2022, by running the model for a reduction of 1,000 daily vehicle trips with average 1-mile trip length. Note that emission rates for a given year (e.g., 2030) will only change when MOVES (or the MOVES rates in the CMAQ tool) are updated.

A state or regional agency planning, or environmental agency may have run MOVES to produce emission factors specific to the county or regional vehicle fleet. The MPO can generally provide guidance on whether such factors are available. For example, Table H-3 provides emission factors for the Atlanta region from the Atlanta Regional Commission CMAQ Calculator. These factors are for light-duty vehicles in 2020, averaged for all road types. The analyst could pick a speed approximating the average speed on facilities from which auto traffic is removed and then apply the emission factors for that speed.

H-4 Estimating Benefits of Closing Gaps in Active Transportation Networks: A Guide**Table H-2. Sample light-duty vehicle emission rates (g/mi).**

Pollutant	2025	2030	2035	2040
Carbon Monoxide (CO)	3.648	2.641	1.989	1.664
Particulate Matter <2.5 µm (PM _{2.5})	0.012	0.011	0.010	0.010
Particulate Matter <10 µm (PM ₁₀)	0.047	0.046	0.045	0.045
Nitrogen Oxide (NO _x)	0.191	0.097	0.057	0.045
Volatile Organic Compounds (VOC)	0.156	0.107	0.086	0.078
Carbon Dioxide (CO ₂)	371	336	N/A	N/A
Carbon Dioxide Equivalent (CO ₂ e)	374	339	N/A	N/A
Total Energy Consumption (MMBTU/day)	5.02	4.55	4.25	4.08

Source:

FHWA CMAQ Emissions Calculator Toolkit, based on national MOVES runs, 2022 (FHWA 2022b). The tool did not provide CO₂ outputs for 2035 and 2040. Rates are provided as examples for this guide. Documentation of emission rates is from U.S. EPA (2024a).

Table H-3. Sample emission rates for Atlanta region, 2020 (g/mi).

Speed (mph)	CO ₂ e	NO _x (annual)	PM _{2.5} (annual)	NO _x (summer)	VOC (summer)
2.5	1,912	0.380	0.018	0.490	0.809
5	1,070	0.283	0.011	0.328	0.430
10	649	0.230	0.008	0.242	0.240
15	513	0.206	0.006	0.205	0.177
20	436	0.191	0.005	0.186	0.142
25	390	0.187	0.005	0.180	0.121
30	357	0.182	0.005	0.174	0.108
35	345	0.190	0.006	0.182	0.100
40	337	0.196	0.006	0.188	0.095
45	331	0.201	0.006	0.193	0.090
50	324	0.204	0.006	0.195	0.086
55	318	0.206	0.006	0.198	0.083
60	316	0.210	0.006	0.201	0.080
65	319	0.221	0.006	0.212	0.081
70	331	0.244	0.006	0.234	0.085
75	348	0.272	0.007	0.261	0.093

Source:

Atlanta Regional Commission, CMAQ Calculator Version 3 (ARC 2024). <https://atlantaregional.org/natural-resources/air-quality/air-quality/> Rates are provided as examples for this guide.

For greenhouse gas emissions, an average carbon dioxide (CO₂) emission factor per mile can also be developed based on the average fuel economy of light-duty vehicles (as reported in the U.S. Department of Energy Annual Energy Outlook [AEO]) and fuel-based emission factors (CO₂ per gallon, CH₄ and N₂O per mile) from a source such as the U.S. EPA or World Resources Institute Greenhouse Gas Protocol (U.S. EIA 2023; U.S. EPA 2024b; World Resources Institute, n.d.).

For example, the 2023 AEO projects a fleet average of 25.8 miles per gallon for light-duty vehicles in year 2025. The CO₂ content of gasoline is approximately 8.9 kg per gallon. This gives an emission rate of 345 grams per mile (8,900 g/gal / 25.8 mi/gal). Other greenhouse gases add 2% to 3% to this value.

5. Calculate emissions changes.

To estimate the emissions changes as a result of the project, multiply the number of new trips each day by the prior drive mode share by the trip length by the emission rate.

$$\text{Avoided Emissions} = \text{new trips/day} \times \text{prior drive mode share} \times \text{trip length} \\ \times \text{emission rate}$$

Illustrated for NO_x emissions displaced by a pedestrian gap closure using regional/state data, where prior drive mode share is taken from Minneapolis, average trip length for Georgia from NHTS, and the 2020 emission rate for Atlanta region from CMAQ tool:

Calculate CO₂ emissions per day:

$$\text{Avoided Emissions} = 300 \text{ new bicycle trips per day} \\ \times 21.9\% \text{ of new bicycle trips had been driving trips previously} \\ \times 2.3 \text{ miles average bike trip length} \\ \times 0.187 \text{ g NO}_x/\text{mile driving in a personal motor vehicle}$$

$$\text{Avoided Emissions} = 28 \text{ g/day} / 1,000 \text{ g/kg} = 0.028 \text{ kg/day}$$

Transform to per year:

$$\text{Avoided Emissions} = 0.028 \text{ kg/day} \times 365 \text{ days/year} = 10 \text{ kg/year}$$

The units described vary between different types of pollutant emissions.

6. Emissions benefits may be reported in a simple table showing emissions benefits by pollutant.

The table and/or accompanying text should specify the units (e.g., kg or tons), time frame (e.g., daily summer, daily winter, or annual), and evaluation year. The text should also describe the key assumptions used. An example table is shown in Table H-4.

7. Monetize benefits. Emissions benefits can be monetized by applying damage costs.

Table H-5 displays the damage costs per metric ton for three types of emissions. To calculate the emissions benefits for your project, multiply the number of metric tons of reduced emissions by the corresponding damage costs. If your emissions benefits are in kilograms, you will need to convert to metric tons by dividing the number of kilograms by 1,000.

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Table H-4. Sample emissions benefits.

Year	CO ₂ eq (metric tons per year)	NO _x (kg/year)	PM _{2.5} (kg/year)	NO _x (kg per summer day)	VOC (kg per summer day)
2020	44.4	10	0.6	0.060	0.037

Table H-5. Values for monetizing emissions benefits.

Year	CO ₂ (value per metric ton)	NO _x (value per metric ton)	PM _{2.5} (value per metric ton)
2025	\$237	\$20,300	\$975,500

Note:

These multipliers are typically updated annually and values are provided for the next 30 years. This example is from the current year in 2022 dollars. Users are encouraged to visit the U.S. DOT website for the most up-to-date guidance and values: <https://www.transportation.gov/mission/office-secretary/office-policy/transportation-policy/benefit-cost-analysis-guidance>

Table Source: U.S. DOT 2023a.

Advanced Application Case Study: Emissions Benefits – Portland, OR

Following are the steps used to calculate emissions benefits in a case study where the team has Advanced Applications. The steps for the Advanced Application Method may depend on the outputs available from the behavior-change analysis.

In this example, based on the *Metro Active Transportation: Return on Investment Study* (ATROI), modeled results included changes in bicycle trips, bicycle miles traveled, auto trips, and auto VMT (Metro 2022). (See *Appendix K* for a description of this study.) If a change in VMT is available, changes in trips and trip lengths are not needed.

1. Estimate change in auto VMT.

If a travel demand model is run for the project analysis, output can be summarized to estimate the change in VMT as a result of the project. This is done by multiplying link-level traffic volumes by the length of each link to get VMT by link, and summing results, for both the “no-project” and “project” model runs.

In the ATROI example, the most impactful project—the Eastbank Esplanade, a 1.5-mile off-street path and bridge in Portland, OR—was modeled to result in a reduction of 2,729 daily VMT.

2. Run MOVES to obtain emissions.

The U.S. EPA MOVES model may be run with a combination of default and local or project-specific inputs as available. When working with travel demand model output covering a network of roadway links, it is recommended to run MOVES at the county scale rather than the project scale, so that network effects can be evaluated (project-scale runs require each travel link to be modeled individually). Aggregate inputs are provided for the modeled area.

The following inputs can be derived from travel demand model output:

- **VMT by vehicle type** for the five Highway Performance Monitoring System (HPMS) vehicle classes included in MOVES. For active transportation projects it can be assumed that only light-duty vehicle VMT (type 25) is affected by the project.
- **Speed distributions.** The speed distribution file includes the percentage of vehicle-hours of travel in 16 speed bins, for each source type (MOVES includes 13 source types which are subdivisions of vehicle types) and road type (MOVES includes 4 road types). It is likely that the impact of any individual active transportation project on the speed distribution will be negligible, and a single speed distribution (which may be taken from existing MOVES inputs if available) can be used for both no-project and project conditions, meaning emissions will only be affected by VMT changes. For an entire network or program of improvements, it is possible that meaningful changes in speed distributions will be observed. Speed distributions can be developed from model output based on link-level travel times and volumes.
- **Road type distributions.** The road type distribution file includes the percentage of VMT, for each source type, taking place on four road types: urban restricted and unrestricted access, and rural restricted and unrestricted access. It is likely that the impact of any

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individual active transportation project on the road type distribution will be negligible, in which case a pre-existing road type distribution file may be used if available. The road type distribution can also be constructed from travel demand model output based on volumes, lengths, and road type information for each link.

Other inputs, such as source type populations, age distributions, and meteorological data, are unrelated to changes in travel activity and pre-developed input files may be used for both the “no-project” and “project” runs.

3. Post-processing and summarizing of model results.

MOVES outputs must be post-processed and summarized to report total emissions for the “no-project” and “project” scenarios. Note that preparing inputs for, running, and post-processing output from the MOVES model requires training and advanced skills that are not detailed in this guide. Users are referred to MOVES documentation and guidance for more detail on preparing inputs for and running MOVES. (See documents available on U.S. EPA’s MOVES website [U.S. EPA 2024a], as well as *NCHRP Web-Only Document 210: Input Guidelines for Motor Vehicle Emissions Simulator Model* [Porter et al. 2015]).



APPENDIX J

Augmented Methods for Enhanced Travel Options Benefit Analysis

The Basic approach applies the benefit only to the length of the gap-closing facility itself. However, in many cases the value of closing a gap extends to providing access to better existing facilities on either side. For example, imagine two stretches of multiuse path that end on either side of a natural or man-made barrier. A gap-closing facility might be very short, generating only a small amount of bicycle miles traveled (BMT) or walk miles traveled (WMT), but it might “unlock” the two connected trail segments, providing many additional miles of high-quality travel for new users. The same could be true for a gap in on-street low-stress facilities as well.

Users that can conduct a GIS least-cost path analysis and have origin-destination (OD) flow estimates can modify the Basic Method calculation as follows to better capture the additional network benefits of a gap closure for new users:

1. Estimate number of additional bike or walk trips on gap facility (after gap is filled, using any of the provided behavior-change methods).
2. Identify additional links directly connected by the gap closure.
Note: connected links should only include those with higher quality facilities versus alternative route options in area.
3. Select all OD pairs within a reasonable bicycling or walking distance of the gap.
 - a. ODs may be TAZs, census geographies, or any similar aggregation.
 - b. Reasonable distance may be based on local or national (e.g., NHTS) trip distance data.
4. Weight each OD pair by expected share of bicycle or walking trips in area.
 - a. If bicycle and walk OD flows exist, use those; if not, apply an appropriate trip distance distribution (from NHTS or local sources) to weight total OD flows before apportioning.
5. Solve least-cost paths between each pair of ODs (guidance provided in *Appendix C*) both:
 - a. Before: without gap closure.
 - b. After: with gap closure.
6. For gap closure link (or links) and connected links, calculate number of additional weighted paths after the gap closure that included each of the links.
7. For each connecting link, multiply the gap user count from Step 1 by the ratio of additional connected link paths to additional gap link paths. This provides an estimate of diversion to higher quality connecting links due to the gap closure.

Figure J-1 shows a graphical representation of this process. Imagine a gap closure with an estimated 100 existing and 100 additional bike trips. Using our network analysis method, we find 1000 additional weighted least-cost paths routing across the gap link, while connected link A sees an increase of 500 paths and link B an increase of 250. Of the 100 additional bike trips using the gap facility, 50 are expected to now use least-cost paths along connected facility A,

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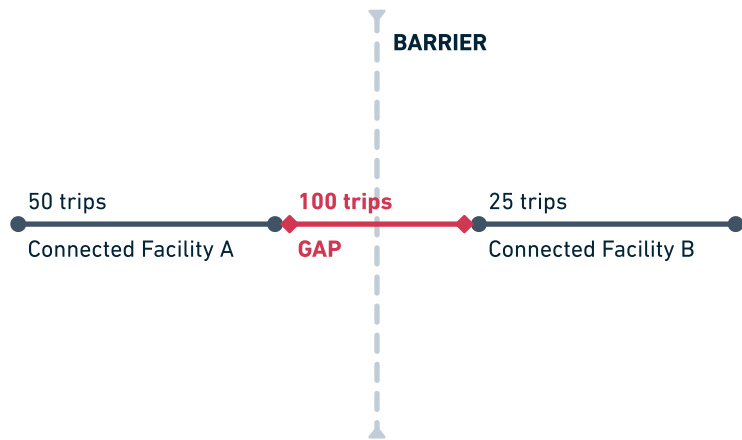


Figure J-1. Additional gap trips accessing connected facilities.

and 25 along connected facility B. The benefit calculation for additional users would thus change from $0.5 \times 100 \times \text{amenity multiplier} \times \text{gap length}$ to:

$$\text{Additional Users Benefit} = 0.5 \times \left(\begin{aligned} &100 \times \text{length}_{\text{gap}} \times \text{amenity multiplier}_{\text{gap}} + 50 \times \text{length}_A \\ &\times \text{amenity multiplier}_A + 25 \times \text{length}_B \times \text{amenity multiplier}_B \end{aligned} \right)$$

The total enhanced travel options benefits (for existing and new users) calculation becomes:

$$\begin{aligned} \text{Total Benefit} = & (100 + 0.5 \times 100) \times \text{length}_{\text{gap}} \times \text{amenity multiplier}_{\text{gap}} \\ & + (0.5 \times 50) \times \text{length}_A \times \text{amenity multiplier}_A \\ & + (0.5 \times 25) \times \text{length}_B \times \text{amenity multiplier}_B \end{aligned}$$

Total benefits will always be equal to or greater using the Augmented Method, reflecting the use of additional data and analytic support to better capture network effects of a gap closure. Maximum benefit extent should still follow U.S. DOT guidance and not exceed NHTS average trip length for biking or walking without further local justification. The “rule of half” multiplier still applies, since the additional network benefits accrue to new users.

Table J-1 can be used as a template for reporting of enhanced travel options benefits.

Table J-1. TEMPLATE for summary of enhanced travel options benefits.

Estimated Daily Bike Trips on Gap	Daily BMT on Gap and Connected Facilities	Total Bicycle Enhanced Travel Options Benefits	Estimated Daily Pedestrian Trips on Gap	Daily WMT on Gap and Connected Facilities	Total Walk Enhanced Travel Options Benefits

Advanced Application Case Study: Enhanced Travel Options Benefits – Portland, OR

A more complete accounting of travel option benefits will typically rely on a regional travel demand model. The magnitude of travel option benefits depends on both the scale of the gap-closing improvement and the quality and cost of existing route and mode options. Regional models can capture impacts on the overall improvement experienced by travelers, typically as a “logsum,” the expected maximum utility (or value) of the best option available to them. Only regional models that include walk or bike models sensitive to network improvements are suitable for this purpose, however, and until recently few U.S. regions had such models available. Most existing Advanced Application travel option analyses were conducted in Europe. More recently, a small but growing collection of U.S. regions have developed regional models capable of estimating travel option value from bicycle or, less frequently, walk-focused improvements. Logsum-based estimates typically include both travel time savings and quality (or facility amenity) improvements, but it is possible to partial out the travel time savings component. In many cases, travel time savings for active travel projects can be negative, since improved facilities often draw users to longer, more comfortable routes.

Background

Metro–Portland, Oregon’s MPO–applied their regional travel demand model to estimate travel option benefits generated by previously completed gap-filling bicycle infrastructure projects. The application was conducted as part of a larger research effort, the ATROI study to support future Regional Flexible Funds Allocation process (Metro 2022). The travel option benefit calculations were performed for three selected bicycle projects as a proof-of-concept exercise.

Metro’s trip-based bicycle travel model was developed in partnership with Portland State University based on revealed preference bicyclist route choice research (Broach et al. 2012). The agency maintains detailed present and future year routable bicycle networks. Maximum utility paths are generated for all TAZ pairs. Route choice utilities then enter the mode and destination choice models. As a result, a number of bicycle facility and intersection attributes enter into modeled decisions of whether and where people are likely to bike. Bicycle trips are then assigned back to the detailed network, generating expected miles of bicycle travel on specific facilities.

From a list of completed projects funded in part by regional flex funds, three were selected for detailed analysis, spanning a range of expected project impacts from local to regional significance:

- **Eastbank Esplanade.** A marquee project that added 1.5 miles of off-street multiuse path to the east side of the Willamette River across from downtown, providing a key north-south connector through an area hemmed in by a freeway, highway, and active freight rail tracks, as well as the river itself. Included a floating portion where sufficient right of way could not be obtained, and a new lower deck on a lift bridge crossing the river to downtown Portland.

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- **SE 17th Avenue Path.** Converted narrow, conventional bike lanes into an off-street multiuse path along a 1-mile stretch of two-lane arterial (35 mph) serving as a connector between Portland and the City of Milwaukie about 5 miles from the Portland city center. No reasonable parallel alternatives existed in the corridor, which is constrained by multiple highways, train tracks, a golf course, and a river. Future years benefited from complementary projects near gap closure.
- **NE Cully Cycle Track.** Converted an intermittent bike lane to raised, protected bike lanes on a 0.5-mile stretch of minor, two-lane arterial street (30 mph) serving residential and commercial areas about 6 miles from the city center. Parallel routes were available, but Cully was a diagonal street that often provided a shortest path route and connected multiple east-west bikeways.

The ATROI study leveraged the regional bike model to calculate travel option benefits for specific project facilities. The following sections describe the specific procedures used and present some results from the application.

Method

Modeling staff created and ran the procedures outlined in Table K-1 for each of the three selected projects. ATROI analysis used version “Kate” of the regional travel demand model (Metro 2020). Most steps are completed within the usual regional model run flow. Custom post-processing scripts were created to calculate the travel option benefits based on standard model outputs.

Figure K-1 shows model representations of the Eastbank Esplanade project. The shift in bicycle traffic is apparent, reflecting shifts to lower cost routes (and, to a lesser extent, mode shifts). The availability of improved route options forms the basis for calculation of travel option benefits.

Results

Table K-2 provides key standard model outputs, and Table K-3 provides travel option benefit estimates.

Travel option benefits varied considerably by gap, based on the scale of each project, the network leverage, and the presence and quality of existing routes. Eastbank Esplanade, the most impactful project, had estimated travel option benefits averaging over \$180,000 annually in 2010 dollars, or about 70% of the value of health benefits estimated in the same study.

Benefit distribution to the three income groups typically tracked the group’s share of trip making in the region. Because lower-income households tend to take fewer trips, the share of benefits to the lowest income group lagged their share of households in the region.

Discussion

The application of a regional travel model that includes bicycle facility preferences was successful in estimating significant travel option benefits for three network gap-filling projects. Results from the larger ATROI analysis agreed with limited existing research that suggests travel option benefits may be substantial, potentially similar in magnitude to health benefits from increased physical activity.

Table K-1. ATROI travel option benefit calculation steps.

Step	Description
Step 1	“Roll back” each project in base year and horizon year model networks (2015 and 2040) by removing links or modifying facility types as needed.
Step 2	Update route utilities and shortest path distances used as model inputs with “next best” routes without project.
Step 3	Updated utilities input into Destination Choice model, updating relative attractiveness of travel between TAZs w/o project. ¹
Step 4	Updated utilities input into Mode Choice model, estimating new mode splits w/o project. ¹
Step 5	Compare resulting travel options between each pair of TAZs by calculating increases in Destination Choice “logsums” with project versus without for both base and future year. ²
Step 6	Convert to per trip dollar benefits using marginal utility of income. ³
Step 7	Aggregate over trip tables (w/ project) to calculate daily benefits, by income group.
Step 8	Apply linear interpolation of benefits for years between 2015 and 2040 model runs.
Step 9	Multiply daily benefits by 365 to approximate annual benefits. ⁴
Step 10	Apply various time discount rates (3%, 5%, 7%) to past and future year benefits to arrive at a stream of annual travel option benefit amounts in 2010 dollars, by income group. U.S. DOT guidance recommends applying a single discount rate to all benefits and costs.

Notes:

¹ Both Destination Choice and Mode Choice models segment households by income category (\$25,000 “low”, \$25,000 to \$100,000 “augmented”, and \$100,000+ “high,” all in 2010 dollars).

² Logsums represent the expected maximum utility in a way that considers the entire package of modal options (driving, transit, walking, biking) for each potential trip. This allows measurement of the combined impact of route and mode shifts due to the project (i.e., without the project, some existing riders would shift to other [worse] routes, while others would shift to other [worse] mode options). Since “utility” is a unitless measure, it captures tradeoffs between disparate trip attributes like transit fares, travel times, and facility comfort.

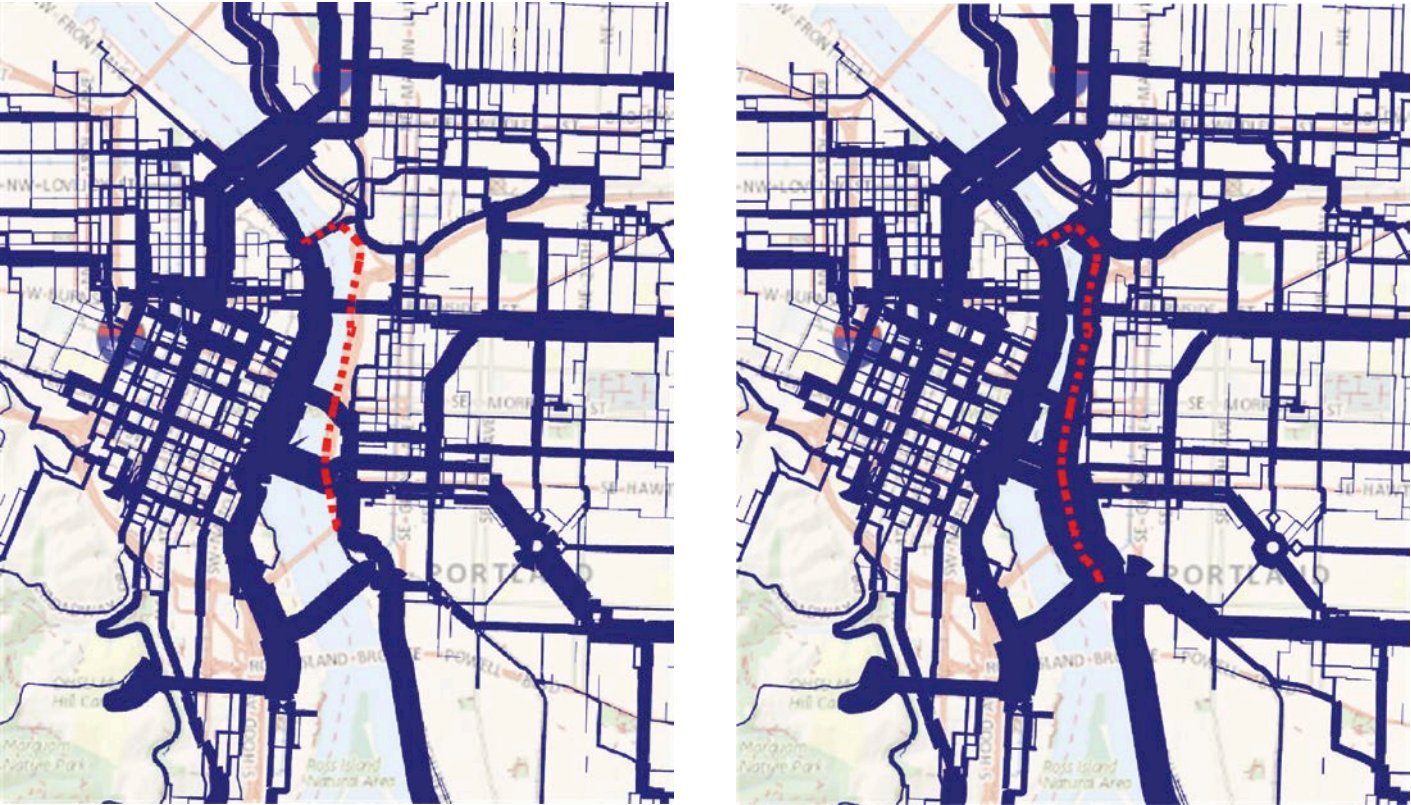
³ Marginal utility of income represents an individual’s exchange rate between travel utility and income. ATROI analysis used the (negative) of the appropriate monetary single occupancy vehicle operating cost to derive the values, which varied by trip purpose and household income group.

⁴ The regional model targets a midweek May or October day, about average seasons for cycling in Portland. While an argument could be made for modifying the annualizing factor to account for lower commuting rates on weekend days, the Portland region lacked data to calculate the appropriate reduction in rates. The use of a straight multiplier was also used to offset the fact that many unmodeled, purely recreational trips excluded from the model estimates likely benefit from improvements also.

A number of challenges and limitations were noted:

- Regional travel demand models are resource intensive, and each project took 1–2 full days of a modeler’s time.
- By design, the travel model captures only a subset of bicycling:
 - Purely recreational trips are omitted.
 - Metro’s school travel model runs separately and is not sensitive to bicycle improvements.
- Intrazonal (within a single TAZ) travel is only crudely represented; somewhat of a problem for bicycling (around 14% of regional bike trips are intrazonal) but would loom larger for pedestrian modeling.
- A key factor in monetizing travel option benefits, the marginal utility of income, is estimated based on drivers’ sensitivity to costs, and the extent this applies to non-driving populations (and non-driving trips) is the subject of some debate; this is an area where more research is needed.
- While using logsums to capture generalized travel cost savings has advantages, it also creates potential overlap with other benefit categories to the extent choices to bicycle include considerations of health, safety, environmental, and other benefits; there is a risk of double counting (or at least overcounting).

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Source: Portland Metro.

Figure K-1. Model representation of bicycle demand before (left) and after (right) gap-filling project shown in red.

Table K-2. Selected model outputs from Metro ATROI application (base year).

Project	Description	Estimated New Daily Bike Trips	Estimated Re-Routed Daily Bike Trips	Est. Add'l Daily Miles Bicycled
Eastbank Esplanade	1.5 mi off-street path and bridge near central business district.	883	11,487	7,052
SE 17th Avenue Path	1 mi off-street path near edge of city; poor existing alternatives.	79	2,675	648
NE Cully Cycle Track	0.5 mi cycle track about 6 mi from central business district.	10	239	54

Table K-3. Selected travel option benefit estimates from Metro ATROI application.

Average Total Annual Benefits (2010\$, 5% Discount Rate, 40-Year Horizon)				
	Low Income	Medium Income	High Income	Total
% of regional households	22%	54%	24%	100%
% of regional trips	15%	54%	31%	100%
Project				
Eastbank Esplanade	\$23,885	\$94,581	\$62,459	\$180,925
SE 17th Avenue Path	\$4,097	\$16,956	\$9,799	\$30,851
NE Cully Cycle Track	\$266	\$864	\$398	\$1,528

Comparison with Basic Method

For this guidebook, we compared results with the Basic Method from the U.S. DOT BCA Guidance (see Enhanced Travel Options Benefits in *Chapter 4: Step 3. Estimating Benefits*). The regional model produces estimates of bicycling on the project facility and compares tradeoffs among all available modes and routes. The resulting benefit estimates were significantly lower than the U.S. DOT guidance (Table K-4).

Differences in methodology and local context likely explain the differences. U.S. DOT guidance does not explicitly consider the quality of existing alternatives and assumes that travel time is equivalent between old and new routes (users can separately account for travel time changes, if they wish). The ATROI study and other international studies that have applied travel models consider the marginal improvement per trip considering all alternative travel options. They also naturally allow for increases in travel time as cyclists divert to an improved route, partly offsetting the benefit of a higher quality facility. Finally, to monetize changes in expected route utility, the ATROI study used a marginal utility of income drawn directly from cost coefficients in the regional destination choice model, instead of standard values of travel time savings. Context may also play a role. The ATROI study was applied to just three locations that each had varying but reasonably dense networks of alternative routes.

No comparable existing research on pedestrian travel option benefits was available. Given the similarity in method between U.S. DOT bike and pedestrian facility guidance, we expect that the provided values are similarly higher than an approach that more fully considers available alternatives.

Table K-4. Daily use and benefit estimates example.

Project	Description	Estimated Daily Bike Trips	Daily BMT on Facility	ATROI Benefit per Daily BMT (2022\$*)	U.S. DOT Benefit per Daily BMT (2022\$)
Eastbank Esplanade	1.5 mi off-street path and bridge near central business district; parallel path available across river for some trips	12,451	9,533	\$0.16	\$1.97
SE 17th Avenue Path	1 mi off-street path w/ at-grade crossings near edge of city; poor existing alternatives	2,754	2,754	\$0.11	\$1.57
NE Cully Cycle Track	0.5 mi cycle track about 6 mi from central business district; more direct than alternatives for some trips	249	125	\$0.10	\$1.86

Note:

* Inflated from original study using Consumer Price Index

Augmented Methods for Health Benefit Analysis

Unit of Measure and Data Needs

As with the Basic Methods, the Augmented Methods in this guide use premature deaths avoided at the population level as the unit of measure. This is because the methods are based on HEAT, which only considers the value of premature deaths avoided (mortality). The Advanced Applications rely on ITHIM, which considers a wider range of outcomes affected by physical activity, including various cardiovascular diseases, depression, dementia, diabetes, and certain cancers (morbidity). Additionally, the tools underlying methods presented here differ in their monetization of health benefits. HEAT estimates reducing mortality risk benefits for individuals, relying on the economic concept of value of a statistical life. Intermediate outputs of HEAT are communicated in simple terms of premature deaths avoided. ITHIM includes a monetization module based on avoided health care costs for treating specific chronic diseases. The tool produces an intermediate, non-monetized health outcome measure in terms of disability-adjusted life years avoided, where one disability-adjusted life year represents a cost equivalent to 1 year of full health.

Methods

1. **Estimate the number of new active trips and trips shifting from vehicular modes.**
 - a. Estimates of these numbers may be obtained using a variety of methods as described in the behavior-change section. Do not include trips that were due to route shifts or shifts from non-vehicular modes. This will be used to estimate additional time walking or biking and serve as an estimate of the affected population.
2. **Gather localized data.**
 - a. Local mortality rate (U.S. DOT suggests values from <https://wonder.cdc.gov/>),
 - i. For walking calculations: Find the all-cause mortality rate (deaths per 100,000 people) for the 20 to 74 age range
 - ii. For biking calculations: Find the all-cause mortality rate (deaths per 100,000 people) for the 20 to 64 age range
 - b. Local age profiles
 - i. For walking calculations: Find the percent of population within 20 to 74 age range
 - ii. For biking calculations: Find the percent of population within 20 to 64 age range
 - c. Local speed and trip distance profiles, if available
3. **Suggested fixed inputs.**
 - a. RR = relative risk of death in underlying studies from HEAT. These numbers quantify the risk of dying for people who walk or cycle regularly relative to people who do not.
 - i. Walking: 0.89
 - ii. Bicycling: 0.90

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- b. Reference level of walking or bicycling from HEAT, defining the level of walking or cycling used to establish the RR rates in 3a (i.e., a population walking 168 minutes per week is expected to have an 11% lower all-cause mortality rate, relative to a population that does not walk at all). (Reference values from HEAT: walking: 168 minutes walking per week; bicycling: 100 minutes cycling per week.) The relationships are assumed to be linear up to a cap (460 minutes of walking per week, 447 minutes of cycling, or about 60 minutes per day), so that, e.g., walking half the baseline rate yields a 5.5% protective benefit, walking twice the baseline rate yields a 22% protective benefit, and so on.
- c. Value of a statistical life: \$12,500,000 (2022 \$) from current U.S. DOT BCA Guidance, update or modify as needed. HEAT provides alternative values.
- 4. **Calculate additional minutes of walking or biking per week.**
 - a. May be calculated from estimated number of new weekly trips multiplied by reference value for average trip length (e.g., NHTS, local survey), using appropriate assumed mode travel speeds. Total additional minutes must be capped so that total minutes do not exceed mode-specific caps.
- 5. **Calculate additional annual deaths avoided.**
 - a. Bike: $\text{Additional Annual Deaths Avoided} = (1 - 0.9) \times \text{Additional Weekly Minutes Biking} / 100 \times \text{Mortality Rate} \times \text{New Weekly Bike Trips}$
 - b. Walk: $\text{Additional Annual Deaths Avoided} = (1 - 0.89) \times \text{Additional Weekly Minutes Walking} / 168 \times \text{Mortality Rate} \times \text{New Weekly Walk Trips}$
- 6. **Monetize the benefits.**
 - a. Multiply the number of additional annual deaths avoided by the value of a statistical life.
 - b. Reduce by age profile identified earlier. Multiply the result by the percentage of the population that is in the appropriate age range (20 to 74 for walking and 20 to 64 for biking). Use Table L-1 to report the monetized benefits.

Analysis Considerations and Limitations

These methods can be used across project scales, land-use contexts, and gap types. For both the Basic and Augmented Methods, the calculated benefits are restricted to reductions in mortality and do not consider morbidity impacts. The Basic approach relies on national data for mortality profiles and age distribution, and applies a constant value per trip. The Augmented Method uses local data, which will be particularly impactful in regions having mortality rates that differ from national averages, or at locations where walking and bicycling trip lengths differ from national averages, based on local or regional survey data or other localized sources. A limitation of both HEAT-based approaches is that they do not include the benefits of increased physical activity for the youngest and oldest adults (benefits are restricted to ages 20 to 74 for walking and 20 to 64 for bicycling due to the underlying research).

Table L-1. TEMPLATE for reporting health benefit summary.

Benefit Category	Value
Reduced mortality from walking	
Reduced mortality from biking ¹	

¹ Users are directed to the HEAT web tool (WHO 2023) for the most up-to-date methods and parameter values. E-bike and bike share options have recently been added.

Offsetting the increased data and technical requirements of the Advanced Application ITHIM-based method, potential advantages over the others include:

- Consideration of both mortality and morbidity benefits
- Option to include variability in outcome calculations
- Disaggregated disease burden and dose-response rates
- Can consider baseline physical activity levels and apply non-linear Dose-Response Functions
- Has been popular among U.S. health studies
- Can cover the benefits to all age groups, if supporting data available for all groups, since age-specific benefits are calculated before aggregation



APPENDIX M

Advanced Application Case Study: Health Benefits – State of Oregon

Background

In the report *Oregon Outdoor Recreation Metrics: Health, Physical Activity, and Value* (Rosenberger 2018), the Oregon State University College of Forestry estimated the public health benefits of outdoor recreation in Oregon using the ITHIM framework.

ITHIM has the capacity to estimate the impact of active transportation on public health arising from three sources: increased physical activity, changes to injury rates, and air pollution reduction. The Oregon study used ITHIM only to determine health benefits from physical activity.

Methods

The study used the Outdoor Recreation Health Impacts Estimator (OR Estimator) tool to estimate health benefits from the selected activities. The OR Estimator tool is built on the ITHIM model and adapted for Oregon specifically.

OR Estimator Inputs:

- The 2017 Oregon Resident Outdoor Recreation Survey provides information on the amount, type, and rate of physical activity among all Oregonians. The survey collected demographic and activity information from a stratified random sample of residents throughout the state. Data was collected on 30 different outdoor activities of at least moderate intensity.
- The Ainsworth Compendium of Physical Activities provides MET values (a standardized metabolic rate) for a variety of physical activities to quantify the intensity of the activity.

OR Estimator Process:

- Extrapolate survey data to calculate aggregate physical activity among all Oregonians participating in the selected activities.
- Use a built-in function to model the relationship between physical activity and reduced risk of certain chronic diseases such as heart disease, diabetes, and cancer.
- Convert reduced disease risk to cost of illness (COI) savings. This includes savings from costs of both direct healthcare and lost productivity.

Results

The study estimated that Oregonians participated in the tracked activities on 794 million separate occasions and expended 503 billion calories in 2017. This resulted in an estimated COI savings of 2–3.6% of all healthcare costs in Oregon, and 9–17% of costs involved in treating heart disease, cancer, diabetes, and depression specifically.

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The top two activities, by total COI savings, were walking on local streets or sidewalks and walking on trails. Data was collected for bicycling as well. Results for the biking and walking activities most related to active transportation are summarized in Table M-1. The study concluded that the social, economic, and physical environment of a community is the largest predictor of health.

Table M-1. Summary results for active transportation activities.

Activity	Total Participants	User Occasions	Total Calorie Expenditure	Total COI Savings
Walking on local streets/ sidewalks	2.7 million	312.7 million	117 billion	\$385–629 million
Walking on local trails/ paths	2.4 million	113.1 million	57.4 billion	\$71–125 million
Bicycling on local streets/ sidewalks	1.2 million	51.2 million	32.1 billion	\$47–78 million
Bicycling on paved trails	983,000	26.1 million	17.7 billion	\$15–18 million

Discussion

The application of the ITHIM was successful in modeling benefits from outdoor physical activity. The selected activities captured the most common activities associated with active transportation – biking and walking. The study cast a wider net than active transportation alone, capturing benefits from a variety of outdoor recreational activities. This case study demonstrates ITHIM could easily be applied to active transportation specifically as long as data are available on the amount of time the general population devotes to active transportation, what types, and at what intensity.



APPENDIX N

Augmented Methods for Local Economic Benefits Analysis

Unit of Measure

There have been many studies of the economic impacts of active transportation facilities, examining impacts of (1) increased spending by active transportation tourists; (2) increased business sales or spending based on mode or destination shifts of local travelers; and (3) impacts of facilities on property values (Table N-1).

These studies have used a variety of methods, most commonly visitor surveys; time-series analysis comparing business activity before and after an improvement and/or in improved vs. non-improved areas; and hedonic analysis of property values. (Hedonic analysis evaluates how one variable—in this case, property values—relates to a variety of other influencing variables, such as lot size, home size, age and condition of structure, and number of bedrooms. Proximity to bike or walk facilities or connectivity of local networks can be included as a variable in the analysis.) These studies have not found consistent impacts across regions or projects. As a result, there is little reliable information on which to base generalized estimates of the expected economic benefits of closing gaps in active travel networks.

An additional caution with the use of economic impact analysis is that increases in local economic activity may simply be shifted from elsewhere within the region or state, rather than representing net increases in economic activity. For example, people may spend more money in one neighborhood or community that is now more attractive, but less in another neighborhood or community that they visit less frequently. Also, there may be some overlap among different categories of local economic benefits: for example, commercial property values may reflect and capitalize other benefits such as business value, or increases in residential and commercial property values may reflect other benefits such as time savings or lower-stress access routes.

Data Needs

Table N-1. Data needs for Augmented and Advanced Application Method economic benefit estimation.

Key Data Item	Augmented Method	Advanced Application Method
Visitor spending	Estimates of increase in active travelers from behavior-change analysis. Share of traffic that is visitors (from studies of other similar facilities). Share of visitors that are encouraged to visit because of the improvement. Spending per visitor from other tourism studies.	Intercept surveys to determine (1) share of travelers who are visitors, (2) influence of project improvements on their decision to visit the area, and (3) spending per visitor. Location-based data to determine share of traffic that is visitor traffic.

(continued on next page)

N-2 Estimating Benefits of Closing Gaps in Active Transportation Networks: A Guide**Table N-1. (Continued).**

Key Data Item	Augmented Method	Advanced Application Method
Business activity – sales, jobs, wages	Discussion of % changes in sales, jobs, or wages observed in studies of similar projects in similar contexts.	Business sales tax receipts, job data, and/or wage data for the project area and control areas without improvements (time-series).
Property values	Discussion of % changes in property values observed in studies of similar projects in similar contexts.	Property values (comparable sales and/or assessed values) in the project area and in control areas without improvements (time-series).

Methods

The Augmented Method for estimating local economic benefits of closing gaps in the active transportation network involves applying factors from other studies to develop rough quantitative estimates of visitor spending, while describing other impacts qualitatively.

1. Describe the type of facility/gap closure, its land-use context, and its expected user characteristics:

- Is the increase in active travel from the project expected to be mainly local traffic, visitors, or a mix of both?
- To what degree are there businesses adjacent to the project that could benefit from pass-by walking and bicycling traffic?
- To what degree are there businesses in the vicinity of the project (not necessarily adjacent) that might benefit from increased visitor activity (e.g., restaurants or lodging)?
- To what degree are there residential properties directly served by the project? Is the project likely to be viewed as an amenity that improves property values?

2. Estimate the share of traffic that is visitors and the share of visitors that were encouraged to visit the area as a result of the improvement.

3. Estimate spending per visitor.

One Montana-based survey found that bicyclists between 2011 and 2013 spent between \$75 and \$102 per person per day (Nickerson et al. 2013).

A Washington State study used data from the U.S. Forest Survey study of spending patterns of visitors in national forests (Rosenberger et al. 2017) as shown in Table N-2.

Apply values for spending per visitor only to the people who traveled from elsewhere specifically to use the facility, not those using the facility near their home.

4. Estimate total spending impacts.

Compute: total spending impact = total traffic × share of traffic that is visitors × share of visitors that were encouraged to visit by the improvement × spending per visitor.

Table N-2. Average visitor spending (2019 dollars).

Type of Trip/Visitor Origin	Local	Nonlocal
Day trip	\$38	\$73
Overnight, staying with National Forest	\$191	\$264
Overnight, staying outside National Forest	\$291	\$636

Source:

Rosenberger et al. 2017. Researchers in a Washington State study applying these data to recreational trails assumed that only 10% of local day visitors to trails in the state spend the amount estimated (\$37.65), while the rest (90%) are using trails near their home and perhaps not making any purchases during their activity (ECONorthwest 2019).

5. **Summarize economic benefit findings for similar projects where business and/or property value benefits were studied.**

See “assemble your data” for potential sources.

Look for a project, or preferably multiple projects, with similar land-use conditions.

Document the findings of the study or studies in terms of which benefits were measured and what impacts were found (business sales, wages, property values, etc.). Different studies or projects may show a range of impacts and these should be reported to illustrate the uncertainty in findings.

Describe ways in which the proposed project might be expected to have similar or different impacts.

A sample benefits distribution table, Table N-3, is shown here and can be used as a template for reporting economic benefits.

Table N-3. Template for reporting economic benefits.

Total Traffic	Traffic Share		Spending per Person	Spend		Total Spend
	Visitor (%)	Local (%)		Visitor	Local	

 APPENDIX O

Advanced Methods for Local Economic Benefits Analysis

The Advanced Method involves preparing an analysis of the economic impact of the gap closure project. This appendix contains the general steps needed for such an analysis. Liu and Shi (2020) provide examples of how the economic impacts of bicycle and pedestrian street improvements in the United States have been evaluated previously using multiple data sources and analytical approaches.

1. Describe the type of facility/gap closure, its land-use context, and its expected user characteristics.

Is the increase in active travel from the project expected to be mainly local traffic, visitors, or a mix of both?

To what degree are there businesses present adjacent to the project that could benefit from pass-by walking and bicycling traffic?

To what degree are there businesses in the vicinity of the project (not necessarily adjacent) that might benefit from increased visitor activity? (E.g., restaurants or lodging)

To what degree are there residential properties directly served by the project? Is the project likely to be viewed as an amenity that improves property values?

2. Determine the types of benefits you want to estimate.

You may choose to focus on one or more types of benefits—visitor spending, local business activity, and/or property values—depending upon the expected effects of the project and the benefits of most interest to the community and decision-makers.

The types of benefits you want to measure may also be determined in part by the availability of data and feasibility of additional data collection.

3. Identify your project (experiment) area, control area(s), and study time frames.

Determine whether you are going to:

- Conduct measurements of the effects of other similar projects that have already been completed, to inform decision-making on future projects; or
- Conduct measurements specific to the project in question.

If you are evaluating other projects, identify projects of similar type, similar roadway and land-use contexts, and similar expected usage characteristics.

Identify “control areas” that are similar to the project study area(s) (land use, demographics, roadway, etc.) except that active transportation improvements have not been made.

The geographic scope of the study and control areas may be determined in part by the scale at which data are available (e.g., individual businesses or properties, CBGs, ZIP codes); therefore, this step may require some iteration with the next step.

At least 3 years of data before and after an improvement is ideal. Consider the possibility of temporary construction-related impacts while the improvement is being constructed.

O-2 Estimating Benefits of Closing Gaps in Active Transportation Networks: A Guide**4. Identify your data sources and define your measurement methods.**

Table O-1 provides examples of potential data sources and their advantages and limitations.

Your preferred data sources may be affected by the project budget as well as the desired information, timeframe, and geography of analysis.

The recommended measurement approach for observed data (employment, sales, etc.) is to compare time-trends before and after project completion, ideally for a period of at least 3 years before and after, for both the project location and control locations.

Table O-1. Potential sources of economic impact data.

Data Source	Information Collected	Geography	Advantages	Limitations
Intercept surveys of people using active travel facility and/or parallel roadway or visiting local businesses	Where they came from (local vs. visitor), impact of facility improvement on travel and spending patterns	Location-specific, customizable	Customizable in terms of data obtained	Can be costly to implement; relies on self-reporting of behavior changes
Business surveys of business owners	Actual or perceived changes in retail traffic and sales volumes, other changes related to facility improvement (e.g., hiring, property improvements)	Location-specific, customizable	Customizable in terms of data obtained	Can be costly to implement; businesses may not be willing to share data
Street observations	Number of active business establishments by type; condition of business (qualitative)	Location-specific, customizable	Provides observed data along facility	Condition assessment is qualitative
Location-based data	Local vs. nonlocal (visitor) origin (home base) of travelers	Location-specific, customizable	May be cheaper than conducting intercept surveys	Requires custom queries. Only provides information on travel patterns, not spending, behavior-change factors, etc.
Sales tax filings from state tax office	Sales receipts/volumes	Establishment	Potentially customizable geographies and time frames	Availability may be limited by confidentiality restrictions; different states may have different policies on sharing
Multiple listing service (MLS)	Value of sold properties	Parcel level	Best to evaluate property values based on comparable sales	May not be enough sales to draw conclusions
City or county tax assessors	Assessed value of properties	Parcel level	Much larger data set than sales	Assessed values are inferior to sales values for inferring facility impacts
Census LEHD	Number of jobs (by sector, wage, education)	Census block	Free dataset with granular detail and annual updates	Lack of industry detail, data distortion at small geographies, time lags
Quarterly Census of Employment and Wages (QCEW)	Wages, employment	Establishment	Micro-level data updated quarterly	May be difficult to gain access due to confidentiality issues
Proprietary establishment & employment data sets	Business establishments by type, jobs, revenues	Establishment	More detailed than public data sets	Expensive

Statistical tests can be applied to evaluate whether differences among locations and/or years are significant.

Intercept and/or business surveys can be used for either a snapshot of stated conditions, or can be collected before and after the improvement. “After” surveys should be conducted at least 1 year after completion to allow changes in travel and spending patterns to stabilize.

5. Develop and execute a data collection plan.

Who will collect and analyze the data? Do you need to procure contractor assistance or schedule time for in-house staff? Can you use creative methods to reduce costs and build local capacity, such as working with a local college or university to develop class projects?

What is the budget/estimated cost?

What is the schedule?

What procedures are in place for quality control/assurance?

What approvals might be needed (e.g., if surveying is to be conducted on private property, or to obtain tax data from a state agency)?

6. Execute the data collection plan and collect, analyze, and report on your data.

The following reference gives an example of setting up an economic impact study for an active transportation facility:

An example for New York Avenue in Washington, DC:

Focus is on business activity along New York Avenue between Mt. Vernon Square and Florida Ave., before and after a complete streets project with bike lanes and traffic calming.

Data sources selected:

1. Retail sales (from District tax receipts).
2. Surveys of business owners, to gather before and after perceptions and validate quantitative findings from retail sales data.
3. Street observations to count number of active establishments by type.
4. Three comparison corridors without complete streets improvements are selected for comparing retail sales data (business surveys will not be conducted in these corridors).

Data collection timeframe:

- Retail sales: 3 full years before improvement, 3 full years after improvement (preliminary findings will be reported after 1 year).
- Street observations: 1 year before and 1, 2, and 3 years after improvements are completed.
- Business surveys: 1 year after improvements are completed.



References

- Alta (Alta Planning + Design). 2022. *Great Springs Project Trails Plan*. Prepared for Great Springs Project. <https://greatspringsproject.org/vision>.
- Alta. 2021. *Economic Benefits of the Great Springs Trail*. Prepared for Great Springs Project.
- Alta. 2015. *Memphis Gets Moving: Health & Economic Impacts of Building the Wolf River Greenway*. Prepared for Wolf River Conservancy.
- American Planning Association. 2020. Urban and Regional Planning Resources. <https://github.com/APA-Technology-Division/urban-and-regional-planning-resources>.
- ARC (Atlanta Regional Commission). 2024. Air Quality. <https://atlantaregional.org/natural-resources/air-quality/air-quality/>.
- Avelar, R., K. Dixon, S. Ashraf, A. Jhamb, and B. Dadashova. 2021. *Development of Crash Modification Factors for Bicycle Lane Additions While Reducing Lane and Shoulder Widths*. Report No. FHWA-HRT-21-013. FHWA, Office of Safety Research and Development. <https://www.cmfclearinghouse.org/detail.php?facid=10738>.
- Bremmer, D., and J.H. Bryan. 2008. “Bridging the Gap between Agencies and Citizens: Performance Journalism as a Practical Solution to Communicate Performance Measures and Results.” *Transportation Research Record* 2046(1), 20–29. <https://doi.org/10.3141/2046-03>.
- Broach, J., E. Rose, M. McDonald, and J. Sherman. 2022. *Metro Micromobility Dashboard Project*. Draft Final Report. Submitted to Oregon Department of Transportation.
- Broach, J., and J. Dill. 2017. “Bridging the Gap: Using Network Connectivity and Quality Measures to Predict Bicycle Commuting.” Paper presented at the 96th Annual Meeting of the Transportation Research Board.
- Broach, J., and J. Dill. 2016. “Using Predicted Bicyclist and Pedestrian Route Choice to Enhance Mode Choice Models.” *Transportation Research Record* 2564(1), 52–59. <https://doi.org/10.3141/2564-06>.
- Broach, J., and J. Dill. 2015. “Pedestrian Route Choice Model Estimated from Revealed Preference GPS Data.” Paper presented at the 94th Annual Meeting of the Transportation Research Board.
- Broach, J., J. Dill, and J. Gliebe. 2012. “Where Do Cyclists Ride? A Route Choice Model Developed with Revealed Preference GPS Data.” *Transportation Research Part A: Policy and Practice* 46(10), 1730–1740. <https://doi.org/10.1016/j.tra.2012.07.005>.
- Broach, J., J. Gliebe, and J. Dill. 2010. “Calibrated Labeling Method for Generating Bicyclist Route Choice Sets Incorporating Unbiased Attribute Variation.” *Transportation Research Record* 2197(1), 89–97. <https://doi.org/10.3141/2197-11>.
- CARB (California Air Resources Board). n.d. MSEI – Modeling Tools. Accessed April 10, 2024. <https://ww2.arb.ca.gov/our-work/programs/mobile-source-emissions-inventory/msei-modeling-tools>.
- Castiglione, J., M. Bradley, and J. Gliebe. 2015. *SSHRP2 Report S2-C46-RR-1: Activity-Based Travel Demand Models: A Primer*. Transportation Research Board of the National Academies, Washington, DC. <https://doi.org/10.17226/22357>.
- Claros, B., A. Bill, and D.A. Noyce. 2022. “High Injury Network: City of Madison, Wisconsin.” International Conference on Transportation and Development.
- Claros, B., M. Chitturi, A. Bill, and D.A. Noyce. 2020. “Roadway Safety Management in Small Municipalities.” *Transportation Research Record* 2674(3), 341–354. <https://doi.org/10.1177/0361198120910147>.
- Committee of the Transport Access Manual. 2020. *Transport Access Manual: A Guide for Measuring Connection between People and Places*. https://ses.library.usyd.edu.au/bitstream/handle/2123/23733/COTAM_Nov2020_GoldMaster_r001.pdf?sequence=3&isAllowed=y.
- Decisio and Transaction Management Centre. 2012. *Maatschappelijke kosten-batenanalyse van de fiets*. [Social Cost Benefit Analysis of the Bicycle]. <https://decisio.nl/wp-content/uploads/MKBA-Fiets.pdf>.
- Deenihan, G., and B. Caulfield. 2014. “Estimating the Health Economic Benefits of Cycling.” *Journal of Transport and Health* 1(2), 141–149.

REF-2 Estimating Benefits of Closing Gaps in Active Transportation Networks: A Guide

- Dill, J., and N. McNeil. 2013. "Four Types of Cyclists? Examination of Typology for Better Understanding of Bicycling Behavior and Potential." *Transportation Research Record* 2387(1), 129–138. <https://doi.org/10.3141/2387-15>.
- Dill, J., N. McNeil, A.N. Pelaez, S. Kothuri, C. Monsere, J. MacArthur, K. Kim, S. Brodie, and F. Proulx. 2021. *AASHTO Council on Active Transportation Research Roadmap: Research Review*. Final deliverable for NCHRP Project 20-123. Transportation Research and Education Center at Portland State University and Toole Design Group. <http://onlinepubs.trb.org/onlinepubs/nchrp/docs/NCHRP20-123-02AASHTOCATResearchReview.pdf>.
- Dixon, K., R. Avelar-Moran, and M.M. Seyedeh. 2023. *Developing Crash Modification Factors for Separated Bicycle Lanes*. Report No. FHWA-HRT-23-078. Federal Highway Administration. Available from: <https://www.cmfclearinghouse.org/detail.php?facid=11303>.
- ECONorthwest. 2019. *Economic, Environmental, & Social Benefits of Recreational Trails in Washington State*. Washington State Recreation and Conservation Office.
- FEMA. 2023. Benefit-Cost Analysis. <https://www.fema.gov/grants/tools/benefit-cost-analysis>.
- FHWA. 2024. Zero Deaths and Safe System. <https://highways.dot.gov/safety/zero-deaths>.
- FHWA. 2023. *Trails as Resilient Infrastructure Guidebook*. FHWA-HEP-24-007. https://www.fhwa.dot.gov/environment/recreational_trails/publications/trails-resilient-infrastructure-guidebook.pdf.
- FHWA. 2022a. *Improving Safety for Pedestrians and Bicyclists Accessing Transit*. FHWA-SA-21-130. https://highways.dot.gov/sites/fhwa.dot.gov/files/fhwasa21130_PedBike_Access_to_transit.pdf.
- FHWA. 2022b. CMAQ Emissions Calculator Toolkit. https://www.fhwa.dot.gov/environment/air_quality/cmaq/toolkit/.
- FHWA. 2018a. *Guidebook for Measuring Multimodal Connectivity*. FHWA-HEP-18-032. https://www.fhwa.dot.gov/environment/bicycle_pedestrian/publications/multimodal_connectivity/.
- FHWA. 2018b. *Guide for Improving Pedestrian Safety at Uncontrolled Crossing Locations*. FHWA-SA-17-072.
- Fitch, D.T., S. Kamalapuram, M. Favetti, and S.L. Handy. 2022. *Caltrans Active Transportation Benefit-Cost Tool*. Technical Documentation, Institute for Transportation Studies, University of California, Davis, Version 0.1.0. Last updated July 31, 2022.
- Fitzpatrick, K., M.J. Cynecki, M.P. Pratt, E.S. Park, and M.E. Beckley. 2019. *Evaluation of Pedestrian Hybrid Beacons on Arizona Highways*. Report No. FHWA-AZ-19-756. Arizona Department of Transportation. Phoenix, Arizona. Available from: <https://www.cmfclearinghouse.org/detail.php?facid=10607>.
- Goughnour, E., D. Carter, C. Lyon, B. Persaud, B. Lan, P. Chun, I. Hamilton, and K. Signor. 2018. *Safety Evaluation of Protected Left-Turn Phasing and Leading Pedestrian Intervals on Pedestrian Safety*. Report No. FHWA-HRT-18-044. Federal Highway Administration. Available from: <https://www.cmfclearinghouse.org/detail.php?facid=9906>.
- Guo, Y., and Y. Zhang. 2021. "Understanding Factors Influencing Shared E-Scooter Usage and Its Impact on Auto Mode Substitution." *Transportation Research Part D: Transport and Environment* 99. <https://doi.org/10.1016/j.trd.2021.102991>.
- Haas, P.M., C. Makarewicz, A. Benedict, and S. Bernstein. 2008. "Estimating Transportation Costs by Characteristics of Neighborhood and Household." *Transportation Research Record* 2077(1), 62–70. <https://doi.org/10.3141/2077-09>.
- Higgins, N., R. Basile, S. Van Hecke, J. Zissman, and S. Gilkeson. 2017. *NCHRP Web-Only Document 226: Data Visualization Methods for Transportation Agencies*. Transportation Research Board, Washington, DC. <https://doi.org/10.17226/24755>.
- Karner, A., R. H. M. Pereira, and S. Farber. 2024. "Advances and Pitfalls in Measuring Transportation Equity." *Transportation*. <https://doi.org/10.1007/s11116-023-10460-7>.
- Kuzmyak, J. R., J. Walters, M. Bradley, and K. M. Kockelman. 2014. *NCHRP Report 770: Estimating Bicycling and Walking for Planning and Project Development: A Guidebook*. Transportation Research Board of the National Academies, Washington, DC. <https://doi.org/10.17226/22330>.
- Lawrie, J., T. Cook, M. Meletiou, and S.W. O'Brien. 2004. *The Economic Impact of Investments in Bicycle Facilities: A Case Study of the Northern Outer Banks, Summary Report*. North Carolina Department of Transportation.
- Li, M., and A. Faghri. 2014. "Cost–Benefit Analysis of Added Cycling Facilities." *Transportation Research Record* 2468(1), 55–63. <https://doi.org/10.3141/2468-07>.
- Litman, T. 2022. *Evaluating Accessibility for Transport Planning: Measuring People's Ability to Reach Desired Services and Activities*. Victoria Transport Policy Institute.
- Liu, J.H., and W. Shi. 2020. *Understanding Economic and Business Impacts of Street Improvements for Bicycle and Mobility: A Multi-City Multi-Approach Exploration*. NITC-RR-1031-1161. April 2020. <https://doi.org/10.15760/trec.248>.
- Marshall, W.E., and N.N. Ferenchak. 2019. "Why Cities with High Bicycling Rates Are Safer for All Road Users." *Journal of Transport & Health* 13, 100539. <https://doi.org/10.1016/j.jth.2019.03.004>.

- McKinsey & Company. 2022. *Rural Rising: Economic Development Strategies for America's Heartland*. Accessed February 27, 2024. <https://www.mckinsey.com/industries/public-sector/our-insights/rural-rising-economic-development-strategies-for-americas-heartland>.
- McNeil, N., C.M. Monsere, and J. Dill. 2015. "Influence of Bike Lane Buffer Types on Perceived Comfort and Safety of Bicyclists and Potential Bicyclists." *Transportation Research Record* 2520(1), 132–142. <https://doi.org/10.3141/2520-15>.
- Mekuria, M., P. Furth, and H. Nixon. 2012. *Low-Stress Bicycling and Network Connectivity*. Mineta Transportation Institute. <https://transweb.sjsu.edu/research/Low-Stress-Bicycling-and-Network-Connectivity>.
- Mendonca, L.T., and M. Miller. 2007. "Crafting a Message that Sticks: An Interview with Chip Heath." *The McKinsey Quarterly*.
- Metro. 2022. *Metro Active Transportation: Return on Investment Study*. <https://www.oregonmetro.gov/sites/default/files/2022/05/15/Active-Transportation-Return-on-Investment-study-2022.pdf>.
- Metro. 2020. *2018 Kate v2.0 Trip-Based Travel Demand Model: Methodology Report*. <https://www.oregonmetro.gov/sites/default/files/2020/05/13/trip-based-travel-demand-model-methodology-report-May-2020.pdf>.
- Mills, K. 2021. "Biden Infrastructure Goals Would Be Achieved With More Strategic Investment in Walking and Biking." *Rails to Trails Conservancy*. <https://www.railstotrails.org/2021/april/13/biden-infrastructure-goals-would-be-achieved-with-more-strategic-investment-in-walking-and-biking/>.
- NACTO. 2022. *Making Bikes Count: Effective Data Collection, Metrics, & Storytelling*. https://nacto.org/wp-content/uploads/2022/03/Making_Bikes_Count_FINAL_March31-2022.pdf.
- NHTS. 2017. 2017 NHTS Datasets. <https://nhts.ornl.gov/downloads>.
- Nickerson, N.P., J. Jorgenson, M. Berry, J. Kwenye, D. Kozel, and J. Schutz. 2013. *Analysis of Touring Cyclists: Impacts, Needs and Opportunities for Montana* (Research Report No. 2013–17). Institute for Tourism and Recreation Research, University of Montana.
- Nordback, K., S. Kothuri, D. Johnstone, G. Lindsey, S. Ryan, and J. Raw. 2019. "Minimizing Annual Average Daily Nonmotorized Traffic Estimation Errors: How Many Counters Are Needed per Factor Group?" *Transportation Research Record* 2673(10), 295–310. <https://doi.org/10.1177/0361198119848699>.
- NYSDOT. 2015. *Electric Bicycle Share Feasibility Study*. Report Number 15–18. June 2015. <https://www.nyserda.ny.gov/-/media/Project/Nyserda/Files/Publications/Research/Transportation/Electric-Bicycle-Share-Feasibility-Study.pdf>.
- ODOT. 2023. "Chapter 14 Multimodal Analysis." *Analysis Procedures Manual*. <https://www.oregon.gov/odot/Planning/Documents/APMv2.pdf>.
- PeopleForBikes, Bennett Midland, Portland State University, and The Summit Foundation. 2020. *Economic Impacts of Bicycle and Pedestrian Street Improvements*. <https://www.peopleforbikes.org/reports/the-national-study-of-the-economic-impact-of-street>.
- Persaud, B., B. Lana, C. Lyon, and R. Bhim. 2010. "Comparison of empirical Bayes and full Bayes approaches for before-after road safety evaluations." *Accident Analysis & Prevention* 42(1), 38–43. Available from: <https://www.cmfclearinghouse.org/detail.php?facid=2841>.
- Porter, C., D. Kall, D. Beagan, R. Margiotta, J. Koupal, S. Fincher, and A. Stanard. 2015. *NCHRP Web-Only Document 210: Input Guidelines for Motor Vehicle Emissions Simulator Model, Volume 1: Practitioners' Handbook: Regional Level Inputs*. Transportation Research Board, Washington, DC. <https://doi.org/10.17226/22214>.
- Rail Trail Conversion Advisory Committee. 2015. Home Sales Near Two Massachusetts Rail Trails. August 25, 2015. Accessed November 7, 2022. <http://www.sudbury.ma.us/documents/dl/2470/HomeSalesNear2MARailTrails.pdf>.
- Rosenberger, R.S. 2018. "Part B: Total Net Economic Value from Residents' Outdoor Recreation Participation in Oregon." *Oregon Outdoor Recreation Metrics: Health, Physical Activity, and Value: 2019–2023 Oregon Statewide Comprehensive Outdoor Recreation Plan Supporting Documentation*. Final Report. (Revised). Oregon State University College of Forestry.
- Rosenberger, R.S., E.M. White, J.D. Kline, and C. Cvitanovich. 2017. *Recreation Economic Values for Estimating Outdoor Recreation Economic Benefits from the National Forest System*. U.S. Department of Agriculture, Forest Service. Gen. Tech. Rep. PNW-GTR-9.
- Ryus, P., E. Ferguson, K. M. Laustsen, R. J. Schneider, F. R. Proulx, T. Hull, and L. Miranda-Moreno. 2014. *NCHRP Report 797: Guidebook on Pedestrian and Bicycle Volume Data Collection*. Transportation Research Board of the National Academies, Washington, DC. <https://doi.org/10.17226/2222>.
- Sanders, R., B. Schultheiss, B. Judelman, R. Burchfield, K. Nordback, D. Gelinne, L. Thomas, D. Carter, C. Zegeer, C. Semler, M. Sanders, H. Steyn, P. Ryus, W. W. Hunter, and P. Koonce. 2020. *NCHRP Research Report 926: Guidance to Improve Pedestrian and Bicyclist Safety at Intersections*. Transportation Research Board, Washington, DC. <https://doi.org/10.17226/25808>.

REF-4 Estimating Benefits of Closing Gaps in Active Transportation Networks: A Guide

- Sciortino, S., M. Vassar, M. Radetsky, and M.M. Knudson. 2005. "San Francisco Pedestrian Injury Surveillance: Mapping, Under-Reporting, and Injury Severity in Police and Hospital Records." *Accident Analysis & Prevention* 37(6), 1102–1113. <https://doi.org/10.1016/j.aap.2005.06.010>.
- Semler, C., M. Sanders, C. Dartnell, M. Alston, S. Semensky, L. Ahramjian, K. Taylor, R. Sanders, M. Elbech, Z. Vanderkooy, and North Carolina State University Institute for Transportation Research and Education. 2023. *NCHRP Research Report 1036: Roadway Cross-Section Reallocation: A Guide*. Transportation Research Board, Washington, DC. <https://doi.org/10.17226/26788>.
- Sevtsuk, A., and R. Kalvo. 2022. "Predicting Pedestrian Flow along City Streets: A Comparison of Route Choice Estimation Approaches in Downtown San Francisco." *International Journal of Sustainable Transportation* 16(3), 222–236.
- Sevtsuk, A., R. Basu, X. Li, and R. Kalvo. 2021. "A Big Data Approach to Understanding Pedestrian Route Choice Preferences: Evidence from San Francisco." *Travel Behaviour and Society* 25, 41–51. <https://doi.org/10.1016/j.tbs.2021.05.010>.
- Smart Growth America. 2024. *Dangerous by Design 2024*. https://smartgrowthamerica.org/wp-content/uploads/2024/05/Dangerous-By-Design-2024_5.30.pdf.
- Sundquist, E., C. McCahill, and M. Brenneis. 2021. *Measuring Accessibility: A Guide for Transportation and Land Use Practitioners*. State Smart Transportation Initiative. <https://ssti.us/wp-content/uploads/sites/1303/2020/12/Measuring-Accessibility-Final.pdf>.
- Thomas, L., L. Sandt, C. Zegeer, W. Kumfer, K. Lang, B. Lan, Z. Horowitz, A. Butsick, J. Toole, and R. J. Schneider. 2018. *NCHRP Research Report 893: Systemic Pedestrian Safety Analysis*. Transportation Research Board, Washington, DC. <https://doi.org/10.17226/25255>.
- TOPS Laboratory. n.d. Traffic Operations and Safety Laboratory. Accessed April 9, 2024. <https://topslab.wisc.edu/>.
- U.S. Bureau of Labor Statistics. 2022. "QCEW Location Quotient Details." Quarterly Census of Employment and Wages. September 13, 2022. <https://www.bls.gov/cew/about-data/location-quotients-explained.htm>.
- U.S. DOT. 2024. Bureau of Transportation Statistics. Distribution of Trips by Distance: National, State, and County level. Accessed March 28, 2024. <https://www.bts.gov/browse-statistical-products-and-data/covid-related/distribution-trips-distance-national-state-and>.
- U.S. DOT. 2023a. *Benefit Cost Analysis Guidance for Discretionary Grant Programs*. <https://www.transportation.gov/sites/dot.gov/files/2023-12/Benefit%20Cost%20Analysis%20Guidance%202024%20Update.pdf>.
- U.S. DOT. 2023b. "Departmental Guidance on Valuation of a Statistical Life in Economic Analysis." Effective date: May 1, 2023. <https://www.transportation.gov/office-policy/transportation-policy/revised-departmental-guidance-on-valuation-of-a-statistical-life-in-economic-analysis>.
- U.S. DOT. 2013. *Systemic Safety Project Selection Tool*. FHWA-SA-13-019. July 2013. <https://highways.dot.gov/safety/data-analysis-tools/systemic/systemic-safety-project-selection-tool>.
- U.S. EIA (U.S. Energy Information Administration). 2023. "Annual Energy Outlook 2023." Analysis and Projections. March 16, 2023. <https://www.eia.gov/outlooks/aeo/>.
- U.S. EPA (U.S. Environmental Protection Agency). 2024a. "Latest Version of Motor Vehicle Emission Simulator (MOVES)." MOVES and Related Models. January 18, 2024. <https://www.epa.gov/moves/latest-version-motor-vehicle-emission-simulator-moves>.
- U.S. EPA. 2024b. "GHG Emission Factors Hub." EPA Center for Corporate Climate Leadership. February 22, 2024. <https://www.epa.gov/climateleadership/ghg-emission-factors-hub>.
- U.S. EPA. 2023a. Final Ecosystem Goods and Services (FEGS) Scoping Tool. Last updated October 4, 2023. <https://www.epa.gov/eco-research/final-ecosystem-goods-and-services-fegs-scoping-tool>.
- U.S. EPA. 2023b. "Ecosystem Services Research." Last updated October 31, 2023. <https://www.epa.gov/eco-research/ecosystem-services-research>.
- U.S. EPA. 2018. Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990–2016.
- U.S. EPA. 2017. Light-Duty Automotive Technology and Fuel Economy Trends: 1975 Through 2017.
- U.S. Surgeon General. 2015. *Step It Up! The Surgeon General's Call to Action to Promote Walking and Walkable Communities*. Executive Summary. September 8, 2015. <https://www.hhs.gov/surgeongeneral/reports-and-publications/physical-activity-nutrition/walking-executive-summary/index.html>.
- Vadali, S., C. J. Kruse, K. Kuhn, and A. Goodchild. 2017. *NCFRP Research Report 38: Guide for Conducting Benefit-Cost Analyses of Multimodal, Multijurisdictional Freight Corridor Investments*. Transportation Research Board, Washington, DC. <https://doi.org/10.17226/24680>.
- Vanasse Hangen Brustlin, Inc. 2011. Investigation of Existing and Alternative Methods for Combining Multiple CMFs/Task A.9 T-06-013, Highway Safety Improvement Program Technical Support. https://cmfclearinghouse.fhwa.dot.gov/collateral/Combining_Multiple_CMFs_Final.pdf.
- Volker, J., and S. Handy. 2021. "Economic Impacts on Local Businesses of Investments in Bicycle and Pedestrian Infrastructure: A Review of the Evidence." *Transport Review* 41(4), 401–431. <https://doi.org/10.1080/01441647.2021.1912849>.

- Volker, J., S. Handy, A. Kendall, and E. Barbour. 2019a. *Quantifying Reductions in Vehicle Miles Traveled from New Bike Paths, Lanes, and Cycle Tracks, Technical Documentation*, California Climate Investments Quantification Methods Assessment, California Air Resources Board, Agreement #16TTD004, April 15, 2019.
- Volker, J., S. Handy, A. Kendall, and E. Barbour. 2019b. *Quantifying Reductions in Vehicle Miles Traveled from New Pedestrian Facilities, Technical Documentation*, California Climate Investments Quantification Methods Assessment, California Air Resources Board, Agreement #16TTD004, April 15, 2019.
- VTa (Santa Clara Valley Transportation Authority). 2018. *Santa Clara Countywide Bike Plan*. May 2018. <https://www.vta.org/projects/santa-clara-countywide-bike-plan-update-2018>.
- Winters, M., K. Teschke, M. Grant, E.M. Setton, and M. Brauer. 2010. “How Far Out of the Way Will We Travel? Built Environment Influences on Route Selection for Bicycle and Car Travel.” *Transportation Research Record* 2190(1), 1–10. <https://doi.org/10.3141/2190-01>.
- WHO. 2023. HEAT v5.2.0. July 2023. <http://heatwalkingcycling.org>.
- World Resources Institute. n.d. “Greenhouse Gas Protocol.” <https://www.wri.org/initiatives/greenhouse-gas-protocol>.
- YCEA (York County Economic Alliance). 2023. “Codus Creek Beautification.” <https://www.yceapa.org/strategic-development/special-initiatives/codus-creek-beautification/>.
- YCEA. 2022. *Codus Greenway: Fulfilling a Century-Long Vision for Mobility, Equity & Economic Opportunity*. FY 2022 U.S. DOT RAISE Grant Application. Submitted by the York County Economic Alliance, York, Pennsylvania.
- Yang, H., Y. Bao, J. Huo, S. Hu, L. Yang, and L. Sun. 2022. “Impact of Road Features on Shared E-Scooter Trip Volume: A Study Based on Multiple Membership Multilevel Model.” *Travel Behaviour and Society* 28, 204–213.
- Zhang, W., R. Buehler, A. Broaddus, and T. Sweeney. 2021. “What Type of Infrastructures Do E-Scooter Riders Prefer? A Route Choice Model.” *Transportation Research Part D: Transport and Environment* 94, 102761.
- Zmud, J. P., J. L. Paasche, M. Zmud, T. J. Lomax, J. Schofer, and J. Meyer. 2009. *NCHRP Report 610: Communication Matters: Communicating the Value of Transportation Research*. Transportation Research Board of the National Academies, Washington, DC. <https://doi.org/10.17226/14259>.



Acronyms and Abbreviations

AADT	annual average daily traffic
ACS	American Community Survey
AEO	Annual Energy Outlook
ATROI	Active Transportation Return on Investment
BCA	benefit cost analysis
BCA Guidance	<i>Benefit Cost Analysis Guidance for Discretionary Grant Programs</i>
BIPOC	Black, Indigenous, People of Color
BMT	bicycle miles traveled
CBG	census block group
CMAQ	Congestion Mitigation and Air Quality
CMF	crash modification factor
CO ₂	carbon dioxide
COI	cost of illness
CRIS	Crash Records Information System
DOT	department of transportation
EB	Empirical Bayes
EPDO	equivalent property damage only
FARS	Fatality Analysis Reporting System
FEGS	Final Ecosystem Goods and Services
FEMA	Federal Emergency Management Agency
HAWK	high-intensity activated crosswalk
HEAT	Health Economic Assessment Tool
HIN	high-injury network
HPMS	Highway Performance Monitoring System
IIJA	Infrastructure Investment and Jobs Act
ITHIM	Integrated Transportation and Health Impact Modeling
KABCO	traffic crash injury classification scale, where K = fatal injury, A = suspected serious injury, B = suspected minor injury, C = possible injury, O = no apparent injury
LBS	location-based services
LEHD	longitudinal employer-household dynamics
LODES	LEHD origin-destination employment statistics
LPI	leading pedestrian interval
LTS	level of traffic stress
MET	metabolic equivalent of task
MNL	multinomial logit
MOVES	Motor Vehicle Emission Simulator

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mph	miles per hour
MPO	metropolitan planning organization
MRS	marginal rate of substitution
N/A	not applicable
NACTO	National Association of City Transportation Officials
NHTS	National Household Travel Survey
NLCD	National Land Cover Database
NYSERDA	New York State Energy Research and Development Authority
OD	origin/destination
ODOT	Oregon Department of Transportation
OSM	OpenStreetMap
PHB	pedestrian hybrid beacon
QCEW	Quarterly Census of Employment and Wages
RAISE	Rebuilding American Infrastructure with Sustainability and Equity
RP	revealed preference
RR	relative risk
SLD	Smart Location Database
SPF	safety performance function
SS4A	Safe Streets and Roads for All
SWITRS/TIMS	Statewide Integrated Traffic Records System/Transportation Injury Mapping System
TAZ	transportation analysis zone
TIGER	Topologically Integrated Geographic Encoding and Referencing System
U.S. EPA	Environmental Protection Agency
U.S. FS	United States Forest Service
U.S. GS	U.S. Geological Survey
VMT	vehicle miles traveled
VRU	vulnerable road user
VTA	Valley Transportation Authority
WHO	World Health Organization
WMT	walk miles traveled

Abbreviations and acronyms used without definitions in TRB publications:

A4A	Airlines for America
AAAE	American Association of Airport Executives
AASHO	American Association of State Highway Officials
AASHTO	American Association of State Highway and Transportation Officials
ACI-NA	Airports Council International-North America
ACRP	Airport Cooperative Research Program
ADA	Americans with Disabilities Act
APTA	American Public Transportation Association
ASCE	American Society of Civil Engineers
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
ATA	American Trucking Associations
CTAA	Community Transportation Association of America
CTBSSP	Commercial Truck and Bus Safety Synthesis Program
DHS	Department of Homeland Security
DOE	Department of Energy
EPA	Environmental Protection Agency
FAA	Federal Aviation Administration
FAST	Fixing America's Surface Transportation Act (2015)
FHWA	Federal Highway Administration
FMCSA	Federal Motor Carrier Safety Administration
FRA	Federal Railroad Administration
FTA	Federal Transit Administration
GHSA	Governors Highway Safety Association
HMCRP	Hazardous Materials Cooperative Research Program
IEEE	Institute of Electrical and Electronics Engineers
ISTEA	Intermodal Surface Transportation Efficiency Act of 1991
ITE	Institute of Transportation Engineers
MAP-21	Moving Ahead for Progress in the 21st Century Act (2012)
NASA	National Aeronautics and Space Administration
NASAO	National Association of State Aviation Officials
NCFRP	National Cooperative Freight Research Program
NCHRP	National Cooperative Highway Research Program
NHTSA	National Highway Traffic Safety Administration
NTSB	National Transportation Safety Board
PHMSA	Pipeline and Hazardous Materials Safety Administration
RITA	Research and Innovative Technology Administration
SAE	Society of Automotive Engineers
SAFETEA-LU	Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (2005)
TCRP	Transit Cooperative Research Program
TEA-21	Transportation Equity Act for the 21st Century (1998)
TRB	Transportation Research Board
TSA	Transportation Security Administration
U.S. DOT	United States Department of Transportation

Transportation Research Board
500 Fifth Street, NW
Washington, DC 20001

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